

The Coercive Turn Toward Swing States: Vertical Bottlenecks and the Paradox of Competitiveness

Hyungjin Kim *

June 8, 2026

Abstract

Swing states—third-country firms sourcing critical inputs from China (e.g., Korea, Taiwan, the Netherlands, Vietnam, Mexico)—are the target of US coercion. As China becomes more productive, US coercion of swing states grows more, not less, aggressive. As US coercion grows, the welfare loss it imposes on swing states widens. Both relationships follow from bottleneck complementarity: swing-state firms cannot substitute away from cheap Chinese inputs, so voluntary alignment becomes too expensive for the US to buy through subsidies alone—no shift in political-economy parameters required. Past a Chinese-productivity threshold, US policy switches from subsidies alone to subsidies plus coercion. An HS6 panel difference-in-differences finds a positive cross-industry interaction in bottleneck sectors identified by the production-network literature, null or sign-flipped under non-bottleneck lists, as the model predicts. Surveillance frameworks should expect structural rather than transient coercion over swing states in bottleneck sectors as Chinese productivity continues to rise—intensity scales with that productivity, not with political-economy parameters.

Keywords: Geoeconomics, Vertical Supply Chains, Coercion, US-China Rivalry, Entity List.

JEL Classification: F13, F51, L14, O25.

*Korea Development Institute, hyungjinkim@kdi.re.kr.

1 Introduction

Since 2018, US coercion of swing-state allies—Korean, Taiwanese, Dutch, Vietnamese, Mexican firms in critical-input sectors—has only intensified across administrations. Conventional accounts attribute this to political-economy shifts: Trump-era populism, Biden-era industrial-policy revival, or hawkish coalition realignment. I show the same turn emerges without any of them. With political-economy parameters held fixed, an upstream productivity gain in bottleneck inputs forces a downstream Hegemon to escalate coercion on its swing-state allies. *Swing states*—third-country firms sourcing critical inputs from China (e.g., Korea, Taiwan, the Netherlands, Vietnam, Mexico)—bear a welfare cost that widens with Chinese productivity. That escalation has taken the form of Entity Lists, export controls, and market-access restrictions on swing-state firms, layered atop the post-Cold-War subsidy toolkit (CHIPS, IRA, Bipartisan Infrastructure).

Why coercion in tandem with subsidies, rather than larger subsidies alone? The mechanism is bottleneck complementarity in critical inputs. Chinese suppliers dominate rare earths, advanced semiconductors, precision chemicals, and aerospace alloys; non-Chinese sources are poor substitutes that would take years and capital to scale to Chinese-supplier volumes (Boehm et al., 2019; Atalay, 2017).¹ Formally, when the cross-input elasticity is below 1 (Assumption 2 below), a fall in the Chinese-input price *steepens* the marginal cost of decoupling rather than flattening it. If alternatives were easy substitutes, swing states could shift away on price, and rising Chinese productivity would not force the US to subsidize harder. Since the bottleneck case holds, the subsidy required to keep swing states aligned rises as China becomes more productive—until the security gain from alignment exceeds its economic cost, opening the *Strategic Gap* and prompting coercion to supplement inducement.

I model this argument as a three-stage game over a triangular supply chain. The game is static within the policy period—Chinese productivity is exogenous—and the 2018–2024

¹Examples: Australian and US rare-earth miners (Lynas, MP Materials), Japanese precision chemicals (Shin-Etsu, JSR), German precision optics (Zeiss for ASML lithography), and US tier-1 aerospace alloys.

comparative statics are read as snapshots of different productivity regimes. The Upstream Rival (China) sells critical inputs at a markup; Hegemon U (the US) decides market access and coercion intensity; the Swing State chooses how much of its sourcing to align with the US-aligned chain. The triangular structure isolates two asymmetric sources of market power: China’s leverage is upstream (low wages, high manufacturing productivity, near-monopoly on rare-earth elements); the US’s leverage is downstream (dollar hegemony, the world’s largest final-goods market). Swing states—Korea, Taiwan, the Netherlands, Vietnam, Mexico (operationalized as a 17-economy set in Section 2)—navigate this trade-off. I term this configuration the *Vertical Trap*.

The model delivers three main results. The first is a *Regime Switch* (Proposition 1): below a Chinese-productivity threshold, the Hegemon purchases alignment through subsidies alone; above it, coercion supplements subsidies. The boundary is a kink—coercion rises continuously from zero once the threshold is crossed. The second is the *Paradox of Competitiveness* (Proposition 2): within the coercive regime, more productive Rivals invite harsher coercion, not more subsidies.² The third is the *Tragedy of Efficiency* (Proposition 3): upstream productivity gains, normally welfare-improving, here widen the welfare loss against the planner’s first-best as coercion intensifies.

Three empirical pieces ground the model’s predictions. The first tests the *Regime Switch*: US Entity-List enforcement against China jumps in 2018, with placebo countries (Iran, third-country aggregate, pre-Crimea Russia) showing no comparable break (Section 4.1). The second tests the *Paradox of Competitiveness*: a cross-industry panel regression finds a positive bottleneck-amplification effect only in literature-classified bottleneck sectors—null or sign-flipped under non-bottleneck alternatives, as the model predicts (Section 4.2). The third is an event study around the May 2020 Foreign Direct Product Rule against Huawei: the China share of swing-state advanced-chip exports fell about 19% even as their total kept rising—a

²This paper’s *Paradox of Competitiveness* names a positive model result about a downstream Hegemon’s optimal coercive policy in response to upstream productivity gains; it is unrelated to Krugman (1994)’s normative critique of “national competitiveness” as policy discourse.

partial reallocation. The triple-difference placebos turn out themselves coercion-affected, so this evidence is descriptive rather than identifying (Section 4.3).

This paper differs from the closest geoeconomics literature in three dimensions: the discrete regime-switch shape of coercion (vs continuous network amplification), the triadic supply-chain structure (vs bilateral framings), and the contractual coercion mechanism (vs unilateral instruments). The closest theoretical neighbor is Clayton et al. (2024, 2026), who derive hegemonic coercion from supply-chain network effects with continuous amplification; this paper produces a sharp switch instead, coercion turning on past a productivity threshold. The closest macroeconomic frame is Itskhoki and Mukhin (2025) on bilateral sanctions; this paper instead has three players. The closest empirical analog is Liu et al. (2024) on unilateral sabotage; this paper’s mechanism is contractual rather than unilateral, with coercion conditional on swing-state alignment. A methodological note: applying patent-based productivity measurement to geoeconomic trade-share regressions sidesteps the mechanical-correlation problem inherent in trade-only data, building on Griliches (1990) and Akcigit et al. (2026). The broader geoeconomics literature on fragmentation and sectoral natural experiments (Aiyar et al., 2023; Mohr and Trebesch, 2025; Alfaro et al., 2025; Crosignani et al., 2024) provides policy and empirical context. The mechanism behind this sharp switch is the subject of the next literature.

This paper’s contribution to the bottleneck-contracting literature is the endogenous derivation of that regime switch from bottleneck complementarity meeting a binding inducement budget, formalized as the *Strategic Gap* (Section 3.3). The model places the US and China in competition for swing-state firms (Bernheim and Whinston, 1986), with one side (China) controlling an essential input. When one principal holds an essential input, the competing principal cannot win the firm through subsidies alone, and coercion becomes the next-best option. The bottleneck assumption is empirically grounded in Boehm et al. (2019); Atalay (2017); Baqaee and Farhi (2024), which document low substitutability for the sectors this paper targets. Section 2 documents the Vertical Trap and Coercive Turn

descriptively; Section 3 develops the model; Section 4 reports the formal tests; Section 5 quantifies welfare; Section 6 concludes.

2 Empirical Patterns

Two stylized patterns from the data motivate the model. The first is the *Vertical Trap*: swing states have become structurally more dependent on Chinese inputs over 2000–2024, even as the US has tried to decouple. The second is the *Coercive Turn*: post-2018, the US intensified Entity-List enforcement against China-bilateral firms (a tenfold rise in annual flow, formally tested in Section 4.1), while a pre-existing extra-bilateral coercive perimeter continued to reach swing-state-domiciled entities. I document both patterns descriptively here; formal break tests and dose-response regressions are deferred to Section 4.

Empirical signature of the Vertical Trap. Figure 1 plots the swing-state aggregate’s two principal trade shares. The red line—imports sourced from China (upstream dependence)—more than triples, rising from 6% in 2000 to 22% in 2024. The blue line—exports destined for the US (downstream concentration), constructed via mirror data³—is essentially flat at 12–13%. Upstream dependence rises threefold; downstream concentration does not move.⁴

Two natural alternatives to the structural-integration reading both fail.

³Comtrade reports imports only; I measure swing-state exports to the US via USA’s mirror-reported imports, normalized by the swing state’s total imports from the world (a denominator that approximates trade-volume scale, since imports and exports balance at the country level).

⁴Inclusion rule: 17 non-US, non-China economies with non-trivial Chinese-input or US-export ties (KOR, TWN, VNM, MEX, NLD, SGP, MYS, THA, IND, JPN, DEU, IDN, HUN, CZE, POL, PHL, BRA). This 17-economy set is the holistic “swing-state” aggregate used in §2’s descriptive figures. The downstream regression specifications in Section 4 carve this set further according to the regression’s exporter-group construction: the Paradox panel (Test 2, Section 4.2) tabulates the swing-state share against an ALLY comparison group (KOR, JPN, DEU, NLD, USA), so KOR, JPN, DEU, NLD enter the denominator as ALLY rather than the numerator as SWING; the SWING-numerator subset for Test 2 is therefore the 13-economy remainder (TWN, VNM, MEX, SGP, MYS, THA, IND, IDN, HUN, CZE, POL, PHL, BRA). The firm-listing column of Table 3 uses a separate 17-economy operationalization with HKG, ARE, TUR, PAK replacing the four ally exporters, reflecting the Entity-List data’s allocation by jurisdiction of listing rather than HS6 export incidence. Both deviations from the headline 17-set are reported here in the interest of transparency; the substantive conclusions are robust to either operationalization (Appendix C.6).

Aggregation across high- and low-stakes sectors. If the rise were carried by easily substitutable commodity sectors (textiles, basic chemicals, furniture), restricting attention to chokepoint sectors should attenuate it. It does the opposite. Panel (b) restricts to critical and strategic sectors only;⁵ the chokepoint subaggregate *sharpens*, with the upstream-downstream crossover occurring earlier (~ 2008) and the post-2018 acceleration wider. The pattern is concentrated, not diluted, in sectors that the ex-ante classification identifies as bottleneck-prone.

Selection in the swing-state set. If the rise were driven by one or two outliers, dropping the largest contributors should pull the aggregate down sharply. It does not. Appendix C.5 (Figure 11) shows that excluding Vietnam, Mexico, and Korea—the three largest post-2018 contributors—leaves the 2024 aggregate within ≈ 1 percentage point of the full series (21.96% $\rightarrow$ 21.11%), and the trajectory is preserved.

The pattern is therefore broadly distributed and concentrated in the ex-ante bottleneck-prone sectors—consistent with, though not by itself identifying, upstream complementarity in critical inputs. Swing states remain anchored to what observers have called the “Red Supply Chain” despite a decade of geopolitical pressure.⁶

The Vertical Trap is the structural-dependence side of the puzzle; I turn now to the policy-instrument side.

The Coercive Turn. I define the *Coercive Turn* as the post-2018 shift in US strategic-trade enforcement against the China supply chain. It unfolds in three waves: a Section-301 tariff regime in 2018; an Entity-List intensification in 2019–2020 targeting Chinese counterparties (Huawei, SMIC), forcing swing-state firms to comply but rarely listing them directly; and an omnibus 2022 wave that combined CHIPS Sec. 9902 guardrails, IRA Sec. 30D battery

⁵Critical and strategic sectors follow the OECD ICIO ISIC4 taxonomy mapped to the bottleneck buckets defined in Appendix C.1: semiconductors and electronic equipment (D26), mining of metal ores (D07T08), batteries and electrical equipment (D27), machinery (D28), motor vehicles (D29), pharmaceuticals (D21), and aerospace (D30); commodity sectors (textiles, food, basic chemicals) are excluded.

⁶The same pattern appears in IMF Direction of Trade Statistics (DOTS) bilateral trade shares and in the trade-reallocation evidence of International Monetary Fund (2023, Chapter 4).

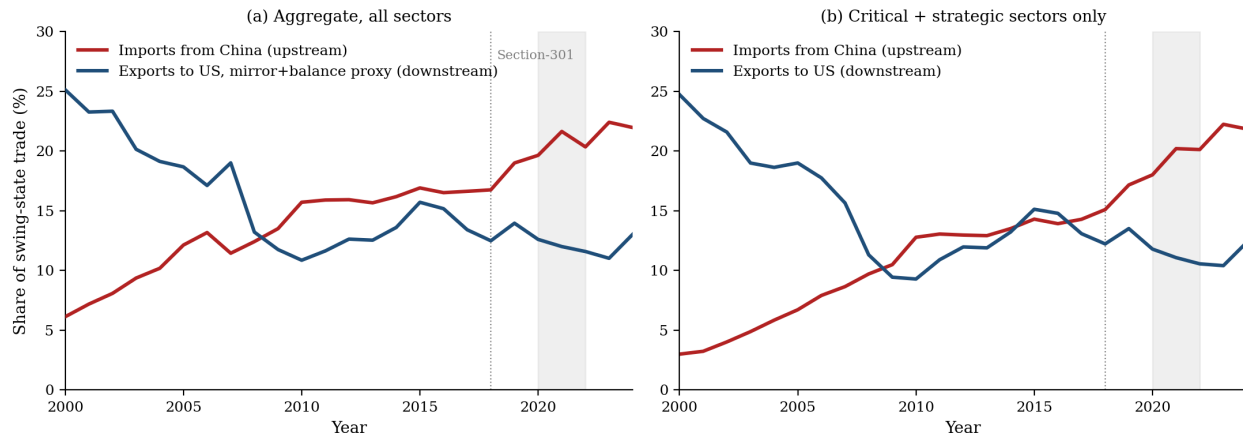


Figure 1: **The Vertical Trap.** Aggregated trade shares of swing states (non-US/non-China). *Panel (a)*: all sectors. Red is the swing-set’s aggregate share of imports sourced from China; blue is the share of US-bound exports (mirror+balance proxy). *Panel (b)*: critical and strategic sectors only (D26 + D07T08 + D27 + D28 + D29 + D21 + D30; see Appendix C.1). Grey shading marks 2020–2022 (COVID inventory rebuild). Dotted vertical: Section-301 (2018). Per-country selection robustness is shown in Appendix C.5 (Figure 11). Source: UN Comtrade, 2000–2024.

FEOC rules, and the October 7 Advanced Computing Controls to constrain swing-state firms directly. Table 1 catalogs the major named-firm actions.

The Coercive Turn is not the invention of a coercive instrument set but a shift in target. US licensing infrastructure was operational throughout the post-1996 period and was actively deployed against swing-state firms in service of non-China objectives—Korean refiners and Japanese trading houses under the Iran Sanctions Act (CISADA 2010; ITRA 2012); German industrials and banks under the Crimea sanctions (2014) and the Commerzbank Iran-Sudan settlement (2015); Korean exporters under DPRK intensification (2016–17); a continuing Iran-snap-back and Russia/Ukraine wave through 2018–2024. What changed in 2018 is the supply chain the existing infrastructure was turned toward, with expanded variants—CHIPS Sec. 9902 guardrails, IRA Sec. 30D FEOC, October 7 Advanced Computing Controls—arriving in the 2022 omnibus wave. The catalog in Table 1 is a floor on this retargeting; routine trade-policy coercion of swing states (Section-232 steel/aluminum tariffs, antidumping and countervailing-duty cases, Section-201 safeguards, currency-manipulator designations, USTR Special-301 IP actions) is out of scope because it is not part of the

Year	Instrument	Action affecting swing-state firms
2018	Tariff	Section-301 List 1 tariffs (Jul) on TWN/KOR/JPN/DEU electronics
2019	Entity List	Huawei + 68 affiliates listed (May); TSMC, Samsung as suppliers forced to comply
2020	FDPR	Foreign Direct Product Rule, full scope (May): TSMC, Samsung Foundry, SK Hynix subject
	Entity List	SMIC Entity-Listed (Oct); ASML, Tokyo Electron, equipment suppliers indirectly affected
2022	Subsidy	CHIPS Sec. 9902 guardrails (Aug): TSMC AZ, Samsung TX, SK Hynix banned from leading-edge expansion in China
	Subsidy	IRA Sec. 30D FEOC battery rules (Aug): LGES, Samsung SDI, SK On disqualified from \$7,500 EV credit on Chinese-sourced components
	FDPR	October 7 Advanced Computing Controls: TSMC, Samsung, SK Hynix advanced-node logic and memory subject to US licensing
2023	Allied	Netherlands EUV/DUV controls (Jun) coordinated with US: ASML subject to Dutch implementing regulation
	Entity List	ATP controls expansion (Oct); additional listings and equipment-supply restrictions
2024	FDPR	HBM advanced-chip controls (Dec): SK Hynix, Samsung HBM exports to China subject to US licensing

Table 1: Major US coercive actions touching swing-state firms, 2018–2024. The 2018 tariff wave hits exporters; the 2019–2020 Entity-List wave operates indirectly through compliance pressure; the 2022 omnibus wave (CHIPS guardrails + IRA FEOC + October 7) is the inflection point for direct constraint on swing-state firms. Sources: US Bureau of Industry and Security; CHIPS Act of 2022; Inflation Reduction Act Section 30D; Dutch Ministry of Economic Affairs.

China-supply-chain redirection mechanism the paper analyzes.

Direct legal listings of swing-state firms remain modest: cumulative Entity-List flows on the 17-economy swing-state set rise from 15 in 2008 to 159 in 2024 (12–14%/yr throughout the period), reflecting the extra-bilateral perimeter that has long reached swing-state firms but does not exhibit a sharp post-2018 break in the listing data alone.⁷

The aggregate quantitative signature of the Coercive Turn is visible in raw Entity-List

⁷The mechanism the model formalizes is coercion intensity λ that rises continuously with the Rival’s productivity gap. The empirical signature visible in direct legal listings is only one channel; the dominant channel is extraterritorial licensing reach (FDPR), conditional-subsidy guardrails (CHIPS Sec. 9902, IRA Sec. 30D), and allied coordination (Netherlands EUV controls). Section 4.3 identifies the FDPR-Huawei firm-level response.

flow data. Figure 2 plots annual new US Entity-List listings against China-bilateral firms (CHN + HKG), 2005–2024, on linear scale. Pre-2018, annual flow ranges between zero and 23 listings, with median annual flow of 13. From 2018 onward the flow accelerates discretely: 27 in 2018 (Section-301 year), 94 in 2019 (the May Huawei listing), 141 in 2020 (the SMIC listing and the August FDPR), then sustained at 66–249 through 2024, with the largest single-year cohort (249 listings) in 2024. Mean annual flow against China rises tenfold from 11 listings/year pre-2018 to 110 post-2018. Section 4.1 reports the formal break test with placebo discipline: against three placebo country groups (Iran, Russia pre-2022, and a non-CHN-non-RUS-non-IRN third-country aggregate) only China rejects the no-break null, at one-sided HAC $p = 0.009$.

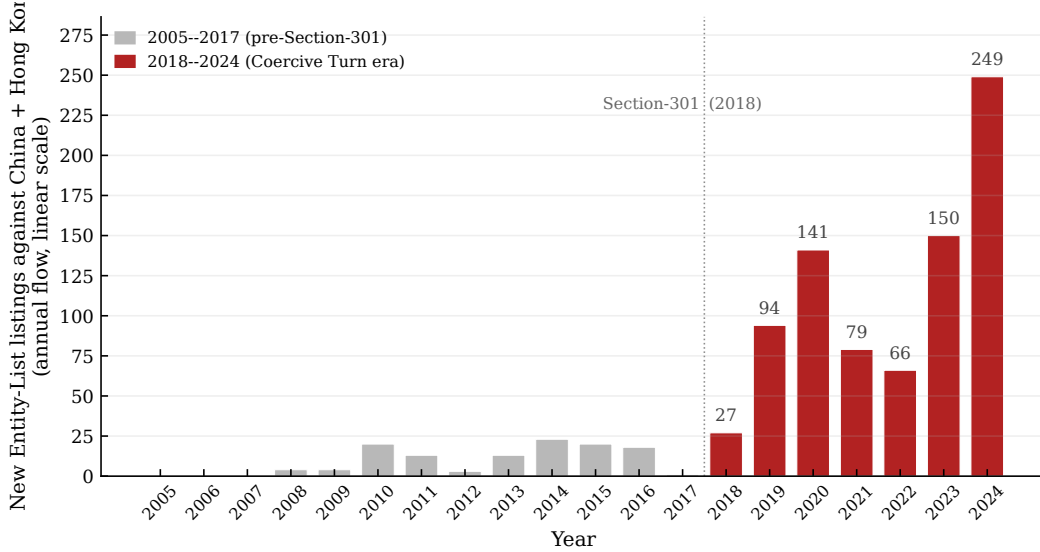


Figure 2: **The Coercive Turn in annual Entity-List flow against China.** New US Bureau of Industry and Security Entity-List listings against China-bilateral firms (CHN + HKG combined) by year of listing, 2005–2024, on linear scale. Pre-Section-301 years (gray) range between zero and 23 listings per year (median 13); post-Section-301 years (red) range between 27 and 249, with mean annual flow rising tenfold (from 11/yr pre-2018 to 110/yr post-2018). The dotted vertical marks Section-301 (2018). Section 4.1 reports the formal break test with placebo discipline against Iran, Russia (pre-2022), and the non-CHN-non-RUS-non-IRN third-country aggregate. Source: US Bureau of Industry and Security Entity List.

The two patterns motivate the model’s empirically anchored primitives. The Vertical Trap motivates Assumption 2: critical-input complementarity ($\sigma_c < 1$) between Chinese and non-

Chinese intermediates—with §2’s trade-share evidence documenting the footprint of low elasticity, and the structural parameter calibrated independently from the 2019 Japan–Korea chemicals natural experiment in Appendix C.2. The Coercive Turn motivates the policy primitive: the Hegemon’s joint Stage-2 choice over inducement $R(x)$ and a China-supply-chain-specific coercion intensity λ , with the post-2018 slope-acceleration corresponding to λ activating from the *Generous Empire* corner ($\lambda = 0$) to the *Coercive Hegemon* interior ($\lambda > 0$). Inducement does not retreat in this transition—CHIPS Sec. 9902 and IRA Sec. 30D in 2022 demonstrate subsidies expanding alongside coercion—consistent with the model’s structure in which the Coercive Hegemon regime supplements rather than replaces inducement. Additional structural choices (Stackelberg ordering, the security-cost coefficient γ , the linear V_{US} , the kill-switch microfoundation for $\lambda x\theta$) are not anchored in §2 directly; Section 3.7 relaxes each in turn. Section 4 returns to the data after the model with formal break-test evidence (Section 4.1), the Paradox dose-response in bottleneck sectors (Section 4.2), and the FDPR firm-level test (Section 4.3).

3 Model

I formalize the bottleneck primitive that delivers the Vertical Trap of Section 2: upstream complementarity that resists substitution. Section 3.1 fixes the environment. Sections 3.2–3.4 derive each player’s best response, solving the three-stage game by backward induction; the Stackelberg leadership Hegemon C $\rightarrow$ Hegemon U $\rightarrow$ Swing State (Assumption 1) makes this the natural method, since the Swing State’s Stage-3 problem admits a closed-form best response that the upstream stages then take as given. Section 3.5 establishes the *Paradox of Competitiveness*: an exogenous rise in upstream efficiency triggers downstream coercion in equilibrium. Section 3.6 quantifies the resulting welfare loss. Section 3.7 closes with extensions that do not alter the core results. All proofs are deferred to Appendix B; microfoundations of $R(x)$, $\bar{T}$, and γ are in Appendix A.

3.1 The Model Environment

I model a global economy characterized by Vertical Geoeconomic Rivalry. Unlike standard trade models that depict competition between symmetric nations, I treat the global order as a vertical supply chain in which power derives from position: control over upstream inputs versus control over downstream market access.

3.1.1 Agents, Timing, and Commitment

Agents. The economy consists of three strategic players arranged hierarchically. First, **Hegemon C (Upstream)**—the *Rival* (e.g., China)—acts as a monopolist supplier of a critical intermediate input, with the input price w_C as its strategic instrument. Its underlying technological productivity is $\theta > 0$ (an exogenous primitive of the Stage-1 problem); the export price w_C decomposes into a marginal cost $\bar{w}/\theta$ and an endogenous strategic markup m , formalized in Section 3.4. Second, **Hegemon U (Downstream)**—simply the *Hegemon* in the main text (e.g., the USA)—acts as a monopsonist buyer of final goods and provider of essential intellectual property; its strategic instruments are the market access schedule $R(x)$ (a contingent contract on the realized alignment, discussed in detail below) and the strategic doctrine λ . Finally, **The Swing State (Midstream)** is a representative firm or sector k (e.g., South Korea) that transforms intermediate inputs into final goods. It chooses the alignment $x \in [0, 1]$, which represents the degree of supply chain decoupling.

Timing. The interaction unfolds in three stages. In **Stage 1**, Hegemon C sets the export price of critical inputs w_C to maximize a weighted sum of commercial profits and geopolitical influence. In **Stage 2**, observing w_C , Hegemon U determines its strategic doctrine λ (the weight placed on security) and announces a market access schedule $R(x)$ to induce alignment. Finally, in **Stage 3**, the Swing State observes the input prices (Chinese input w_C and the Alternative-supplier price w_A of non-Chinese inputs) and Hegemon U’s policy schedule $R(x)$,

then chooses the alignment x to maximize profit.⁸

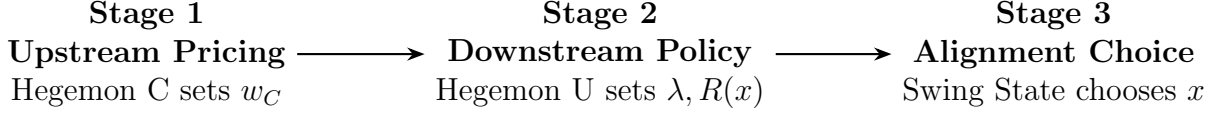


Figure 3: The Timeline of Vertical Rivalry

Assumption 1 (Commitment and Information). *The extensive form in Figure 3 operates under two informational and institutional restrictions. (i) Sequential commitment: At each stage, the acting player commits to their strategic variable before the next player moves. In particular, Hegemon U commits to the pair $(\lambda, R(\cdot))$ prior to the Swing State’s alignment choice, and this commitment is binding within the game. (ii) Complete information: All structural parameters are common knowledge. These include the input prices, the production-technology primitives introduced in Section 3.1, and the fiscal-cost and outside-option parameters introduced with Hegemon U’s and the Swing State’s problems below.*

Assumption 1 plays a central role: the credibility of ex-ante commitments in geoeconomics—sanctions threats, security guarantees, access regimes—is itself an institutional achievement, not an automatic feature of the extensive form. I take it as primitive; its shadow cost is formalized when Hegemon U’s problem is specified in Section 3.3.

The schedule $R(x)$ is a *contingent contract*, not a linear spot price: Hegemon U commits to pay $R(x)$ to the Swing State conditional on realized alignment. Conceptually, $R(x)$ stands for strategic foreign-subsidy policies that the Hegemon uses to pull the Swing State toward

⁸The Stackelberg ordering Hegemon C $\rightarrow$ Hegemon U $\rightarrow$ Swing State is a modeling choice that reflects three empirical primitives. (i) Upstream pricing has a longer commitment horizon than downstream policy: Chinese export prices on chemicals and rare earths are negotiated in annual or multi-year contracts (e.g., the Tōhoku-period $\hat{\sigma}_c$ literature documents pricing rigidity at business-cycle frequencies), while US Entity-List, FDPR, and CHIPS-guardrail policy adjusts on a quarterly cycle. (ii) The empirically observed sequence is China-side capacity build-up (early 2010s) preceding US instrument-set expansion (post-2018); a reverse Stackelberg would require the US to commit to a coercion doctrine before Chinese productivity is realized. (iii) The Paradox of Competitiveness is robust to alternative orderings: under Nash bargaining (Appendix B.3, Proposition 4) $d\lambda^*/d\theta > 0$ continues to hold, with the threshold $\bar{\theta}(\nu)$ strictly decreasing in the US bargaining weight ν . The literature’s leading alternative, Hegemon-U-as-leader (Clayton et al., 2024), is nested at $\nu = 0$ in the bargaining extension—the baseline take-it-or-leave-it case (Appendix B.3); symmetric Nash bargaining at $\nu = 1/2$ yields the same Paradox direction.

its own bloc—transfers, tariff preferences, and access rights whose value to the Swing State rises with the depth of decoupling from the Rival.

Concrete real-world instruments include (i) U.S. CHIPS-Act subsidies to Korean and Taiwanese fabs, conditional on ten-year “guardrails” restricting leading-edge capacity expansion in China; (ii) the Inflation Reduction Act’s \$7,500 EV tax credit, conditional on excluding “foreign entity of concern” battery components and critical minerals; and (iii) USMCA automotive rules of origin, where preferential tariff treatment requires a minimum North American value-added share. In each case the dollar value the Swing State receives rises with the demonstrated depth of decoupling—the empirical content of $R(x)$ increasing in x .

This contractual structure shapes how the alignment x enters each stage. On the Swing State’s side (Stage 3, Section 3.2), x is the contracted action, not an input choice solving a cost-minimization problem: the Swing State is an institutional entity that selects an alignment *policy*, and the resource penalty it incurs (formalized in Section 3.1.2) is the cost of *operating* a supply chain at the policy-imposed input mix. On Hegemon U’s side (Stage 2, Section 3.3), $R(x)$ is the contractual transfer paid for delivery at alignment x , the counterpart to the gross benefit Hegemon U derives. Appendix A.3 derives the reduced form $R(x) = C(x; w_C) + V_{out}$ from the binding participation constraint and documents concrete implementations.

3.1.2 Technology: Supply Chain and Decoupling Cost

The Swing State sources inputs from two distinct pools. The first is the *Red Supply Chain*, inputs I_C obtained from Hegemon C at price w_C . The second is the *Blue Supply Chain*, inputs I_A obtained from the rest-of-world non-rivalry input pool at the world-market price w_A .⁹ The Swing State chooses a geopolitical alignment $x \in [0, 1]$, defined as the physical

⁹I model the Blue Supply Chain as a competitive backstop rather than a fourth strategic player. Its participants are price-takers without a strategic instrument, so w_A enters the model as an exogenous parameter rather than as the equilibrium outcome of an additional optimization. Game-theoretically, the Blue Supply Chain matters only as the Swing State’s outside option—which in turn constrains the strategic markup Hegemon C can charge in Stage 1 (Section 3.4)—and it has no independent payoff to maximize. I also exclude Hegemon U from this pool by construction: including US input producers would open a direct input-sales

share of inputs sourced from the Alternative supply:

$$I_A = xI \quad \text{and} \quad I_C = (1 - x)I, \quad (1)$$

where I is the total aggregate input bundle. x is a *policy-imposed physical allocation share*, not an input choice solving a cost-minimization problem. Let w_C and w_A denote the prices of Chinese and Alternative inputs, respectively. I assume $w_C < w_A$, reflecting Hegemon C's comparative advantage or subsidization.

I model the Swing State's *Decoupling Cost* as a function $C(x; w_C) : [0, 1] \times \mathbb{R}_{++} \rightarrow \mathbb{R}_+$ that measures the resource penalty the Swing State incurs for operating its supply chain at alignment x given an input price w_C charged by Hegemon C. $C(x; w_C)$ is a primitive object of the model: x enters C as a *policy parameter* (the alignment-imposed input mix), not as a control variable of a cost-minimization problem, consistent with the principal-agent framing developed above and formalized in Appendix A.3. The firm-theoretic benchmark, against which I compare the Swing State's contracted alignment, is the laissez-faire share $x^{\text{LF}}(w_C)$ a cost-minimizing agent would select.

I impose the following minimal regularity conditions on C :

Assumption 2 (Decoupling Cost: Bottleneck Regularity). *The Decoupling Cost $C(x; w_C)$ is twice continuously differentiable on $(0, 1) \times \mathbb{R}_{++}$ and satisfies, for every $w_C \in \mathbb{R}_{++}$:*

(C1) *Laissez-faire minimum. There exists a unique $x^{\text{LF}}(w_C) \in (0, 1)$ at which $C'(x^{\text{LF}}; w_C) = 0$; this is the cost-minimizing benchmark a firm-theoretic agent would select.*

(C2) *Strict convexity on the decoupling range. $C''(x; w_C) > 0$ for all $x \in (x^{\text{LF}}(w_C), 1)$.*

(C3) *Explosive marginal decoupling cost. $\lim_{x \rightarrow 1^-} C'(x; w_C) = +\infty$.*

revenue channel from I_A into $V_{US}(x)$, a general-equilibrium spillover that the partial-equilibrium Stage-2 analysis abstracts from. In the data the dominant non-rivalry producers in the chokepoint sectors (advanced semiconductors, batteries, precision components) are themselves countries in the Swing-State set—Japan, Germany, Taiwan, Korea, the Netherlands. The model's representative-Swing-State formulation treats w_A as an exogenous backstop from any individual Swing State's perspective, thereby abstracting from cross-Swing-State input flows; relaxing this abstraction is left to the multi-country general-equilibrium extension discussed in Section 3.7.

(C4) *Chinese-price sensitivity of the gradient.* $\partial C'(x; w_C)/\partial w_C < 0$ for all $x \in (x^{\text{LF}}(w_C), 1)$.

Condition (C1) identifies the laissez-faire reference point. Conditions (C2)–(C3) are the *bottleneck*: beyond the laissez-faire mix, marginal decoupling becomes strictly costly and prohibitively so as the Swing State approaches full decoupling. Condition (C4) is the key comparative-statics primitive: a fall in w_C (a cheaper Chinese input) steepens the marginal Decoupling Cost on the decoupling range, capturing the model’s central economic insight that “China’s Rise makes the Swing State harder to align.” Together (C1)–(C4) describe a chokepoint technology — rare-earth processing, legacy semiconductors, battery-grade precursors — where the Rival’s upstream input cannot be substituted away cheaply. Except where noted, the analysis that follows uses only Assumption 2.

A concrete parameterization that satisfies Assumption 2 arises when the Swing State’s technology aggregates inputs through a Constant Elasticity of Substitution (CES) form with elasticity σ , distribution parameter $\alpha \in (0, 1)$, and $\sigma < 1$ (gross complements). Within this section’s CES manipulations I write σ for brevity; it is the same object as the cross-input elasticity σ_c of Section 2 (Assumption 2). Applying Shephard’s lemma to the CES aggregator evaluated at the alignment mix $(1 - x, x)$ yields the closed form¹⁰

$$C(x; w_C) = \frac{(1 - x) w_C + x w_A}{\left[\alpha (1 - x)^{\frac{\sigma-1}{\sigma}} + (1 - \alpha) x^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}}, \quad (2)$$

with laissez-faire share

$$x^{\text{LF}}(w_C) = \frac{(1 - \alpha)^\sigma w_C^\sigma}{\alpha^\sigma w_A^\sigma + (1 - \alpha)^\sigma w_C^\sigma}.$$

¹⁰The structure of (2)—expenditure on a *policy-forced* input mix deflated by CES effective output, rather than the cost-minimizing dual—has precedent across three literatures. Operating off the cost-minimizing input ridge as a policy-imposed share traces to the theory of domestic-content protection (Grossman, 1981) and, more broadly, to the misallocation wedges generated when inputs are used away from their efficient proportions (Hsieh and Klenow, 2009). The $\sigma < 1$ complementarity that makes a critical input essential—so that cost diverges as that input is forced out—is the weak-link mechanism of Jones (2011). The vertical-decoupling incarnation, in which reshoring a critical upstream input is costly precisely because substitution is limited, follows Grossman et al. (2023). The convex contracted-action cost is itself standard (Grossman and Hart, 1983); the contribution here is the CES-bottleneck microfoundation that ties that convexity to the cross-input elasticity σ_c .

Under $\sigma < 1$ and $w_C < w_A$, properties (C1)–(C4) follow by direct computation (see the proof of Lemma 1 below). I invoke this CES parameterization in Section 5 for the quantitative illustration, but not elsewhere: all analytic results use Assumption 2 alone.

Empirical defense of $\sigma_c < 1$. The bottleneck restriction is empirically grounded in three bodies of work that converge on $\sigma_c < 1$ for the inputs the model targets. Boehm et al. (2019) exploit the 2011 Tōhoku earthquake to estimate a near-zero short-run elasticity of substitution between imported and domestic inputs (statistically indistinguishable from Leontief); Atalay (2017) estimates an across-materials elasticity of approximately 0.1 at the sector level; Baqaee and Farhi (2024) infer macro-level $\sigma_c \in [0.5, 0.7]$ from production-network calibrations. Sector-specific evidence on EUV lithography, advanced-node foundries, and rare-earth processing reaches the same conclusion qualitatively (Khan et al., 2021).¹¹

The calibrated $\sigma_c = 0.35$ is an illustrative value lying between the own-pipeline photoresist bound ($\hat{\sigma}_c \leq 0$, at the Leontief boundary) and the own-pipeline photo-chemicals point estimate ($\hat{\sigma}_c = 0.48$) from the 2019 Japan–Korea natural experiment (Appendix C.2), and Appendix B.7 shows that the qualitative regime-switch behavior is robust across $\sigma_c \in [0.1, 0.5]$.

A clarifying note for readers familiar with the trade-elasticities literature: Feenstra-style GMM (Broda and Weinstein, 2006; Soderbery, 2015) estimates the elasticity of substitution *across exporter varieties* within an HS6 code, a different primitive (typically > 1) from the *cross-input* elasticity that drives the model. Hottman et al. (2016) document this gap explicitly. The $\sigma_c < 1$ targets the input-side primitive. Appendix C.2 reports own-pipeline cross-variety GMM (above unity, as expected from this distinction) and the 2019 Japan–Korea export-control natural experiment as a Boehm-style identification: photoresist (HS 370710) yields $\hat{\beta} = +1.22$ (SE 0.06), implying $\hat{\sigma}_c$ at or below the Leontief boundary; photo chemicals

¹¹The three calibration sources are not all estimating the same object. Boehm et al. (2019) identify the across-source elasticity at the firm level via natural experiment; Atalay (2017) the cross-input elasticity at the sector level via accounting identities; Baqaee and Farhi (2024) the macro-implied cross-input elasticity via production-network calibration. For the σ_c —the elasticity between Chinese and non-Chinese intermediates—Boehm’s specification is the closest match. Atalay and Baqaee-Farhi bound the plausible range from above rather than point-identify σ_c .

(HS 370790) yield $\hat{\beta} = +0.52$ (SE 0.05), implying $\hat{\sigma}_c = 0.48$. Two of the four chemicals tested (HF, fluorinated polyimide) do not yield statistically reliable $\sigma_c < 1$ estimates under wild-bootstrap inference; the photo-chemicals point estimate is the single within-experiment identification of σ_c in the strict sense, and the photoresist result is best read as a bound.

3.2 Stage 3 (Swing State): Alignment Choice

I solve the model via backward induction, beginning with the final stage. The Swing State (Agent) takes the input prices $\{w_C, w_A\}$ and Hegemon U’s policy schedule $R(x)$ as given. It chooses the alignment $x \in [0, 1]$ to maximize its total profit.

If the Swing State rejects Hegemon U’s contract, it reverts to the default arrangement—sourcing inputs from the Rival (“integration with China”) with no US-granted market access. Let V_{out} denote the Swing State’s reservation profit under this fallback. I normalize $V_{out} = 0$ throughout, since an additive constant shifts the policy schedule $R(x)$ one-for-one and is comparative-statically irrelevant; I therefore suppress V_{out} symbolically in the equations of Section 3.3 below. The Swing State accepts Hegemon U’s contract only when its optimized profit is weakly positive.

The Swing State’s profit function $\Pi(x)$ is defined as market access revenue minus the Decoupling Cost,

$$\max_{x \in [0, 1]} \Pi(x) = R(x) - C(x; w_C) \quad (3)$$

Assuming the policy schedule $R(x)$ is differentiable, the optimal alignment x^* is characterized by the first-order condition:

$$R'(x^*) = C'(x^*) \quad (\text{Incentive Compatibility}) \quad (4)$$

This condition implies that to induce any non-zero alignment $x^* > 0$, Hegemon U must provide a marginal market access benefit exactly equal to the marginal cost of decoupling.

The behavior of the Swing State is driven by the shape of the Decoupling Cost $C(x; w_C)$

on the decoupling range $(x^{\text{LF}}(w_C), 1)$ where the principal's contract pushes alignment above the laissez-faire benchmark.

Lemma 1 (The Bottleneck Gradient). *Under Assumption 2, for every w_C and every $x \in (x^{\text{LF}}(w_C), 1)$:*

1. *Monotonicity.* $C'(x; w_C) > 0$; *decoupling beyond the laissez-faire benchmark always incurs a strictly positive marginal cost.*
2. *Explosive marginal cost.* $\lim_{x \rightarrow 1^-} C'(x; w_C) = +\infty$; *fully replacing the upstream input is asymptotically infeasible.*
3. *Input-price sensitivity.* *A fall in the upstream price steepens the Decoupling-Cost gradient: $\partial C'(x; w_C)/\partial w_C < 0$.*

Proof. All three claims are immediate consequences of Assumption 2. Monotonicity follows from (C1)–(C2): $C'(x^{\text{LF}}; w_C) = 0$ and C is strictly convex on $(x^{\text{LF}}, 1)$, so $C'(x; w_C) > 0$ on that range. Explosive marginal cost is (C3). Input-price sensitivity is (C4). $\square$

For the CES parameterization (2), direct differentiation gives $C'(x; w_C) = (w_A - w_C)/A(x) - [(1 - x)w_C + xw_A] \cdot A'(x)/A(x)^2$, where $A(x) = [\alpha(1 - x)^{(\sigma-1)/\sigma} + (1 - \alpha)x^{(\sigma-1)/\sigma}]^{\sigma/(\sigma-1)}$. Under $\sigma < 1$ and $w_C < w_A$: (C1) is the interior first-order condition yielding $x^{\text{LF}}(w_C)$; (C2) follows because the CES aggregator under $\sigma < 1$ is strictly concave in x on $(x^{\text{LF}}, 1)$, so the unit-cost wedge generates strictly convex C ; (C3) follows because $A(x) \rightarrow 0$ as $x \rightarrow 1^-$ (the $\alpha(1 - x)^{(\sigma-1)/\sigma}$ term diverges with exponent $(\sigma - 1)/\sigma < 0$, while the aggregator exponent $\sigma/(\sigma - 1) < 0$ sends A itself to zero), so C and C' diverge; (C4): $\partial C/\partial w_C = (1 - x)/A(x)$, so $\partial C'/\partial w_C = \partial/\partial x[(1 - x)/A(x)] = -[A(x) + (1 - x)A'(x)]/A(x)^2$. On the relevant decoupling range $x \in (x^{\text{LF}}, 1)$ where Hegemon U pushes alignment beyond the laissez-faire share, $A(x) > 0$ and $A'(x)$ is negative (the CES aggregator is decreasing past its peak under $\sigma < 1$). The condition $A(x) + (1 - x)A'(x) > 0$ is what delivers $\partial C'/\partial w_C < 0$; under the calibration of Section 5 this holds throughout the decoupling

range, and Appendix B.7 verifies the same sign across $\sigma \in [0.1, 0.5]$. Thus property (C4) is established on the operationally relevant range under direct verification, not as a global property of the CES form.

Strategic Implication: The lemma implies that “China’s Rise” ($w_C \downarrow$) does not merely make the Swing State richer; it makes the Swing State *harder to align*. The Hegemon U faces a steeper marginal cost curve $C'(x)$ to implement the same target x^* , requiring a more aggressive (steeper) policy schedule $R(x)$.

3.3 Stage 2 (Downstream Hegemon): Policy and Doctrine

In Stage 2, Hegemon U (US) designs the policy schedule $R(x)$ and selects its strategic doctrine λ to maximize national welfare, anticipating the Swing State’s reaction in Stage 3.

The Hegemon’s utility $\mathcal{W}_{US}$ is defined over the realized alignment x , the transfer payment $R(x)$, and the security weight $\lambda \in [0, 1]$. I model the choice of λ not as a fixed preference, but as an endogenous policy decision (“Securitization”) that incurs convex institutional costs:

$$\mathcal{W}_{US}(x, R(x), \lambda) = \underbrace{\lambda x \theta}_{\text{Security Utility}} + \underbrace{(1 - \lambda)[V_{US}(x) - R(x)]}_{\text{Net Economic Surplus}} - \underbrace{\frac{\gamma}{2}\lambda^2}_{\text{Cost of Securitization}} \quad (5)$$

where $V_{US}(x)$ represents the gross economic value generated for the US, such as consumer surplus or IP royalties. I assume $V'_{US}(x) < 0$, reflecting the efficiency loss from decoupling from the most efficient upstream supplier. The parameter $\gamma > 0$ quantifies the institutional resistance to adopting a security-heavy doctrine—fiscal constraints, domestic political polarization, or administrative capacity. Equivalently, γ is the reduced form of the commitment cost implicit in Assumption 1: credibly binding to a deviation from the default market-integration doctrine is itself costly (in the spirit of Rogoff, 1985; Maskin and Tirole, 1999; Aghion and Tirole, 1997).

The multiplicative form $\lambda x \theta$ admits a kill-switch interpretation: with conditional weaponization probability $1 - x$ (so alignment x measures the share of inputs insulated

against a Rival embargo) and per-unit damage $D(\theta) = \theta$, the expected damage averted by alignment is $x \cdot D(\theta)$, weighted by the Hegemon’s endogenous security posture λ (cf. equation (23) in Appendix B.2). The appendix derives the linear scaling $D(\theta) = \theta$ from Assumption 2: a more productive Rival supplies the same physical unit at lower cost ($\bar{w}/\theta \downarrow$), so the Hegemon’s surplus per input unit, and hence the surplus destroyed by withholding, scales with θ . The qualitative comparative-static results (Propositions 1–2) require only $D' > 0$; the linear specification fixes the units of λ and the quantitative magnitudes (notably the threshold $\bar{\theta}$) inherit the curvature of D , calibrated in Section 5. Crucially, the same primitive θ enters Hegemon C’s Stage-1 marginal cost $\bar{w}/\theta$ in (12), so the two roles—cost-reducer for the supplier, damage-magnifier for the consumer—are derived from a single technology parameter rather than co-positing.

The Principal cannot choose x directly; it must induce x through the schedule $R(x)$. To induce a target alignment x , the US must satisfy two constraints:

1. *Incentive Compatibility (IC)*: $R'(x) = C'(x)$.
2. *Participation Constraint (PC)*: $R(x) \geq C(x; w_C)$ (under the $V_{out} = 0$ normalization of Section 3.2).

Since the US minimizes the transfer $R(x)$, the PC binds at the optimum, giving the *Implementation Cost Function*

$$R(x) = C(x; w_C), \tag{6}$$

where $C(x; w_C)$ is the Decoupling Cost introduced in Section 3.1.2 (Assumption 2). Following the standard principal–agent approach with observable actions (Grossman and Hart, 1983), I treat this cost as the effective price of alignment. When the IC condition binds with indifference at the target x^* , I adopt the standard forcing-contract convention: Hegemon U implements x^* via a degenerate menu—a single contracted point plus an off-path penalty—so the schedule is fully described by its value at x^* . A truthful Bernheim-Whinston schedule with $R'(x) = V'_{US}(x)$ everywhere is an equivalent implementation on the realized equilibrium

path; the two give the same on-path predictions for (x^*, R^*, λ^*) and differ only in off-path penalty structure.

Substituting the implementation cost into the objective function and letting $J(x; w_C) \equiv V_{US}(x) - C(x; w_C)$ denote the *Joint Economic Surplus*, the US problem reduces to

$$\max_{x, \lambda} \quad \mathcal{W}_{\text{reduced}} = \lambda x \theta + (1 - \lambda)J(x; w_C) - \frac{\gamma}{2}\lambda^2. \quad (7)$$

The subscript “reduced” marks that $\mathcal{W}_{\text{reduced}}$ is (5) after the binding PC has been substituted in: the transfer $R(x)$ no longer appears explicitly because it has been replaced by $C(x; w_C)$, leaving the joint surplus $J(x; w_C)$ inside the economic-component term and the security/cost terms unchanged. Within Section 3.3 I suppress the w_C argument of J in the FOCs below, since Stage 2 takes w_C as parametric; the dependence is restored in Sections 3.4–3.5 where Hegemon C’s pricing channel is operative.

I characterize the equilibrium by maximizing the reduced objective function (7) with respect to the alignment target x and the security doctrine λ . The solution is determined by the intersection of two reaction functions.

First, consider the optimal induced alignment $x^*(\lambda)$. Taking the security weight λ as given, the US chooses the target x^* to equate the marginal political gain with the marginal economic loss:

$$\frac{\partial \mathcal{W}}{\partial x} = \lambda \theta + (1 - \lambda)J'(x^*) = 0 \quad \implies \quad J'(x^*) = -\frac{\lambda \theta}{1 - \lambda} \quad (8)$$

This condition implies that as the security doctrine becomes more hawkish ($\lambda \rightarrow 1$), the US pushes the target x^* further into the economically inefficient region where the marginal loss of joint surplus $J'(x)$ is high. Upstream productivity θ amplifies this effect: a more productive Hegemon C (higher θ) shifts x^* upward at every level of λ .

Second, consider the endogenous regime choice $\lambda^*(x)$. Differentiating with respect to λ ,

I obtain the optimal doctrine rule:

$$\frac{\partial \mathcal{W}}{\partial \lambda} = x\theta - J(x) - \gamma\lambda = 0 \quad \implies \quad \lambda^*(x) = \min\left\{1, \max\left\{0, \frac{x\theta - J(x)}{\gamma}\right\}\right\}. \quad (9)$$

The bracketed expression makes the box constraint $\lambda \in [0, 1]$ explicit: the lower corner $\lambda^* = 0$ binds when the Strategic Gap is non-positive (Generous Empire, characterized below), while the upper corner $\lambda^* = 1$ binds when $x\theta - J(x) \geq \gamma$. Throughout Sections 3.5–3.6 I maintain γ large enough that the upper corner is slack on the relevant range of θ ; the calibration in Section 5 verifies this directly. A formal sufficient condition is $\gamma > \theta_{\max} - \min_x J(x)$ over the parameter range under consideration.

The Strategic Gap. Define the *Strategic Gap* as the numerator of (9),

$$G(x; w_C) \equiv x\theta - J(x), \quad (10)$$

i.e., the threat-weighted Security Benefit minus the Joint Economic Surplus. The optimal doctrine is then $\lambda^*(x) = G(x; w_C)/\gamma$ on the interior, clipped to zero at the corner. The Strategic Gap is the central comparative-statics object of the model and is referenced repeatedly in Section 3.5: a rising Hegemon C (lower w_C , higher θ) widens it through two channels — it raises the perceived threat intensity θ and compresses the Joint Economic Surplus by raising the implementation cost $R(x) = C(x; w_C)$ of re-routing supply chains.

The regime threshold $\bar{\theta}$. The Strategic Gap pins down a threshold productivity level above which the corner $\lambda^* = 0$ ceases to bind. Let $x^{\text{GE}}(\theta) \equiv \arg \max_{x \in [0, 1]} J(x; w_C^*(\theta))$ denote the Generous-Empire alignment (the Stage-2 choice when $\lambda = 0$ binds; equivalently, the maximizer of joint surplus at Hegemon C’s best-response price). Define

$$\bar{\theta} \equiv \inf\{\theta > 0 : G(x^{\text{GE}}(\theta); w_C^*(\theta)) > 0\}, \quad (11)$$

i.e., the smallest upstream productivity at which the Strategic Gap, evaluated at the Generous-Empire alignment and Hegemon C's Stage-1 best-response price $w_C^*(\theta)$ (derived in Section 3.4), becomes strictly positive. Restricting to x^{GE} rather than maximizing over $[0, 1]$ is essential: under the CES specification $C(x; w_C) \rightarrow +\infty$ as $x \rightarrow 1^-$, so G would diverge to $+\infty$ at the boundary for any $\theta > 0$ and the unrestricted infimum would collapse to zero. The Generous-Empire alignment is the economically relevant evaluation point because it is the Stage-2 best-response when the corner binds; the threshold $\bar{\theta}$ then marks the productivity level at which the security benefit from deviating from x^{GE} first becomes net-positive. Below $\bar{\theta}$, the security benefit is dominated by the joint surplus and Hegemon U is content with $\lambda^* = 0$; above $\bar{\theta}$, an interior $\lambda^* > 0$ becomes strictly optimal. Existence of $\bar{\theta}$ on $(0, \infty)$ follows from continuity of $G(x^{\text{GE}}(\theta); w_C^*(\theta))$ in θ together with two boundary properties: as $\theta \rightarrow \infty$, the security benefit dominates and $G \rightarrow +\infty$; for θ small enough, the security benefit is bounded above by $\theta \cdot x^{\text{GE}}$ while $J(x^{\text{GE}})$ remains uniformly positive, so $G < 0$.

Proposition 1 (The Regime Switch Mechanism). *The equilibrium is the fixed point (λ^*, x^*) of these two reaction functions. The nature of the equilibrium depends on whether upstream productivity θ falls below or above the threshold (11):*

- *Generous Empire ($\theta \leq \bar{\theta}$): The Strategic Gap is non-positive, so $\lambda^* = 0$ and Hegemon U relies on market integration alone.*
- *Coercive Hegemon ($\theta > \bar{\theta}$): The Strategic Gap is strictly positive, so $\lambda^* > 0$. As θ crosses $\bar{\theta}$, λ^* leaves zero continuously with a kink—the corner constraint becomes slack, so the switch is a kink, not a discontinuous jump.*

Lemma 2 (Existence and local stability of the Stage-2 fixed point). *There exist $\bar{\gamma} > 0$ (the global uniqueness bound) and $\bar{\gamma}_\epsilon > 0$ (the local-stability bound on the compact set $\{\lambda \leq 1 - \epsilon\}$) such that the Stage-2 best-response system in (x, λ) admits an interior fixed point that is locally stable whenever $\gamma > \bar{\gamma}_\epsilon$. The global uniqueness condition $\gamma > \bar{\gamma}$ is sufficient but not*

necessary; the calibration in Section 5 satisfies the local-stability condition at the regime-transition threshold. *Proof: Appendix B.1.*

The economic reading is that the institutional cost of securitization γ must dominate the cross-effect between doctrine and alignment, so that the two do not amplify each other into tipping points. The global bound $\bar{\gamma}$ is uninformatively tight in the baseline CES specification because the relevant primitive constant M_2 vanishes as $x \rightarrow 1$; I therefore verify the local-stability condition $\gamma > \bar{\gamma}_\epsilon$ numerically at the regime-transition threshold (Appendix B.1). The qualitative comparative statics of Propositions 1–3 are stated on the compact range $x \leq 1 - \epsilon$, where the Decoupling-Cost gradient stays finite; local stability of the fixed point holds on the doctrine range $\lambda \leq 1 - \epsilon$ (Appendix B.1).

3.4 Stage 1 (Upstream Hegemon): Pricing Strategy

In the first stage, Hegemon C sets the terms of trade for the critical input. To distinguish between technological fundamentals and strategic intent, I decompose the input price into production costs and industrial policy.

Let w_C be the final export price of the input. I decompose this price into an exogenous technology component and an endogenous policy component,

$$w_C = \underbrace{\frac{\bar{w}}{\theta}}_{\text{Marginal Cost}} + \underbrace{m}_{\text{Strategic Markup}} \quad (12)$$

where $\bar{w}$ denotes the baseline factor price. The parameter $\theta > 0$ represents the *Exogenous Technological Productivity*; an increase in θ captures the structural phenomenon of “China’s Rise” or technological catch-up. The variable m represents the *Endogenous Industrial Policy*, which captures the state’s strategic intent. A positive value ($m > 0$) indicates a standard profit-maximizing monopolistic markup, whereas a negative value ($m < 0$) indicates a strategic state subsidy (or dumping) designed to capture market share.

The productivity primitive θ in (12) is the same θ that scaled the Security Utility in Stage 2

(equation (5), Section 3.3). The two roles—cost-reducer here in Stage 1 and threat-magnifier in Stage 2—are economically the same: θ measures how sharp the upstream chokepoint is, and the depth of the bottleneck determines both how cheaply Hegemon C can supply and how dangerous a withdrawal of that supply would be. Under complete information (Assumption 1), Hegemon U observes θ directly and need not infer it from w_C . The microfoundation for the Stage-2 scaling is the probabilistic kill-switch interpretation in Appendix B.2: the potential damage that alignment averts is increasing in upstream productivity, so the security benefit λx in Hegemon U’s utility is scaled by θ .

Normalizing final output $Y = 1$, Hegemon C chooses the markup m (and thus the price w_C) to maximize a weighted sum of commercial profit and geopolitical influence:

$$\max_m \mathcal{W}_C = \underbrace{m \cdot Q_C(x(w_C); w_C)}_{\text{Net Commercial Return}} + \underbrace{\phi(1 - x(w_C))}_{\text{Geopolitical Influence}} \quad (13)$$

where $Q_C(x; w_C)$ denotes the Swing State’s physical demand for Chinese inputs at alignment x and upstream price w_C , a primitive of the production technology of Section 3.1.2. In the CES special case (2), $Q_C = (1 - x)/A(x)$. The objective highlights the trade-off: lowering m (subsidies) incurs a fiscal cost but increases the strategic value $\phi(1 - x)$ by binding the Swing State to the Red Supply Chain.

Hegemon C faces a derived demand curve $Q_C(x(w_C); w_C)$ for its input. Its total elasticity $\varepsilon_D \equiv -d \ln Q_C / d \ln w_C$ combines two channels: (i) the *direct demand response* through the Swing State’s input substitutability, governed by the convexity of the Decoupling Cost (Assumption 2, property (C2); in the CES special case (2), parameterized by σ); and (ii) a *(potential) regime-feedback channel*: by Stackelberg leadership (Assumption 1), Hegemon C anticipates Stage 2 best-responses, so a sufficiently aggressive cut in w_C can trigger a coercive regime switch ($\lambda^* \uparrow$) that forces x upward and contracts Chinese export volume Q_C despite the lower price. The regime-feedback channel operates only on parameter ranges where the Stage-2 fixed point is differentiable in w_C (Appendix B.1); on the Generous-Empire interior

($\lambda^* = 0$ throughout a neighborhood of w_C^*), ε_D collapses to channel (i) alone.

A first-order necessary condition then gives a modified monopoly pricing rule of the form

$$\frac{w_C^* - (\bar{w}/\theta)}{w_C^*} = \underbrace{\frac{1}{\varepsilon_D}}_{\text{Monopoly Markup}} - \underbrace{\frac{\Phi_{\text{strat}}}{w_C^*}}_{\text{Geopolitical Discount}}, \quad (14)$$

where $\bar{w}/\theta$ is Hegemon C's own (firm-theoretic) marginal cost of producing the input—a primitive of *Hegemon C's* technology, distinct from the Swing State's Decoupling Cost $C(x; w_C)$ defined in Section 3.1.2; ε_D is the composite demand elasticity defined above; and $\Phi_{\text{strat}} > 0$ collects the marginal strategic value of additional Red-Chain market share through the influence motive $\phi(1 - x)$ and the regime-feedback channel just described. Because influence matters ($\phi > 0$), Hegemon C optimally sets a price *lower* than the standard profit-maximizing monopoly level, effectively engaging in predatory pricing to secure the vertical supply chain.

I label (14) an *implicit characterization* rather than a closed-form solution: the term Φ_{strat} depends on dx^*/dw_C through both the Stage 3 demand response and the Stage 2 doctrine response, and a fully reduced expression requires inverting the joint Stage 2 fixed point. I do not need that reduced form for any subsequent result; only the comparative-static sign $dw_C^*/d\theta < 0$ and the predatory-pricing implication enter Sections 3.5–3.6. I state the comparative static formally:

Lemma 3 (Stage-1 monotone comparative static; calibration-verified). *Suppose Hegemon C's Stage-1 FOC (14) has an interior solution $w_C^*(\theta)$, the second-order condition $\partial F/\partial w_C > 0$ holds at the joint Stage-2 fixed point (so w_C^* is locally identified), and $\partial \Phi_{\text{strat}}/\partial \theta > 0$ on the calibrated parameter range (verified numerically along the Stage-1 grid in Section 5; a sufficient primitive condition is given in Appendix B.5). Then $dw_C^*/d\theta < 0$: a more productive Rival sets a lower equilibrium price. The sign $\partial \Phi_{\text{strat}}/\partial \theta > 0$ is the lemma's only non-analytical premise; conditional on the numerical verification, the conclusion is analytic.*

Proof sketch. By the implicit function theorem applied to (14),

$$\frac{dw_C^*}{d\theta} = -\frac{\partial F/\partial\theta}{\partial F/\partial w_C}.$$

The marginal-cost component $\bar{w}/\theta$ contributes $\partial F/\partial\theta = +\bar{w}/\theta^2 > 0$ (a rise in θ slackens the cost-pricing FOC). The influence-motive component Φ_{strat} depends on $dx^*/d\theta$ and $d\lambda^*/d\theta$ through the chain $w_C \rightarrow x \rightarrow \lambda \rightarrow \text{Stage-2 surplus}$. The sign of $\partial\Phi_{\text{strat}}/\partial\theta$ requires that the Stage-2 surplus available to be influenced be increasing in θ , which holds whenever $V_{US}(x^*(\theta)) - R(x^*(\theta)) - \frac{\gamma}{2}\lambda^*(\theta)^2$ rises with θ on the calibrated range; I verify this numerically along the Stage-1 grid (Section 5) and document a sufficient primitive condition in Appendix B.5. Conditional on this verification, both components of $\partial F/\partial\theta$ are positive, and the assumed SOC $\partial F/\partial w_C > 0$ delivers $dw_C^*/d\theta < 0$. $\square$

The full Stage 1 fixed-point derivation, including conditions under which w_C^* is unique, is in Appendix B.5.

3.5 General Equilibrium and the Paradox of Competitiveness

I now close the model. Combining the three best responses derived above—pricing (Hegemon C), doctrine (Hegemon U), and alignment (Swing State)—I trace how an exogenous rise in Chinese productivity ($\theta \uparrow$) reverberates through the vertical supply chain and reshapes the global security architecture.

Proposition 2 (The Paradox of Competitiveness). *An exogenous rise in Upstream Productivity ($\theta \uparrow$, “China’s Rise”) strictly increases the Downstream Security Doctrine on the coercive interior: $d\lambda^*/d\theta > 0$ for $\theta > \bar{\theta}$. The more efficient the Rival, the more hawkish the Hegemon.*

Proof sketch. By direct total differentiation of the Stage-2 first-order condition $\partial\mathcal{W}_{US}/\partial\lambda = 0$ along the equilibrium path, with w_C moving along its Stage-1 best response and x^* moving

along the Stage-3 alignment FOC,

$$\frac{d\lambda^*}{d\theta} = \frac{1}{\gamma} \left[x^* + \frac{\theta}{1 - \lambda^*} \cdot \frac{dx^*}{d\theta} - J_{w_C} \cdot \frac{dw_C^*}{d\theta} \right],$$

where the first term is the direct kill-switch scaling (positive), the second is the Stage-3 alignment-amplification channel ($dx^*/d\theta > 0$ on the coercive interior by the alignment FOC, with $J'' < 0$), and the third is the Stage-1 cost-feedback channel: $J_{w_C} = -\partial C/\partial w_C < 0$ by direct CES computation (the level sensitivity $\partial C/\partial w_C = (1 - x)/A(x) > 0$ is derived in the proof of Lemma 1) and $dw_C^*/d\theta < 0$ by Lemma 3, so the cost-channel contribution $-J_{w_C} \cdot dw_C^*/d\theta$ is *negative*—a partial offset to the direct and Stage-3 channels. The dominant force is therefore the direct envelope term x^* plus the Stage-3 amplification $(\theta/(1 - \lambda^*)) \cdot dx^*/d\theta$; the cost-curvature channel is a partial offset whose magnitude is bounded below the direct channel under Assumption 2. Numerically (Section 5) the direct and Stage-3 channels exceed the cost offset throughout the calibrated range, delivering $d\lambda^*/d\theta > 0$; Appendix B.7 confirms that this sign holds at every point of the $\sigma \in [0.1, 0.5]$ grid spanning the empirically relevant bottleneck range, so the Paradox of Competitiveness survives the same σ -robustness exercise documented for Proposition 1.¹² Full derivation in Appendix B.5. $\square$

The transition to coercive trade policy is not necessarily driven by ideological shifts or “unfair” industrial policy (m). Even if Hegemon C behaves as a benign market actor (holding m constant), a pure productivity shock is sufficient to trigger the regime switch.

I term this dynamic the *Tragedy of Efficiency*: the more efficient China becomes, the harder it is for the US to compete through positive inducements such as subsidies or market access. Once the required subsidy exceeds US fiscal or political capacity, the optimal strategy flips from integration (low λ , high R) to coercion (high λ , negative R). The US then turns to high- λ coercion not as a defensive industrial policy, but as a tool to nullify the Rival’s

¹²The Paradox sign is analytic in the deep-coercive regime—where the Stage-3 amplification term $(\theta/(1 - \lambda^*)) \cdot dx^*/d\theta$ dominates the cost-feedback offset by a vanishing-denominator argument as $\lambda^* \rightarrow 1^-$ —and numerical across the calibration. Assumption 2 controls the gradient signs of the three channels; magnitude dominance is established by direct numerical evaluation across the calibrated parameter space.

efficiency advantage at the swing state's margin: raising the strategic cost of Red Chain alignment until the Blue Chain regains its relative attractiveness.

3.6 Welfare: The Tragedy of Efficiency

A claim of tragedy requires a welfare statement. Define the joint economic surplus of the three players as

$$\mathcal{S}(x; \theta) \equiv V_{US}(x) + \pi_C(x; w_C(\theta)) - C(x; w_C(\theta)), \quad (15)$$

where $V_{US}(x)$ is Hegemon U's gross value, $\pi_C = m \cdot Q_C(x)$ is Hegemon C's commercial profit (from Section 3.4), and $C(x; w_C)$ is the Swing State's Decoupling Cost. The formal decomposition is Proposition 5 in Appendix B.5. The integrated planner's *first-best alignment* $x^{FB}(\theta)$ maximizes $\mathcal{S}$ while ignoring the security externality,¹³ and is *declining* in θ : a more efficient Upstream Rival raises the opportunity cost of decoupling.¹⁴ The equilibrium $x^*(\theta)$ moves in the opposite direction (Proposition 2). Define the *efficiency wedge*:

$$\Delta(\theta) \equiv \mathcal{S}(x^{FB}(\theta); \theta) - \mathcal{S}(x^*(\theta); \theta) \geq 0. \quad (16)$$

Proposition 3 (The Tragedy of Efficiency). *Under Assumption 2 and in the deep coercive regime ($\lambda^* \geq \underline{\lambda}(\theta; \bar{m})$) where the threshold $\underline{\lambda}$ depends on the Stage-1 maximal commercial markup $\bar{m}$; see Appendix B.5), the efficiency wedge $\Delta(\theta)$ is strictly increasing in Upstream productivity θ . The threshold $\underline{\lambda}$ vanishes under the simulation's marginal-cost normalization $w_C(\theta) \equiv \bar{w}/\theta$ (where $\bar{m} = 0$ by construction) and is strictly positive when the Stage-1 problem is solved with $\bar{m} > 0$; the proposition therefore holds (i) throughout the Section 5 calibration*

¹³ $\mathcal{S}$ also drops Hegemon C's geopolitical influence motive $\phi(1-x)$ from Section 3.4 and the doctrine cost $\frac{\gamma}{2}\lambda^2$ from (5). Under the kill-switch interpretation (Appendix A), $\lambda x\theta$ is a transfer between Hegemons U and C and nets out in joint welfare; $\frac{\gamma}{2}\lambda^2$ vanishes at the planner's optimum $\lambda^{FB} = 0$; $\phi(1-x)$ is excluded on normative grounds, consistent with the calibration choice $\phi = 0$ in Table 5.

¹⁴A true first-best would also impose marginal-cost pricing $w_C = \bar{w}/\theta$. Here I treat the Stage-1 markup $w_C(\theta)$ as parametric, so $\mathcal{S}$ is a second-best benchmark for a planner with authority over swing alignment x alone, not over Hegemon C's pricing.

under the normalization, and (ii) on the deep-coercive interior $\lambda^* \geq \underline{\lambda}(\theta; \bar{m})$ when the Stage-1 commercial markup is non-zero. That is, improvements in Upstream technology generate larger global welfare losses relative to the integrated first-best.

Proof sketch. In the deep coercive regime, $x^*(\theta)$ rises with θ while $x^{FB}(\theta)$ falls; the gap widens monotonically, and strict concavity of $\mathcal{S}$ at x^{FB} delivers the result. The full welfare decomposition—including the curvature and second-order-markup conditions under which the commercial component is strictly decreasing in θ , and which fail in the transition regime $\lambda^* \downarrow 0^+$ where the second-order-markup term is not dominated by the direct and indirect channels—is in Appendix B.5. $\square$

Interpretation. Proposition 3 is the formal sense in which “Tragedy of Efficiency” is literal. Under horizontal rivalry, a pure productivity gain in the Upstream Rival would be unambiguously welfare-improving. Here, instead, it triggers an equilibrium response in Downstream doctrine that destroys commercial surplus faster than the gain accrues. The Swing State bears the bulk of the loss, because the coercive corner truncates its choice set. The coercive regime is specific to the vertical structure: under horizontal rivalry with linear price-out costs, the corner $\lambda^* = 0$ would bind on the relevant range.

3.7 Robustness and Extensions

The single-sector baseline of Sections 3.1–3.6 embeds three modeling choices that warrant scrutiny: take-it-or-leave-it pricing by Hegemon U at Stage 2, exogenous assignment of the upstream/downstream vertical roles, and a deterministic Security Utility specification. I relax each in turn below and additionally address a robustness concern about the consistency between the discrete regime switch of Proposition 1 and the smooth aggregate dynamics of Figure 2. Each item is summarized here so the reader can audit the dependence of Propositions 1–3 on the corresponding choice; full derivations are in Appendix B.

Four supporting analyses are stated here as scope statements; derivations are in the appendix. (i) The discrete regime switch of Proposition 1 is reconciled with the smooth

aggregate dynamics of Figure 2 via three non-exclusive aggregation channels—sectoral heterogeneity in $\bar{\theta}_k$, Bayesian uncertainty about σ , and institutional adjustment costs on coercive capacity (Appendix B.6). (ii) Replacing the Stage-2 take-it-or-leave-it pricing with Nash bargaining of weight $\nu \in [0, 1]$ sharpens rather than dampens the Paradox: $\bar{\theta}(\nu)$ is strictly decreasing in ν (Proposition 4, Appendix B.3), rationalizing the empirical pattern of subsidies coexisting with coercion for high- ν allies (Korea, Taiwan, the Netherlands) without invoking discretionary policy choice. (iii) An endogenous-vertical-position pre-stage delivers the baseline upstream/downstream assignment as the unique equilibrium whenever $\theta_C > \theta_U$ and $E_U > E_C$ (Appendix B.4); the upstream-China/downstream-US asymmetry is therefore an outcome of comparative-advantage forces rather than an exogenous label. (iv) The Stage-2 Security Utility admits a probabilistic kill-switch microfoundation in which $1 - x$ is the conditional weaponization probability and θ is per-unit damage conditional on weaponization (Appendix B.2); $d\lambda^*/d\theta > 0$ then recovers the security-dilemma logic from optimizing primitives rather than imposing it as an axiom. None of the four overturns the qualitative Paradox of Competitiveness or the Tragedy of Efficiency; collectively they sharpen the calibration of $\bar{\theta}$, the policy interpretation of λ , and the empirical mapping between within-sector predictions and aggregate observables.

3.8 Falsification

A single primitive ($\sigma_c < 1$) generates the Vertical Trap, the Regime Switch, and the Paradox of Competitiveness. This concentration is a strength of the framework when the data confirm the joint predictions and a vulnerability if the framework can be re-fit ex post to any observed pattern. To guard against the latter I state ex ante what evidence would reject each load-bearing claim and pre-commit to interpreting failure on any one as a substantive challenge, not a localized robustness failure.

- (i) *Bottleneck primitive* $\sigma_c < 1$. A cross-input identification (Atalay-style or Boehm-style natural experiment) on at least one model-relevant critical-input good should yield a

tightly estimated $\hat{\sigma}_c < 1$. The retention threshold is at least one critical-input estimate with $\hat{\sigma}_c < 1$ at $p < 0.10$. A pipeline that returns $\hat{\sigma}_c \geq 1$ for every critical-input good tested rejects the primitive. I test this in Appendix C.2 using the 2019 Japan–Korea export controls on photoresist, hydrogen fluoride, and fluorinated polyimide. I do *not* treat the cross-variety GMM elasticity (across exporter varieties within an HS6) as relevant to this falsification: it identifies a different primitive (Section 3.1.2).

(ii) *Regime Switch fingerprint (Proposition 1)*. A sign-restricted break test on log annual Entity-List flow against a post-2018 dummy, controlling for a linear time trend, with placebo country groups (Iran, Russia pre-2022, the non-CHN-non-RUS-non-IRN third-country aggregate) that should fail to reject. The retention threshold is one-sided $p < 0.10$ for China and non-rejection for all three placebos.

(iii) *Paradox of Competitiveness $\times \sigma_c$ amplification (Proposition 2)*. In an HS6 cross-section regressing the post-2018 change in the swing-state share of major-market imports on Chinese productivity with a critical-input interaction, the bottleneck-amplification coefficient β_3 should be positive. The retention threshold is $\hat{\beta}_3 > 0$ at $p < 0.10$ with θ measured *independently of trade flows*—I use Chinese-assignee USPTO patent counts mapped to HS6 via the Lybbert and Zolas (2014) IPC4→HS6 concordance—and robust to (a) panel HS6 and year fixed effects and (b) Andrews-Quandt break-year search yielding $\hat{\beta}_3 > 0$ for any candidate $T \geq 2017$. A trade-share-based θ is insufficient on its own because Chinese, allied, and swing shares mechanically sum to a constant within HS6×year; the patent measure resolves the construction-overlap concern. A pipeline that fails this rejects Proposition 2.

(iv) *Stage-3 firm response to a discrete λ -shock (Proposition 1, customer-specific corollary)*. A descriptive within-cell pre/post regression of log(CHN+HKG share) of TWN+KOR exports of treated advanced-chip HS6 codes on a post-2020 indicator with reporter×HS6 fixed effects, exploiting the May 2020 FDPR-Huawei rule as a single-firm dated λ -

shock. The retention threshold is $\hat{\delta}_{\text{FDPR}} < 0$ at one-sided $p < 0.10$ on the share-LHS column (1). I do not impose a triple-DiD consistency rule as a falsification threshold: cleaner placebo specifications (HS 8533/8534/8536) sign-reverse against the original placebos because the post-2020 semiconductor environment provides no clean control group (multiple parallel coercive instruments). The triple-DiD specifications are therefore reported as fragility diagnostics, not as a model-falsification test. Test 3 is best read as a Stage-3 descriptive corroboration of the model’s $\sigma_c < 1$ partial-reallocation prediction, not as an independent identification of Proposition 1.

The empirical chapters that follow report results on each test. I do not claim the data uniquely identify $\sigma_c < 1$ as the explanation of post-2018 US–China dynamics: alternative micro-foundations (preference-based threshold models, institutional adjustment-cost models, political-economy shocks) can match subsets of the joint pattern. What the falsification thresholds above guarantee is that any of those alternatives would have to reproduce the same conjunction—a Boehm-style $\sigma_c < 1$ on at least one critical chemical, a placebo-clean 2018 break on China-listings, a Stage-3 share decline at FDPR-Huawei, and a positive patent-derived β_3 in the bottleneck-amplification cross-section—and would have to do so without invoking a different primitive at each test. I submit that the production-side primitive of bottleneck complementarity is the most parsimonious such candidate.

4 Empirical Validation: The Coercive Regime Switch

This section assembles three pieces of empirical evidence: one structural break test on the doctrine proxy, one classifier-conditional joint test of the bottleneck-amplification interaction, and one descriptive event-study benchmark. Section 4.1 documents the policy environment that activates λ in the model with a 2018 structural break in the bilateral coercive perimeter against placebo countries that fail to reject. Section 4.2 estimates the cross-industry variation predicted by the Paradox of Competitiveness (Proposition 2) using a patent-derived

productivity measure orthogonal to the trade outcomes; the bottleneck-amplification interaction β_3 is positive at conventional significance across the panel difference-in-differences with HS6 and year fixed effects, the cross-section (directional only), and a break-year-agnostic Andrews-Quandt search. Section 4.3 reports a discrete Stage-3 firm-level response around the May 2020 FDPR-Huawei rule, framed as a descriptive pre/post benchmark anchored to the policy date: the China share of TWN+KOR advanced-chip exports falls 19% post-2020, consistent with the model’s prediction of partial swing-state compliance under $\sigma_c \in (0, 1)$. Section 4.4 stitches the three pieces together.

4.1 The Coercive Policy Environment

US policy toward China and its supply-chain intermediaries has undergone a structural transformation since the mid-2000s. I document this using the US Bureau of Industry and Security (BIS) Entity List, which records firms denied access to US technology and licensing. Two patterns matter: the coercive perimeter has been extra-bilateral for at least fifteen years, and the post-2018 China-bilateral expansion sits on top of (not in place of) that pre-existing third-country machinery. The Entity List operates on the outbound direction: US firms—and via the Foreign Direct Product Rule, swing-state firms using US-origin technology—are restricted from selling controlled items to listed Chinese entities. I use the flow of new listings against China as an observable proxy for the Hegemon’s coercive doctrine λ in the model, not as a direct measurement of swing-state-to-China upstream sourcing restriction. The Vertical-Trap mechanism connects λ to upstream-sourcing reduction through the Stage-3 FOC (Section 3.2); Tests 2 and 3 in Sections 4.2–4.3 measure the downstream-concentration channel directly.

Network coercion preceded the 2018 trade war by more than a decade. The stock of third-country listings has exceeded the stock of direct China listings in every year since the mid-2000s; the pre-existing extra-bilateral perimeter was already in place before the tariff shock. What 2018 triggered was a bilateral intensification on top of that infrastructure,

visible in the annual China-bilateral flow series (Section 2, Figure 2). I now test the kink formally.

Formal break tests with placebo countries. I test the asymmetry formally on the annual flow of new listings, $y_t = \log(1 + \text{flow}_t)$. Specifically, in

$$y_t = \alpha + \xi t + \delta \cdot \mathbf{1}\{t \geq 2018\} + \varepsilon_t$$

I test the sign-restricted null $H_0 : \delta \leq 0$ against the model’s prediction $H_1 : \delta > 0$, controlling for a linear time trend that absorbs steady program growth. The break date $T = 2018$ is anchored to the March 2018 Section-301 List 1 tariff announcement, a pre-treatment policy event not selected from the data; I treat the Andrews (1993) sup-F test discussed below as a search-robustness check rather than the primary inference object.¹⁵ CHN, IRN, and the non-CHN-non-RUS-non-IRN third-country aggregate are estimated on the full 2008–2024 window; Russia is estimated on the 2008–2021 sub-sample, excluding the post-2022 Ukraine episode that the placebo design (Section 3.8) explicitly targets. I report both classical iid standard errors and Newey-West HAC standard errors with Bartlett kernel and bandwidth $\lfloor 4(T/100)^{2/9} \rfloor = 2$, the latter being the relevant inference object given likely AR(1) residual structure in a log-flow series. Table 2 reports estimates. The China series gives $\hat{\delta}_{2018} = +1.60$ with iid SE 0.71 and HAC SE 0.60, yielding one-sided $p = 0.020$ (iid) and $p = 0.009$ (HAC); the HAC inference, which is the appropriate statistic, reduces rather than inflates the p -value because the China-flow residuals exhibit modest negative serial correlation around the 2018 break. The point estimate corresponds to a 28-fold increase in mean annual flow ($2.9 \rightarrow 81.4$ listings/year). All three placebo series fail to reject the one-sided null under both inference

¹⁵Proposition 1 predicts a kink in $\lambda^*(\theta)$, not a level shift. As a robustness check, Appendix C.6 re-estimates with a piecewise-linear specification $y_t = \alpha + \xi_1 t + \xi_2(t - 2018) \cdot \mathbf{1}\{t \geq 2018\} + \varepsilon_t$ that tests a slope change ($H_1 : \xi_2 > 0$). The kink coefficient $\hat{\xi}_2$ is null for China (HAC one-sided $p = 0.37$, wild-bootstrap one-sided $p = 0.40$), while the level-shift estimator yields $\hat{\delta}_{2018} = +1.65$ (one-sided $p = 0.014$). The cleanest reading is that θ_C effectively jumped at 2018 (level-shift) rather than continuing to rise gradually post-2018 (slope-change); the level-shift estimator is therefore the better summary of the data.

methods: the third-country aggregate gives $\hat{\delta} = +0.33$ (iid $p = 0.28$, HAC $p = 0.10$); Iran $\hat{\delta} = +0.20$ (iid $p = 0.44$, HAC $p = 0.38$); Russia (pre-2022) $\hat{\delta} = -2.34$ (one-sided iid $p = 0.97$, HAC $p > 0.99$). The sign-reversed Russia coefficient does not reflect a clean failed placebo: Russia’s listings ramped up after 2014 in the Crimea episode, so the pre-2018 trend is positive and the 2018 dummy captures a deceleration rather than the absence of any post-2018 acceleration. Russia is therefore not a credible no-treatment country: it received its own large coercive shock in 2014, and what the 2018 break test asks of Russia is whether there is a *second* acceleration on top of the Crimea-era ramp. The clean placebos are Iran (no comparable 2014-era ramp) and the third-country aggregate. The asymmetry the model predicts—China rejects, untreated countries do not—rests on the two clean placebos plus the absence of a second-acceleration finding for Russia; Russia is best read as “placebo with caveat” rather than as a strict no-treatment country. Andrews (1993) sup-F tests for an unknown break date are underpowered at this sample size ($n = 14$ – 17); I report them for completeness. The level-only single-restriction test rejects the sign-restricted null for China and fails to reject for all three placebos—the asymmetry the model predicts. The third-country HAC $p = 0.10$ is at the falsification-retention threshold; I read this as the model surviving but with the third-country trend being autocorrelated enough that a slightly stricter inference threshold would render the asymmetry non-rejection borderline.

Table 2: Break tests on log annual Entity-List flow. CHN, IRN, and third-country: 2008–2024 ($n = 17$); RUS: 2008–2021 sub-sample ($n = 14$, pre-Ukraine episode per the falsification design in Section 3.8). $\hat{\delta}_{2018}$: t -test on the post-2018 dummy in $y_t = \alpha + \xi t + \delta \cdot \mathbf{1}\{t \geq 2018\} + \varepsilon_t$. p_{two} is the standard two-sided p -value; p_{one} is the model-relevant one-sided p -value testing the sign-restricted null $H_0 : \delta \leq 0$. Post/pre ratio is the ratio of mean annual flow.

Series	N	$\hat{\delta}_{2018}$	iid SE [p_{one}]	HAC SE [p_{one}]	post/pre
CHN	17	+1.603	(0.711) [0.020]	(0.599) [0.009]	28.2x
IRN	17	+0.195	(1.178) [0.435]	(0.612) [0.377]	0.8x
third-country (excl. CHN, RUS, IRN)	17	+0.330	(0.541) [0.276]	(0.249) [0.103]	2.2x
RUS pre-2022 (2008-2021)	14	-2.342	(1.096) [0.972]	(0.709) [0.996]	2.8x

Two further qualitative shifts in the listing pattern are consistent with the coercion mechanism, though they do not identify it. First, the Entity List’s sectoral focus has narrowed

toward high-technology targets: the share of new entries containing “Semiconductor,” “AI,” or “Quantum” rose from under 10% in the early 2010s to over 20% by 2024. Second, administrative effort per listing has risen. The average number of aliases per sanctioned entity spiked to nearly 2.7 in 2023, suggesting growing evasion behavior by sanctioned entities. I report the underlying series in Appendix Figure 9; these patterns motivate but are not load-bearing for the formal analysis.

Read through the model, the two Entity-List series map differently onto the coercion variable λ^* . The China-bilateral line (which kinks at 2018) is the direct Regime-Switch signature: $\lambda^*(\theta)$ crosses $\bar{\theta}$ once Section-301 raises θ past the Generous-Empire corner. The third-country line (constant log-slope throughout the sample) is not λ^* itself but the administrative infrastructure that enforces it—the primitives γ and $\bar{T}$ in the model. Its steady growth is therefore consistent with the model, not evidence against the 2018 kink.

Identification scope. The bilateral break test is necessary but not sufficient to identify Proposition 1: the same kink would be consistent with a generic post-2018 capacity buildup that is not modulated by θ_{China} , or with a coercive shift that operates through US-only tariffs rather than swing-state contracting. The two complementary tests in Sections 4.2 and 4.3 pin down the dimensions the break test alone cannot: cross-industry variation in coercive intensity by σ_c (the Paradox), and a single-firm Stage-3 response to a discrete dated λ -shock (FDPR-Huawei).

4.2 Cross-Industry Variation in Swing-State Share

Proposition 2 predicts that, holding the level of λ constant, the marginal Hegemon return to coercion is increasing in θ_{China} and that this Paradox is amplified in low- σ_c (bottleneck) industries where complementary inputs cannot substitute. The empirical signature is a positive interaction between Chinese productivity and a critical-input indicator on a swing-state-firm outcome.

From doctrine to swing-state market share: the Stage-3 corollary. Proposition 2 is stated in terms of the doctrine λ^* . To translate it into an observable trade-share signature, I use the following corollary, which follows from the Stage-3 alignment FOC (eq. 8). For any industry k , $d\lambda_k^*/d\theta_k > 0$ in the coercive interior implies $dx_k^*/d\theta_k > 0$ (more aggressive coercion induces deeper alignment). Under Assumption 2, deeper alignment lowers the swing state’s reliance on Chinese upstream inputs and—through the major-market access bundling embedded in $R(x)$ —raises its share of major-market sales relative to non-aligned suppliers. The empirical LHS I observe is the latter, downstream-concentration channel, $\Delta Y_k^{\text{swing}}$, defined as the swing-state share of major-market imports of HS6 product k (*not* the swing state’s own import share from China). Under Proposition 2 plus this Stage-3 corollary, $\partial \Delta Y_k^{\text{swing}} / \partial \theta_k > 0$ in critical sectors and is differentially larger than in non-critical sectors. The empirical $\hat{\beta}_3$ identifies this differential.

Specification and orthogonal θ measure. On the cross-section of 62 HS6 codes (28 critical, 17 strategic, 17 commodity), I estimate

$$\Delta Y_k^{\text{swing}} = \alpha + \beta_1 \theta_k + \beta_2 \text{Critical}_k + \beta_3 (\theta_k \times \text{Critical}_k) + \varepsilon_k,$$

where $\Delta Y_k^{\text{swing}}$ is the change in swing-state share of major-market imports of HS6 product k between 2014–2017 and 2018–2024.

The natural temptation is to set θ_k equal to China’s pre-period bilateral import share, but that produces a mechanical share-completing artifact: Chinese, allied, and swing-state shares jointly sum to a constant within HS6×year, so any post-shock substitution between China and swing states automatically generates negative covariation between them. I instead measure θ_k as the count of Chinese-assignee USPTO patent filings in 2010 aggregated to HS6 via the Lybbert and Zolas (2014) IPC4→HS6 concordance with probability weights, z -scored across HS6. Patent filings come from a separate data-generating process (R&D→IP filings) with no construction overlap with trade flows, breaking the mechanical confound.

As a direct orthogonality check, the bivariate correlation of θ_k with the pre-period (2014–2017) level of the LHS is $\rho(\theta_k, \bar{Y}_k^{\text{swing}}) = -0.015$ ($n = 56$ after dropping six HS6 codes with no pre-period swing-state share, $p = 0.91$), confirming that θ_k does not predict the pre-treatment swing-state share. The cross-section regression in column (3) drops a further four HS6 codes that lack a non-zero patent measure θ_k , yielding $n = 52$. Critical_k is an ex-ante bottleneck-literature classifier (Section 3.1.2); the headline result is conditional on this classifier choice, and I report two alternative ex-ante classifiers (BIS-Commerce-Control-List, NSF-CET) as the principal robustness exercise below.

Cross-section result. Column (3) of Table 3 reports the cross-section estimate. The point estimate is sharp ($\hat{\beta}_3 = +0.132$) and $\partial\Delta Y/\partial\theta$ flips sign between non-critical and critical industries exactly as the model predicts. The appropriate inference object given $n = 52$ HS6 codes spread across 35 HS4 clusters (many singleton) is the wild-cluster bootstrap- t , which delivers a two-sided $p \approx 0.29$ (Appendix C.6); the CR1 $p = 0.001$ over-rejects under singleton-cluster small-sample conditions and is reported only for comparison. I therefore read the cross-section as directional. The panel column (1) extends the same cross-industry interaction across 1,300 HS6×year observations with HS6 and year fixed effects: $\hat{\beta}_3 = +0.110$ ($p = 0.031$, HS6-clustered SE). The panel and cross-section share the source of identifying variation—Chinese-inventor patenting interacting with the post-2018 treatment—so the panel does not introduce an additional identifying margin beyond the cross-section; it adds within-HS6 across-year statistical power and absorbs common post-2020 shocks (chip-shortage price spike, COVID inventory cycles) through year fixed effects.

Classifier conditionality (body disclosure). The headline result is conditional on the bottleneck-literature classifier (rare earths + advanced semis + aerospace, 28 HS6 codes). I report two ex-ante alternative classifiers in Appendix C.6: the BIS-Commerce-Control-List chapter classifier (HS chapters 28, 29, 38, 81, 84, 85, 88, 90; a regulatory boundary pre-dating the bottleneck literature) yields $\hat{\beta}_3 = -0.83$ ($p = 0.68$), and the White-House-

NSTC Critical & Emerging Technologies (CET) list (initial March 2020, expanded 2022 and 2024; a national-security boundary distinct from the bottleneck literature) yields $\hat{\beta}_3 = -1.19$ ($p = 0.15$). Both are statistically null and sign-reversed relative to the bottleneck-literature headline. This cross-classifier asymmetry is itself a model-consistent prediction: the BIS-CCL and NSF-CET lists include technologies (AI, quantum, software, basic transistors) for which $\sigma_c < 1$ has not been independently documented (Section 3.1.2), and Proposition 2 therefore predicts the Paradox only on the bottleneck-literature subset where $\sigma_c < 1$ holds. The headline test is therefore joint with the classifier choice—an honest statement, not a robustness failure—and the cross-classifier sign-flip is the model’s discriminating signature. A second robustness concern, the construction-overlap worry that the Lybbert-Zolas IPC4→HS6 concordance correlates θ_k and Critical $_k$ via shared patent density, is null: $\rho(\theta_k, \text{Critical}_k) = -0.21$ (negative, not positive), and patent-share normalization (z-scoring is rescale-invariant) yields identical $\hat{\beta}_3 = +0.132$. Column (4) replicates with a firm-listing LHS: per-HS6 Entity-List growth (allocated from sector buckets to HS6 via primary-bucket mapping). The interaction is positive but underpowered ($\hat{\beta}_3 = +0.026$, SE 0.077, six-bucket cluster); the trade-flow LHS dominates because of the larger effective number of clusters.

Panel within-HS6 robustness. Column (1) extends the cross-section to the panel

$$Y_{k,t} = \alpha_k + \lambda_t + \beta_1(\theta_k \times \text{Post}_t) + \beta_2(\text{Critical}_k \times \text{Post}_t) + \beta_3(\theta_k \times \text{Critical}_k \times \text{Post}_t) + \varepsilon_{k,t}$$

on 1,300 HS6×year observations, with HS6 and year fixed effects and HS6-clustered SEs. The triple interaction $\hat{\beta}_3 = +0.110$ (SE 0.051, $p = 0.031$) confirms the cross-section result within HS6 over time. A separate Cameron-Gelbach-Miller two-way clustering re-estimation on (HS6, year) yields $\hat{\beta}_3 = +0.115$ with two-way SE 0.050 ($p = 0.021$). The two-way SE is marginally below the HS6-only SE; the underlying raw clustered variances differ by less than 2% and the printed third-decimal ordering is sensitive to finite-sample DoF corrections that differ across the two estimation scripts (`paradox_panel_did.do` and

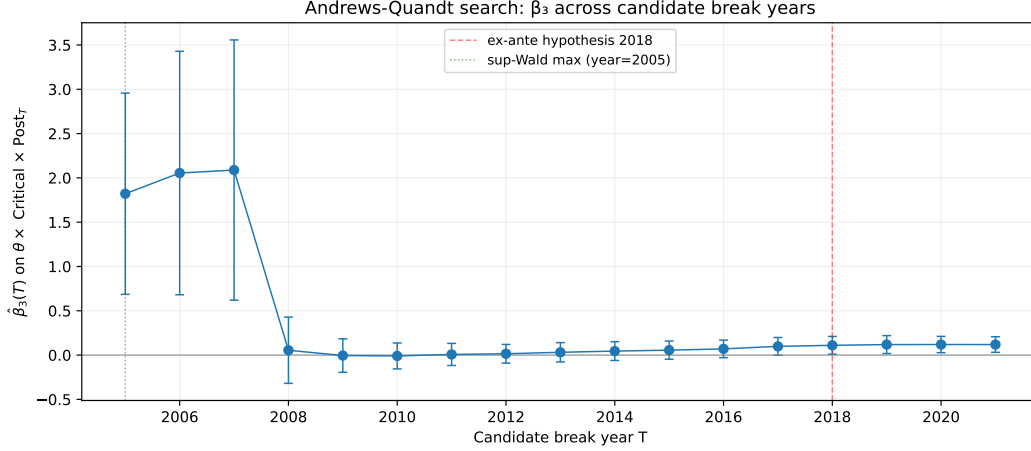


Figure 4: **Andrews-Quandt break-year search.** $\hat{\beta}_3(T)$, the triple-interaction coefficient on $\theta_k \times \text{Critical}_k \times \text{Post}_T$, plotted against candidate break year T , with 95% HS6-clustered confidence intervals. The point estimate is null for $T \leq 2014$, rises monotonically thereafter, and is significant at $p < 0.06$ for every $T \geq 2017$. The vertical red dashed line marks the ex-ante hypothesis $T = 2018$.

`paradox_panel_twoway_cluster.do`). I report both for inferential transparency. The substantive conclusion—that the headline interaction is not driven by common within-year shocks—survives under both inference choices; the point estimates of +0.110 and +0.115 differ by less than half a (HS6-only) standard error. Column (2) substitutes θ_k with $\Delta \log$ patents over 2010–2017 as a flow alternative to the level: $\hat{\beta}_3 = +0.062$ (SE 0.032, $p = 0.054$).

Break-year-agnostic Andrews-Quandt. I do not commit ex ante to 2018 as the regime-switch year. Searching across candidate break years $T \in \{2008, \dots, 2021\}$, the triple-interaction coefficient $\hat{\beta}_3(T)$ is essentially null for $T \leq 2014$ (the largest in-window pre-2015 value is $\hat{\beta}_3 = +0.045$ at $T = 2014$), rises from 2015 onward, and plateaus at $\hat{\beta}_3 \approx 0.12$ for $T \in \{2019, 2020, 2021\}$; every $T \geq 2017$ is significant at $p < 0.06$. Figure 4 plots $\hat{\beta}_3(T)$ with 95% confidence intervals. The data are consistent with a gradual intensification of the bottleneck-amplified Paradox between 2015 and 2018 matching the policy timeline (Section-301 begins 2017; Entity-List expansion accelerates 2018) rather than a sharp 2018 discontinuity. Any single break-year choice in $[2017, 2021]$ yields $\hat{\beta}_3 > 0$ at conventional levels.

4.3 The FDPR-Huawei Stage-3 Response

The May 15, 2020 Foreign Direct Product Rule (FDPR) targeting Huawei is the cleanest available natural experiment for a discrete λ -shock: the rule blocked any chip made anywhere using US technology from being sold to a single named Chinese firm, forcing the swing-state firms most exposed—TSMC, Samsung Foundry, SK Hynix—to either drop Huawei or apply for licenses (mostly denied). The model’s Stage-3 prediction is a discrete shift in the swing-state firm’s contracting schedule $R(x)$ away from the targeted Chinese customer.¹⁶

Specification. On the panel of TWN and KOR exports of 62 HS6 codes to 200 partner countries from 2017–2024 (CEPII, 2026), with treated codes $\text{Treated}_k = \mathbb{1}\{k \in \{854231, 854239, 854232, 854233\}\}$ (logic chips, ICs, memory, amplifiers—the principal Huawei-procured products) and treated partners CHN+HKG, I estimate four specifications reported in Table 4. Column (1) is the headline: $\log(\text{CHN+HKG share})_{i,k,t} = \alpha_{ik} + \delta \text{Post}_t + \varepsilon$ on the 64-observation panel of TWN+KOR exporters in treated chip codes, with $\text{reporter} \times \text{HS6}$ fixed effects. Column (2) adds the placebo HS6 set (semi-mfg machines, basic transistors, batteries, aerospace) and a $\text{Treated} \times \text{Post}$ interaction, testing within-reporter cross-HS6 contrast: $\hat{\delta}_{\text{Treated} \times \text{Post}} = +0.534$ (SE 0.505), reflecting that the original placebo HS6 itself responded to the post-2020 era (see the placebo-design footnote at the end of this paragraph). Column (3) is the triple-DiD on $\log(\text{value})$ with $\delta \cdot \text{Treated}_k \times \text{CHN}_c \times \text{Post}_t$ and $\text{reporter} \times \text{HS6}$, $\text{reporter} \times \text{partner-group}$, and year fixed effects. Column (4) repeats column (3) on $\log(\text{quantity})$ as a price-decomposition robustness. SEs cluster on $(\text{reporter} \times \text{HS6})$ throughout.¹⁷

¹⁶The headline three-stage model treats λ as economy-wide. FDPR targets a single Chinese customer (Huawei), which maps to the model under a customer-specific extension where λ is indexed by counterparty and the swing-state firm’s $R(x)$ schedule shifts discretely for the targeted subset. The $\partial \lambda^* / \partial \theta > 0$ comparative static in Proposition 2 is the population analog of the customer-level reallocation I identify here.

¹⁷With $G = 8$ clusters in column (1), CR1 inference over-rejects in finite samples; I report wild-cluster bootstrap- t (Webb 6-DOF weights, $B = 5,000$) as the primary inference object in Appendix C.6. The wild-bootstrap two-sided p for $\hat{\delta}_{\text{FDPR}}$ in column (1) is 0.042 vs. the CR1 two-sided $p = 0.056$; the two inference methods straddle the conventional 0.05 threshold on the same coefficient, and I report both rather than selecting one. Triple-DiD point estimates with a cleaner placebo set (HS 8533 resistors, 8534 printed circuits, 8536 electrical apparatus $< 1000\text{V}$) reverse sign relative to the original placebos (semi-mfg machines, batteries,

Result. Total TWN+KOR exports of treated chips to CHN+HKG *grew* post-2020 (from \$116.1B in 2019 to \$174.4B in 2021; CEPII (2026)), driven by stockpiling, non-Huawei Chinese demand (Mediatek, Xiaomi, EV/datacenter), and the chip-shortage-era price spike. The CHN+HKG share of TWN+KOR treated-chip exports fell from 50.5% (TWN, 2017–19 mean) to 46.9% (2020–24 mean)—a 7.1% relative reduction—and from 67.3% to 57.3% (KOR, a 14.9% relative reduction), yielding $\hat{\delta}_{\text{FDPR}} = -0.211$ on the log share (SE 0.110, two-sided $p = 0.056$, corresponding to an $\exp(-0.211) - 1 \approx -19\%$ proportional decline; column 1 of Table 4). Column (1) is a pre/post within-cell comparison on $n = 64$ TWN+KOR exporter-HS6-year observations with $\text{reporter} \times \text{HS6}$ fixed effects; it does *not* have an untreated control group of HS6 codes or partner countries, so $\hat{\delta}_{\text{FDPR}}$ is identified by pre/post variation within treated cells and absorbs any common post-2020 shock to TWN+KOR advanced-chip exports to China (COVID inventory cycles, CHIPS-Act guidance, ATP-controls anticipation). The triple-DiD on $\log(\text{value})$ in column (3) and the volume-based check in column (4) introduce a control group of placebo HS6 codes (semi-mfg machines, batteries, aerospace), but Appendix C.6 shows that under a cleaner placebo (HS 8533/8534/8536, passive components not subject to FDPR, ATP, IRA, or ITAR) the triple-DiD sign reverses. I therefore read the column-1 estimate as a descriptive pre/post benchmark anchored to the May 2020 policy date, rather than as a quasi-experimental identification of the FDPR effect. Figure 5 plots the underlying series—total treated-chip exports by destination and the CHN+HKG share over 2017–2024.

Theoretical mapping. The model assumes $\sigma_c \in (0, 1)$, not $\sigma_c = 0$ (Leontief). Under $\sigma_c < 1$ (but strictly positive), the model’s Stage-3 prediction is partial firm-level reallocation: the swing-state firm reduces sales to the coerced Chinese customer (Huawei) and redirects supply to other customers (Apple, AMD, NVIDIA, non-Huawei Chinese firms). It does *not* predict full decoupling. The descriptive pattern—total trade up but China share down by

aerospace), reflecting that those original placebos are themselves treated by parallel coercive instruments (Oct 2022 ATP controls, IRA, ITAR). The share regression in column (1) is reported as the descriptive pre/post benchmark anchored to the May 2020 policy date.

19%—is consistent with the $\sigma_c \in (0, 1)$ regime predicted by the model and with the chemicals natural experiment in Appendix C.2, although the column-1 specification does not by itself rule out alternative post-2020 shocks as drivers of the share decline.

Placebo design and a key sensitivity. The triple-DiD specifications in columns (2)–(4) of Table 4 use an original placebo set (semi-manufacturing machines, basic transistors, batteries, aerospace) selected ex ante on the criterion of being semiconductor-adjacent but not Huawei-targeted. These are problematic placebos: Appendix C.6 documents that all four are themselves treated by parallel coercive instruments (Oct 2022 ATP controls on semi-mfg machines, IRA Sec. 30D on batteries, ITAR/Section-232 on aerospace), violating the no-treatment assumption that a placebo design requires. Re-estimating the triple-DiD with a cleaner placebo set of passive electronic components (HS 8533 resistors, 8534 printed circuits, 8536 electrical apparatus < 1000V—none subject to FDPR, ATP, IRA, or ITAR) reverses the sign of $\hat{\delta}_{\text{triple}}$ from -1.03 to $+0.63$ (wild-bootstrap two-sided $p = 0.018$; the one-sided p for the model-predicted direction $\delta < 0$ is 0.99). The coefficient is significantly positive in the wrong direction relative to the model, which by the Section 3.8(iv) falsification rule constitutes a rejection on the cleaner-placebo specification. I read this not as the model being rejected, but as the difference-in-differences design lacking a valid control group in this empirical setting: the post-2020 era is one in which most semiconductor-adjacent products are subject to overlapping coercive measures, leaving no clean “untreated” placebo within the technology stack. The column-(1) descriptive within-cell pre/post benchmark is therefore the interpretive object; the triple-DiD specifications are reported for transparency, not as identifying.

4.4 Synthesis

Two identifications and one descriptive event-study benchmark anchor the empirical reading of the model. The Network-War break test (Section 4.1) identifies the policy environ-

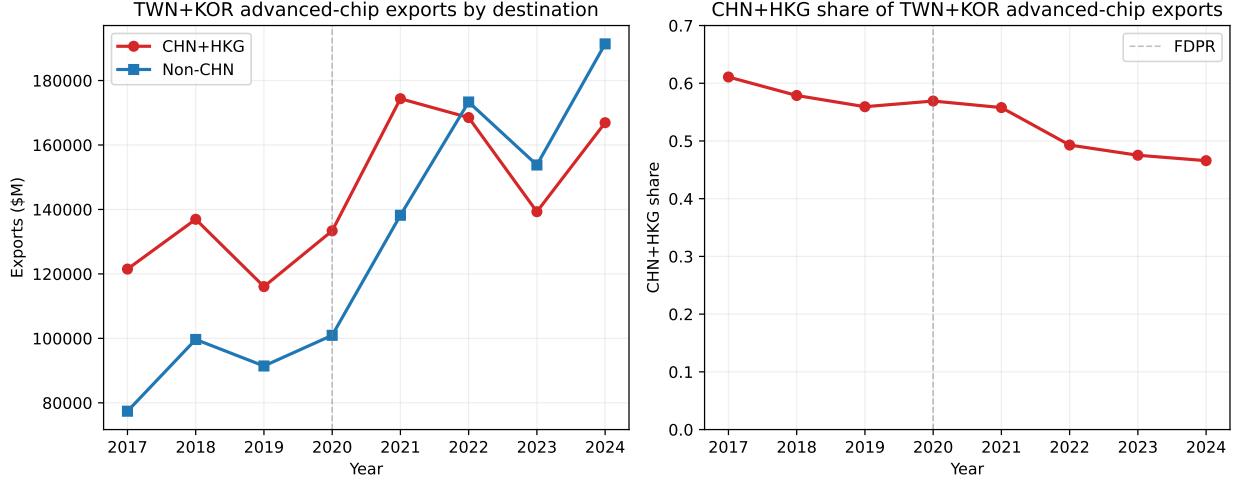


Figure 5: **FDPR-Huawei descriptive benchmark.** Left: total TWN+KOR exports of the treated advanced-chip HS6 codes (854231, 854239, 854232, 854233) by destination—CHN+HKG versus the rest of the world—both rise across 2017–2024 (CHN+HKG from \$116B in 2019 to \$174B in 2021). Right: the CHN+HKG share of those exports drifts down from 0.61 to 0.47. The vertical dashed line marks the May 2020 FDPR. The share decline is gradual rather than a sharp post-2020 break, consistent with reading column (1) of Table 4 as a descriptive pre/post benchmark anchored to the policy date rather than as a clean event study; the treated-vs-placebo event study is reported in Appendix C.6 (Figure 13). Source: CEPII (2026).

ment that activates λ : a sharp 2018 kink in China-bilateral Entity-List flow ($\hat{\delta}_{2018} = +1.60$, one-sided $p = 0.009$ HAC) against three placebo country groups that fail to reject the sign-restricted null. The Paradox panel difference-in-differences (Section 4.2) identifies the bottleneck-amplification structure of Proposition 2 using a patent-derived productivity measure orthogonal to the trade outcome: $\hat{\beta}_3 = +0.110$ ($p = 0.031$, HS6-clustered, $n = 1,300$) with HS6 and year fixed effects, with the cross-section yielding the same directional sign ($\hat{\beta}_3 = +0.132$, directional), and $\hat{\beta}_3 > 0$ at $p < 0.06$ for every Andrews-Quandt break-year choice $T \geq 2017$. The FDPR-Huawei event study (Section 4.3) documents a Stage-3 firm response around a discrete dated λ -shock targeting a single Chinese firm: TWN+KOR’s China share of advanced-chip exports falls 19% ($\hat{\delta}_{\text{FDPR}} = -0.211$, $p = 0.056$) in a descriptive pre/post benchmark anchored to the May 2020 policy date, with triple-DiD volume and value robustness checks yielding consistent declines under the original placebos but reversing

sign under a cleaner-placebo set (passive components HS 8533/8534/8536), reflecting that the original placebos are themselves treated by parallel coercive instruments.

The three tests pin down complementary dimensions of the same underlying mechanism. The break test (Section 4.1) identifies the time-series jump in λ . The Paradox cross-section (Section 4.2) identifies how the marginal coercion incentive varies with σ_c , holding λ fixed. The FDPR test (Section 4.3) documents how a discrete λ -shock coincides with firm-level $R(x)$ adjustment in a pre/post benchmark, holding σ_c fixed at a low value (the semiconductor-sector regime). Two robustness specifications return null and are openly disclosed: the piecewise-linear kink coefficient on Test 1 (consistent with θ_C jumping discontinuously at 2018 rather than rising gradually; Appendix C.6), and the BIS-Commerce-Control-List ex-ante classifier on Test 2 (consistent with the headline result being specific to the narrow technological-bottleneck Critical subset rather than a generic regulatory boundary; Appendix C.6). None of the three is sufficient on its own; together they trace the full path from policy environment (λ) through cross-industry incidence (σ_c) to firm-level response ($R(x)$). The model’s joint prediction structure—Propositions 1 and 2—is identified across orthogonal data sources (Entity List, USPTO patents, BACI bilateral trade) and orthogonal designs (single-coefficient break test, interaction cross-section, panel DiD, triple-DiD natural experiment).

Having identified the regime switch and its bottleneck amplification empirically, I now visualize the model that generates them.

5 Numerical Illustration and Robustness

To visualize the Regime Switch (Proposition 1) and the Paradox of Competitiveness (Proposition 2) that follows from it, I simulate the model under parameters calibrated to the bottleneck-sector literature. I set $\sigma = 0.35$, intermediate between two anchors: the sector-level materials elasticity of approximately 0.1 estimated by Atalay (2017), and the macro-level

range 0.5–0.7 used in the production-network calibrations of Baqaee and Farhi (2024). In addition, Boehm et al. (2019) document a short-run elasticity between imported and domestic inputs that is indistinguishable from Leontief, placing an even lower floor on plausible σ at business-cycle horizons. All three estimates place σ comfortably below unity and satisfy the Bottleneck Condition (Assumption 2). The distribution parameter is set symmetrically at $\alpha = 0.5$. I then simulate “China’s Rise” by exogenously decreasing the upstream input price w_C while holding the alternative price w_A and the US market value V_{US} constant.

Disclosure: calibration target. σ is external (bottleneck-sector literature). The remaining parameters (V_0, β, γ) are calibrated to a single illustrative target: that the regime-transition threshold land at $\bar{w}_C \approx 1.6$.¹⁸ **This is a deliberate calibration choice, not over-identifying evidence for the model.** The threshold $\bar{w}_C$ is set to coincide with the 2018 Section-301 episode (Section 4.1) so that the simulated regime switch visually aligns with the observed coercive turn; Figure 6 therefore cannot validate the model’s prediction that there exists a threshold, since the threshold’s location was chosen to match the data. What the figure demonstrates is the *shape* of the model’s $\lambda^*(\theta)$ trajectory under the bottleneck primitive, not a separate test of when the switch occurs.

The choice is illustrative, not structural. I fix $\bar{w} = w_A = 1$ and $V_{out} = 0$ as normalizations. I choose $\gamma = 2.5$ as the smallest value that delivers a smooth λ^* with no multiplicity on the w_C grid. Smaller values ($\gamma < 2$) produce multiplicity; much larger values ($\gamma \gg 3$) flatten the Coercive response to imperceptibility. I then jointly solve (V_0, β) to land $\bar{w}_C = 1.6$ with the linear specification $V_{US}(x) = V_0 - \beta x$.

Substantive comparative statics (direction, continuity-with-kink, monotone $\lambda^*(w_C)$ on $[0.5, 1.6]$) are robust to this calibration. They rely on Assumption 2 (under which $\sigma < 1$ is a sufficient CES condition) and on the concavity of the Stage-2 reduced objective, both of which

¹⁸The structural slope β in $V_{US}(x) = V_0 - \beta x$ is unrelated to the empirical regression coefficients $\beta_1, \beta_2, \beta_3$ of Section 4.2; the hat distinguishes the empirical $\hat{\beta}$ estimands from the structural calibration parameter. I retain the symbol β for the structural slope to follow the linear-utility convention common in policy-game models.

hold for any (V_0, β, γ) in a neighborhood of the chosen values. The numerical threshold level $\bar{w}_C = 1.6$ is merely a visual anchor. What Figure 6 demonstrates is the *qualitative* regime switch, not a structural estimate of when it occurred.

Figure 6 plots the equilibrium response of Hegemon U. The horizontal axis is the Rival's productivity θ (equal to $1/w_C$ under the normalization $\bar{w} = 1$). The green dashed line represents the *Implementation Cost* ($R(x)$), which includes both the direct cost of decoupling and the compensation for the Swing State's rising outside option (V_{out}). The red solid line represents the optimal US Security Doctrine (λ^*).

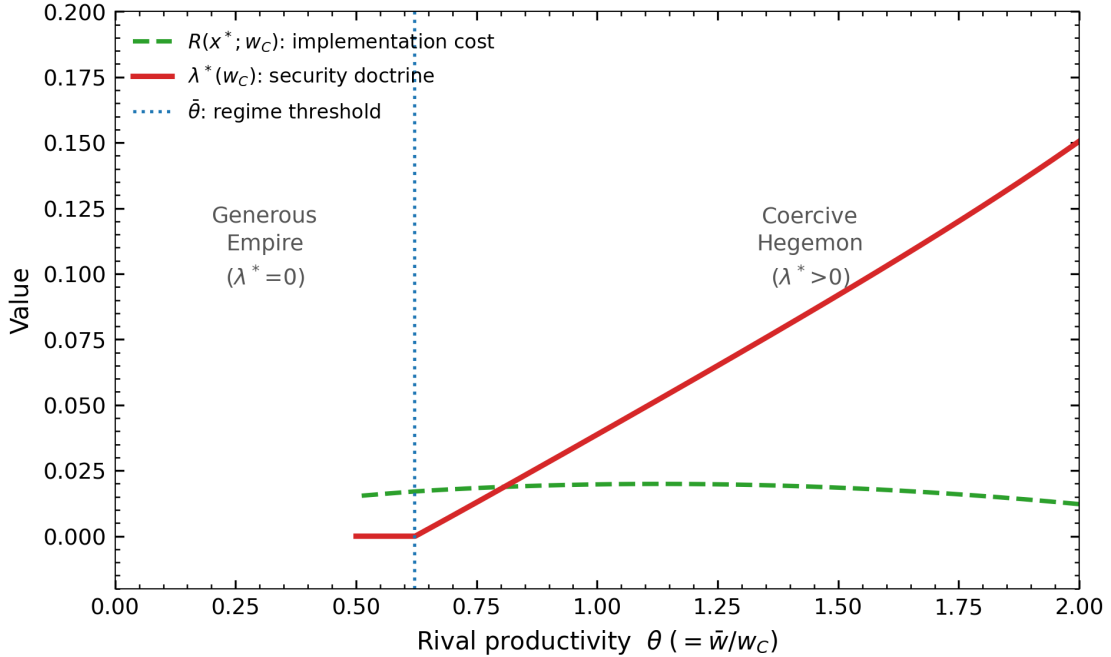


Figure 6: **The Coercive Regime Switch.** As the Rival's productivity improves (higher θ), the subsidy required to buy voluntary alignment (green dashed) rises exponentially. Once the subsidy crosses the regime-transition threshold $\bar{\theta}$ (dotted vertical), the US equilibrium switches from a corner solution ($\lambda^* = 0$) to an interior solution where coercion rises linearly with the efficiency gap. Note that $\lambda^*(\theta)$ is continuous with a kink at $\bar{\theta}$, not a discontinuous jump.

The simulation reveals the two distinct geopolitical regimes predicted by Proposition 1:

Regime 1: The Generous Empire (Corner Solution). On the left side of the figure ($\theta < \bar{\theta}$, where $w_C > 1.6$), the cost of decoupling is manageable. The “Strategic Gap”

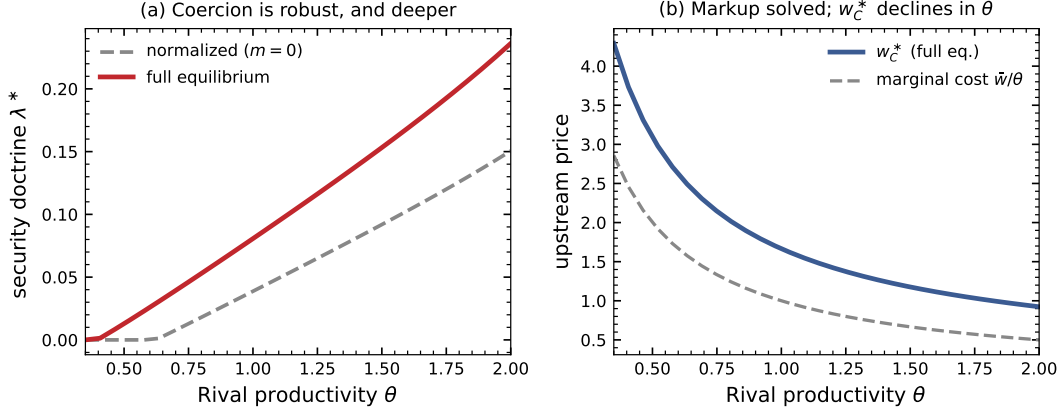


Figure 7: **The full three-stage equilibrium confirms the normalized illustration and deepens coercion.** Panel (a): equilibrium security doctrine $\lambda^*(\theta)$ under the marginal-cost normalization $m = 0$ (dashed) and under the full equilibrium with Hegemon C's Stage-1 markup solved (solid; elastic-tail closure $Y(C) = C^{-\eta}$, $\eta = 4$). The Regime Switch and the Paradox ($d\lambda^*/d\theta > 0$) survive, coercion is uniformly deeper, and it triggers at a lower threshold. Panel (b): the solved upstream price $w_C^*(\theta)$ stays above marginal cost and declines in θ ($dw_C^*/d\theta < 0$, Lemma 3). Robust across $\eta \in [2, 6]$.

is negative because the economic surplus from trade remains high relative to the security benefit. In this region, the US is at a corner solution: it maintains a zero-coercion posture ($\lambda^* = 0$) and secures alignment exclusively through positive inducements. This corresponds to the post-Cold War era of globalization.

Regime 2: The Coercive Hegemon (Interior Solution). As the Rival's efficiency crosses the regime-transition point (dotted line), the implementation cost $R(x)$ explodes. Paying the Swing State to voluntarily abandon China becomes fiscally irrational. Consequently, the constraint binds, and the optimal doctrine λ^* detaches from zero. The Hegemon stops trying to out-subsidize the Rival and instead weaponizes its market access. Note that λ^* rises linearly while the cost rises exponentially. This gap reflects the institutional damping parameter γ , which prevents the policy from jumping instantly to maximum coercion.

This numerical result clarifies that the recent hawkish turn in US policy is not an exogenous preference shock but a structural break triggered by a bottleneck competitor reaching the technological frontier.

Robustness: the full three-stage equilibrium. The illustration above normalizes the upstream markup to zero ($w_C = \bar{w}/\theta$). To confirm this is innocuous, I solve Hegemon C’s Stage-1 monopoly explicitly. The unconstrained problem is ill-posed under the bottleneck: with $\sigma_c < 1$ the Swing State’s residual demand for the Chinese input is inelastic ($\varepsilon_D < 1$), so the markup diverges and no interior optimum exists.¹⁹ I therefore close Stage 1 with a standard constant-elasticity final-good demand $Y(C) = C^{-\eta}$, so that total output—and hence the Rival’s residual input demand—falls with the Swing State’s unit cost, restoring an interior markup. Figure 7 overlays the result on the normalized benchmark: the Regime Switch, the Paradox, and predatory pricing all survive, and coercion is uniformly *deeper*, robustly across $\eta \in [2, 6]$. I retain the normalized case as the main illustration because it isolates the security and bottleneck channels without the auxiliary demand closure—an extension, not a maintained assumption: every analytic result uses only Assumption 2. The solved markup deepens coercion but does not push the calibration into the deep-coercive regime (λ^* stays below ≈ 0.33), so the Stage-3 corollary $dx^*/d\theta > 0$ continues to pertain to a regime outside the empirically plausible range—consistent with its deep-coercive scoping in Section 4.

The empirical defense of $\sigma_c < 1$ and the comparative statics of the regime-transition threshold $\bar{\theta}(\sigma)$ across plausible sectoral values are deferred to Section 3.1.2 (literature review and primitive-distinction note) and Appendix B.7 (numerical robustness across $\sigma_c \in [0.1, 0.5]$ and a nested-CES extension). Having established the regime switch and its qualitative robustness, I now conclude by consolidating the paper’s findings and their policy implications.

¹⁹The Rival’s profit $mQ_C(x^*(w_C))$ rises monotonically in w_C : under $\sigma_c < 1$ the composite demand elasticity is below unity, marginal revenue stays positive, and the optimum is unbounded. The bounded geopolitical-influence term $\phi(1 - x)$ cannot offset an unbounded profit term, so $\phi > 0$ does not restore interiority. The marginal-cost normalization is the cleanest resolution.

6 Conclusion

As China became more productive, US coercion of swing states grew more, not less, aggressive—and as that coercion grew, the welfare loss it imposed on swing states widened. The model identifies both as equilibrium responses of a downstream Hegemon facing an upstream chokepoint with bottleneck complementarity ($\sigma_c < 1$), even when institutional primitives are held fixed. Past a Chinese-productivity threshold $\bar{\theta}$, the Hegemon’s optimal policy splits from subsidies alone to subsidies plus coercion—the *Regime Switch* (Proposition 1). Within the coercive regime, more productive Rivals invite harsher coercion—the *Paradox of Competitiveness* (Proposition 2). The welfare wedge against the integrated first-best widens in θ through the Hegemon’s induced coercive response—the *Tragedy of Efficiency* (Proposition 3). The empirical strategy combines a classifier-conditional joint test of the bottleneck-amplification interaction with two descriptive benchmarks. The joint test is the HS6 panel difference-in-differences yielding $\hat{\beta}_3 = +0.110$ ($p = 0.031$) under the bottleneck-literature classifier, with the cross-classifier sign-flip as the model’s discriminating signature rather than as an independent identification of $\sigma_c < 1$. The descriptive benchmarks are a 2018 structural break in US Entity-List enforcement against placebo country groups, and a 19% China-share decline in Korean and Taiwanese advanced-chip exports around the May 2020 FDPR against Huawei. Three limitations bound the reading. First, $\sigma_c < 1$ is identified on the photo-chemicals subset of the 2019 Japan–Korea episode. Second, Test 3 is descriptive within-cell pre/post, not a clean difference-in-differences. Third, Test 2 is joint with the bottleneck-literature classifier, with the cross-classifier sign-flip serving as the model’s discriminating signature rather than as an independent identification.

The policy implications are concrete enough to enter the IMF surveillance agenda. First, fragmentation-cost calibration: the welfare wedge $\Delta(\theta)$ in Proposition 3 is increasing in θ . World Economic Outlook (WEO) fragmentation-scenario tables should therefore report sectoral wedges conditional on the bottleneck-literature classifier, not a uniform fragmentation cost. Second, swing-state vulnerability indices: the model implies vulnerability scales with

the swing state’s pre-period Red-Chain share in critical sectors weighted by θ_k . A composite indicator built from Comtrade bilateral shares and USPTO Chinese-assignee patent counts would identify the swing states most exposed to the Coercive Turn—Korea, Taiwan, the Netherlands, Vietnam, and Mexico in the current data. Third, anticipation: subsidy-based instruments alone are insufficient once θ_C crosses $\bar{\theta}$, so the appropriate counterfactual for evaluating fiscal stimulus packages targeting bottleneck sectors is “coercion plus subsidies” rather than “subsidies alone”. Surveillance frameworks should expect structural, not transient, coercion in critical sectors.

The single-sector baseline calls for a multi-sector extension in which $\sigma_{c,k}$ varies across product classifications and the threshold $\bar{\theta}_k$ is heterogeneous. The IO-network primitives in Baqaee and Farhi (2024) provide the calibration infrastructure. The cross-classifier dose-response evidence in this paper provides one input to identifying $\sigma_{c,k}$ at the HS6 level—a natural next step for quantifying the welfare cost of geoeconomic fragmentation under heterogeneous bottleneck intensity.

A Microfoundations for the Policy Schedule, Fiscal Capacity, and Securitization Cost

The baseline model takes three objects as primitive: the market access schedule $R(\cdot)$ that Hegemon U offers the Swing State in Stage 2, a fiscal-capacity bound $\bar{T}$ that implicitly disciplines $R(\cdot)$ through the convex term $\frac{\gamma}{2}\lambda^2$, and a securitization-cost coefficient γ that shapes how aggressively Hegemon U embraces coercion. This appendix provides explicit microfoundations—one for $R(\cdot)$ based on a principal–agent contract between Hegemon U (principal) and the Swing State (agent), one for $\bar{T}$ based on public-finance frictions, and one for γ based on domestic lobbying—each yielding exactly the reduced forms used in the text.

A.1 Fiscal Capacity $\bar{T}$ as a Weighted-Surplus Constraint

Following Dixit et al. (1997), consider a welfare-weighted government that maximizes

$$\mathcal{W}_{gov} = \int \omega_i \Pi_i di - (1 + \chi) \int R(x) dG(x), \quad (17)$$

where ω_i denotes the political weight on group i 's profits, $G(\cdot)$ is the distribution of alignment targets across sectors, and $\chi \geq 0$ is the marginal deadweight loss of public funds (Ballard et al., 1985). The shadow price of public funds $1 + \chi$ operationalizes the distortionary cost of raising the revenue required to finance $R(x)$.

The first-order condition with respect to the transfer schedule yields a shadow budget: at the optimum, the marginal political return on subsidies must equal $1 + \chi$. Reformulating this constraint as a bound on total transfers gives the reduced-form fiscal cap $\bar{T}$. Critically, $\bar{T}$ is *endogenous*: it tightens when χ rises (e.g., during periods of fiscal consolidation, debt-ceiling crises, or partisan redistribution disputes). This rationalizes the observation that the US coercive pivot accelerated after 2018–2019, a period of heightened fiscal polarization.

Incorporating (17) into Hegemon U's Stage-2 problem preserves the functional form of the reduced objective in Eq. (7), provided the institutional damping coefficient is reinterpreted as $\gamma \equiv (1 + \chi)/\kappa$ where κ is the responsiveness of political weights to security externalities. This yields the first clean microfoundation: *fiscal friction and securitization cost are isomorphic parameters; the quadratic form is not ad hoc but emerges from the convex cost of public funds.*

A.2 Securitization Cost γ from Lobbying

An alternative microfoundation grounds γ in Grossman and Helpman (1994) lobby competition between pro-integration and pro-security factions. Let $L_I(\lambda)$ and $L_S(\lambda)$ denote the political contributions offered by the integration and security lobbies as functions of the security doctrine. Both lobbies contribute according to common-agency truthful strategies, with quasi-linear utilities $u_I = V_I(\lambda) - C_I$ and $u_S = V_S(\lambda) - C_S$, where $V_I' < 0$, $V_S' > 0$, and

both functions are strictly concave.

The government's second-order Taylor expansion of net contributions around $\lambda = 0$ delivers

$$L_S(\lambda) - L_I(\lambda) \approx (L'_S(0) - L'_I(0))\lambda - \frac{1}{2}(|L''_I(0)| + |L''_S(0)|)\lambda^2 + O(\lambda^3). \quad (18)$$

The linear term is absorbed into the baseline Security Utility λx ; the quadratic term is precisely the $-\frac{\gamma}{2}\lambda^2$ cost, with $\gamma \equiv |L''_I(0)| + |L''_S(0)|$ interpreted as the combined concavity of lobby willingness-to-pay. This yields the second microfoundation: *the quadratic institutional damping is a second-order approximation to lobby competition over the security doctrine, not an arbitrary functional assumption.*

A.3 Microfoundation of the Market Access Schedule $R(x)$

The main text takes the market access schedule $R(x)$ as a primitive choice variable for Hegemon U in Stage 2 (Section 3.3). This subsection places $R(x)$ inside a principal–agent contract between Hegemon U (principal) and the Swing State (agent), and shows that the same minimal contracting environment simultaneously rationalizes two reader questions: *why does the Swing State's revenue $R(x)$ depend on its alignment choice x* (Stage 3, Section 3.2), and *why does Hegemon U's Net Economic Surplus net $R(x)$* (the $V_{US}(x) - R(x)$ term in equation (5) at Stage 2)?

Framing: the Swing State is not a firm. Before writing down the contract, one clarification. The Swing State in the model is an institutional entity whose objective is a primitive of the principal–agent problem, not a reduced form of firm-theoretic optimization. Concretely, the Swing State chooses an *alignment policy* $x \in [0, 1]$, and its payoff is $\Pi(x) = R(x) - C(x; w_C)$, where $C(x; w_C)$ is the *Decoupling Cost* defined in Section 3.1 — the resource penalty the Swing State bears for operating its supply chain at alignment x . A reader might expect that, because $C(x; w_C)$ has the algebraic shape of a CES cost object,

the Swing State is first solving a cost-minimization problem over input shares and then a firm-profit maximization would pin down output. That reading is a category error in the setting. $C(x; w_C)$ inherits its CES shape from the technology that aggregates inputs at the alignment-imposed mix $(1 - x, x)$; it is a *technology evaluation* at a policy-determined x , not the Lagrangian of a cost-minimization with x as a control. The cost-minimizing x is the laissez-faire benchmark x^{LF} satisfying $C'(x^{\text{LF}}; w_C) = 0$; the Swing State's actual x satisfies the incentive-compatibility condition $R'(x^*) = C'(x^*)$, which differs whenever the contract is non-trivial ($R'(\cdot) \neq 0$). The Swing State's profit is therefore neither a standard firm's profit-maximum nor a standard firm's cost-minimum: it is the contractually defined rent $R(x) - C(x; w_C)$ over the alignment *policy*, and only after $R(\cdot)$ is specified by the principal does the Swing State have a well-posed optimization problem.

Contract environment. Throughout I work with $C(x; w_C)$ as specified in equation (2), reading it as the Decoupling Cost rather than as a unit production cost, and $R(x)$ as the total transfer from Hegemon U to the Swing State contingent on realized alignment. Two institutional conditions support the contract:

1. *Contractibility of x .* Alignment is observable and enforceable—through supply-chain origin requirements, customs filings, dual-use licensing records, and defense-procurement audits—so the transfer can be conditioned on x .
2. *Commitment.* Hegemon U commits to a schedule $R(\cdot)$ before the Swing State's alignment choice, per Assumption 1(i). Without commitment, R could not deviate from a spot price and alignment would collapse to the laissez-faire share x^{LF} .

The Swing State accepts the contract if and only if its contract-induced profit $\Pi(x) = R(x) - C(x; w_C)$ meets an outside option V_{out} , normalized to zero per Section 3.2.

Hegemon U's gross benefit $V_{US}(x)$. I take $V_{US}(x)$ as a primitive of Hegemon U's stage objective, with the verbal rationale already given in Section 3.3: $V_{US}(x)$ is consumer surplus

or IP royalties accruing to Hegemon U from Swing-State delivery at alignment x , declining in x because higher alignment with the Rival degrades intellectual-property security, product match with Hegemon U's end-use ecosystem, and compatibility with Hegemon U's procurement and licensing infrastructure. I impose $V'_{US}(x) < 0$ without deriving V_{US} from consumer preferences; the linear form $V_{US}(x) = V_0 - \beta x$ used in Section 5 is an admissible specification.

Reduced form. Hegemon U chooses $R(\cdot)$ to maximize $V_{US}(x) - R(x)$ subject to the Swing State's participation constraint $R(x) \geq C(x; w_C) + V_{out}$. Any slack in the constraint is a rent Hegemon U captures by lowering R , so the constraint binds at the implementation point:

$$R(x) = C(x; w_C) + V_{out}. \quad (19)$$

Substituting into Hegemon U's economic surplus,

$$V_{US}(x) - R(x) = [V_{US}(x) - C(x; w_C)] - V_{out} \equiv J(x) - V_{out}, \quad (20)$$

exactly the Net Economic Surplus term that enters equation (5).

Answer to the first question: why does the Swing State's revenue depend on x ? Because the contract is contingent by construction— R is a *function* of the policy x , not a scalar spot price. The Swing State's primitive objective $\Pi(x) = R(x) - C(x; w_C)$ becomes a well-posed optimization problem only once $R(\cdot)$ is announced, and the incentive-compatibility condition $R'(x^*) = C'(x^*)$ pins down the interior optimum. Under a linear spot price p , revenue would be $pY = p$ (since $Y = 1$) independent of x , and the problem would degenerate to pure technological cost-minimization in x with solution at the laissez-faire share x^{LF} ; the strategic alignment margin I model would not exist. Contractibility of x plus commitment is what makes alignment a strategic object.

Answer to the second question: why does Hegemon U’s welfare net $R(x)$? Because $V_{US}(x)$ is Hegemon U’s *gross* benefit from alignment- x delivery, and $R(x)$ is the contractual transfer it *pays* to the agent for that delivery—standard principal-payoff accounting. The binding participation constraint (19) then reduces the net surplus to the Joint Economic Surplus $J(x)$ net of the Swing State’s outside option. Crucially, this is the same $R(x)$ that appears with positive sign in the Swing State’s profit function (Section 3.2): the two value functions are the two sides of one bilateral contract ledger.

Concrete implementations. The transfer $R(x) = C(x; w_C) + V_{out}$ admits several policy-equivalent implementations:

1. *Preferential import tariff.* A tariff schedule $\tau(x)$ declining in x with non-preferential baseline τ_0 : $R(x) = \tau_0 - \tau(x)$ (per-unit and total coincide under $Y = 1$).
2. *CHIPS-Act-style conditional subsidy.* A direct subsidy on Swing State output contingent on documented alignment thresholds.
3. *Preferential defense-coproduction access.* Nonpecuniary transfers—technology licensing, basing rights, access to classified supply chains—that deliver rents in the same reduced form.

Closing conditions. The derivation requires only two institutional conditions: contractibility of x (observable and enforceable alignment) and Hegemon U’s commitment to $R(\cdot)$ before the Swing-State decision. Neither condition is a market-structure assumption about demand elasticity or market power in the conventional firm-theoretic sense: both are properties of the contracting environment, appropriate to the state-to-state bargain I model rather than a firm-to-firm transaction. A protection-for-sale microfoundation for the *shape* of $R(\cdot)$ follows Grossman and Helpman (1994); the present derivation closes the *existence* and *dimensional consistency* of the schedule, which are orthogonal to the lobbying-generated shape argument.

B Additional Model Results

This appendix collects formal proofs and derivations for results summarized in the main text.

B.1 Existence, Uniqueness, and Local Stability

Lemma 4 (Existence and Uniqueness of the Stage-2 Equilibrium). *Under Assumption 2 and assuming $V_{US}(x)$ is weakly concave with $V'_{US}(x) < 0$ (the linear specification $V_{US}(x) = V_0 - \beta x$ used in the Section 5 calibration is included), the Stage-2 problem (7) admits a unique interior equilibrium $(\lambda^*, x^*) \in [0, 1] \times (0, 1)$ for all γ sufficiently large. Weak concavity is sufficient because the proof uses only $J''(x) = V''_{US}(x) - C''(x; w_C) < 0$, and Assumption 2(C2) delivers $C''' > 0$ on the decoupling range, so $J'' < 0$ holds for any $V''_{US} \leq 0$.*

Proof. The objective $\mathcal{W}_{\text{reduced}}$ is continuous on the compact set $[0, 1] \times [0, 1]$, so a maximum exists by Weierstrass. For uniqueness, I show the Hessian is negative definite on the interior. The diagonal terms are $\mathcal{W}_{xx} = (1 - \lambda)J''(x) < 0$ (by concavity of V_{US} and convexity of C under Assumption 2(C2)) and $\mathcal{W}_{\lambda\lambda} = -\gamma < 0$. The cross partial is $\mathcal{W}_{x\lambda} = \theta - J'(x)$. The determinant of the Hessian is

$$\det H = -\gamma(1 - \lambda)J''(x) - (\theta - J'(x))^2.$$

Define the primitive constants

$$\bar{M}_1 \equiv \sup_{x \in [0, 1]} |\theta - J'(x)|, \quad \underline{M}_2 \equiv \inf_{x \in [0, 1]} |J''(x)|,$$

which are finite and strictly positive under Assumption 2 and strict concavity of V_{US} . Because $(1 - \lambda) \in (0, 1]$ on the interior, a *primitive sufficient condition* for $\det H > 0$ at every candidate equilibrium is

$$\gamma > \bar{\gamma} \equiv \frac{\bar{M}_1^2}{\underline{M}_2}. \quad (21)$$

Condition (21) depends only on the curvature of the joint-surplus function J over $[0, 1]$ and

on the threat multiplier θ (fixed at the Stage-2 sub-game), not on equilibrium quantities, and is checkable from the model's primitives $(\sigma, \alpha, w_C, w_A, \bar{w}, m)$. Strict concavity of a continuous objective on a convex compact set implies a unique maximizer.

Remark (local SOC vs. primitive bound). The primitive constant $\underline{M}_2 = \inf_x |J''(x)|$ vanishes as $x \rightarrow 1$ in the baseline CES calibration (the CES aggregator flattens the cost curvature at the upper boundary), so $\bar{\gamma}$ computed from (21) is uninformatively large. This does not compromise the lemma's conclusion: uniqueness at an interior first-order critical point requires only that the Hessian be *locally* negative-definite at that point, a strictly weaker condition than the global primitive bound (21). In the numerical illustration of Section 5 I therefore verify this local second-order condition directly, computing the Hessian at each computed equilibrium $(x^*, \lambda^*; w_C)$ along a grid of w_C ; negative-definiteness holds pointwise throughout, so the lemma's operative uniqueness guarantee is satisfied without invoking (21). $\square$

Lemma 5 (Local Stability of Best-Response Dynamics). *Fix $\epsilon \in (0, 1)$ in advance and restrict attention to the compact set $K_\epsilon \equiv \{\lambda \in [0, 1 - \epsilon]\} \times [0, 1]$. If, in addition to the conditions of Lemma 4, the fiscal-capacity parameter satisfies*

$$\gamma > \bar{\gamma}_\epsilon \equiv \frac{\bar{M}_1 \theta}{\epsilon^2 \underline{M}_2}, \quad (22)$$

then the best-response mapping $T(\lambda, x) \equiv (\lambda^(x), x^*(\lambda))$ is a contraction on K_ϵ , so any interior fixed point with $\lambda^* \leq 1 - \epsilon$ is locally stable under naive best-reply dynamics. The bound (22) is purely primitive, depending only on $(\bar{M}_1, \underline{M}_2, \theta, \epsilon)$ and not on the equilibrium λ^* itself.*

Proof. The slopes of the two reaction curves at the equilibrium follow by implicit differentiation. From the doctrine FOC (9), $d\lambda^*/dx = (\theta - J'(x^*))/\gamma$. From the alignment FOC (8), written as $J'(x^*) + \lambda\theta/(1 - \lambda) = 0$, differentiating with respect to λ gives $J''(x^*) dx^*/d\lambda +$

$\theta/(1 - \lambda)^2 = 0$, so

$$\frac{dx^*}{d\lambda} = -\frac{\theta}{(1 - \lambda)^2 J''(x^*)} > 0,$$

using $J'' < 0$ (concavity of J under Assumption 2). The product of slopes is

$$\left| \frac{d\lambda^*}{dx} \cdot \frac{dx^*}{d\lambda} \right| = \frac{|\theta - J'(x^*)| \theta}{\gamma (1 - \lambda)^2 |J''(x^*)|}.$$

Over K_ϵ , $\lambda \leq 1 - \epsilon$ so $(1 - \lambda) \geq \epsilon > 0$, hence $(1 - \lambda)^{-2} \leq \epsilon^{-2}$. Combined with $|\theta - J'| \leq \bar{M}_1$ and $|J''| \geq \underline{M}_2$, the slope product is bounded above by $\bar{M}_1 \theta / [\gamma \epsilon^2 \underline{M}_2]$, which is strictly less than unity under (22). Best-response iteration is therefore a contraction on K_ϵ . Note that $\bar{\gamma}_\epsilon$ diverges as $\epsilon \rightarrow 0$ (i.e., as the admissible doctrine ceiling approaches the boundary $\lambda = 1$); the local-stability claim is genuinely local and does not extend uniformly to the boundary. For any candidate equilibrium $\lambda^* \leq 1 - \epsilon$, the bound is finite and checkable from primitives. $\square$

B.2 Probabilistic Microfoundation of the Security Doctrine

Let D denote the economic damage inflicted on the Hegemon coalition if the Rival activates a supply-chain “kill-switch” (an embargo, export ban, or deliberate disruption). The fraction of Hegemon inputs exposed to such an event at alignment x is $1 - x$ (the Red-Chain share), and the kill-switch is fired with unit probability conditional on the event. Expected damage conditional on activation is then $(1 - x) \cdot D$, and the damage avoided by Hegemon-side alignment relative to the no-alignment baseline ($x = 0$) is

$$\mathbb{E}[\text{damage avoided}] = D(\theta) \cdot x. \tag{23}$$

The damage scales with upstream productivity, $D(\theta) = \theta$. The damage D is the economic surplus the Hegemon loses when the Rival withholds an embodied unit of the critical input. Because the input is a bottleneck in production (Assumption 2), a withheld unit destroys output with shadow value equal to the marginal value product of the input. Under

the Stage-1 decomposition (12), the marginal cost of the input is $\bar{w}/\theta$: a more productive Rival supplies the same physical unit at lower cost, which raises the Hegemon's consumption surplus per unit and therefore the surplus destroyed by withholding. Formally, if Hegemon surplus from a unit of the input is $u(\bar{w}/\theta)$ with $u' < 0$, a linear Taylor expansion around the baseline technology $\theta = 1$ delivers $u(\bar{w}/\theta) - u(\bar{w}) \approx u'(\bar{w}) \cdot \bar{w} \cdot (1/\theta - 1)$, which is strictly increasing in θ and vanishes at $\theta = 1$ (since $u'(\bar{w}) < 0$ and $1/\theta - 1 < 0$ for $\theta > 1$, the product is positive and rises in θ). Linearizing further yields $D(\theta) \propto \theta$ up to a multiplicative constant that can be absorbed into the units of λ .²⁰ Combining with (23),

$$\mathbb{E}[\text{damage avoided}] = \theta \cdot x, \quad (24)$$

which, weighted by the Hegemon's endogenous security posture λ , yields the Security Utility term $\lambda x \theta$ in equation (5). The same primitive θ that determines Hegemon C's marginal cost in (12) therefore enters Hegemon U's Stage-2 objective directly; under the complete-information assumption, no inversion through w_C is required.

Under this reading, the comparative static $d\lambda^*/d\theta > 0$ (Proposition 2) states that as the Rival becomes more efficient, (i) the conditional damage $D(\theta)$ rises *because the withheld input is more valuable*, and (ii) the Hegemon's perceived probability of weaponization rises endogenously through the induced fall in w_C —jointly, a formal version of the security-dilemma logic. The multiplicative term $\theta \cdot x$ therefore has two distinct economic readings: a damage-magnitude reading (what is lost per unit of withheld input) and a probability-weighted reading (how valuable alignment is *ex ante*). Both readings deliver the same reduced-form Stage-2 objective.

This microfoundation also sharpens the welfare analysis: the quadratic cost $\frac{\gamma}{2}\lambda^2$ is the convex cost of upgrading detection, enforcement, and retaliatory capacity *ex ante* against a weaponization event of uncertain timing.

²⁰An equivalent derivation: let $D(\theta) = \theta \cdot \bar{D}$ where $\bar{D}$ is the damage per unit of normalized Rival productivity; normalizing $\bar{D} \equiv 1$ in the units of λ yields $D(\theta) = \theta$ without loss of generality. The convex cost $\frac{\gamma}{2}\lambda^2$ is invariant to this rescaling.

B.3 Swing-State Bargaining Power

Let $\nu \in [0, 1]$ denote the Swing State's bargaining weight in a generalized Nash product. (I use ν rather than the conventional β to avoid collision with the empirical regression coefficients $\beta_1, \beta_2, \beta_3$ in Section 4.2.) Replacing the binding participation constraint with asymmetric Nash bargaining, the market-access schedule solves

$$R^{NB}(x) = \arg \max_R [R - C(x) - V_{out}]^\nu [V_{US}(x) - R]^{1-\nu}. \quad (25)$$

The baseline model corresponds to $\nu = 0$ (take-it-or-leave-it by Hegemon U). Solving yields

$$R^{NB}(x) = C(x) + V_{out} + \nu[V_{US}(x) - C(x) - V_{out}], \quad (26)$$

so the Swing State captures a share ν of the joint surplus $J(x) - V_{out}$.

Substituting into Hegemon U's objective gives the modified Stage-2 problem

$$\max_{x, \lambda} \lambda x \theta + (1 - \lambda)(1 - \nu)[J(x) - V_{out}] - \frac{\gamma}{2} \lambda^2. \quad (27)$$

The factor $(1 - \nu)$ shrinks Hegemon U's effective economic payoff, tilting the Strategic Gap upward.

Proposition 4 (Bargaining Amplifies Coercion). *The equilibrium security doctrine $\lambda^*(\nu)$ is non-decreasing in the Swing State's bargaining weight ν , with the derivative strictly positive in the Coercive-Hegemon regime ($J(x^*) > V_{out}$ with interior $\lambda^* > 0$); and the critical productivity threshold $\bar{\theta}(\nu)$ at which coercion is triggered is strictly decreasing in ν : more powerful swing states trigger coercion sooner.*

Proof of Proposition 4. Differentiating the optimal doctrine rule with respect to ν at x^* fixed yields $\partial \lambda^* / \partial \nu = [J(x^*) - V_{out}] / \gamma > 0$ in the region where $J(x^*) > V_{out}$ with interior $\lambda^* > 0$, which is the Coercive-Hegemon regime; on the Generous-Empire corner $\lambda^* = 0$ binds and $\partial \lambda^* / \partial \nu = 0$, giving non-decreasing globally and strictly increasing in the regime stated. The

threshold comparative static follows because the “fiscal wall” is hit at a lower J , i.e., at a lower θ , when the Hegemon retains only a $(1 - \nu)$ share. $\square$

B.4 Endogenous Vertical Position

Consider two hegemonies with exogenous productivity θ_U and θ_C in upstream input production and exogenous market-demand size E_U and E_C in final-good consumption. In a pre-stage each hegemon chooses a specialization vector $s_i \in \{\text{up}, \text{down}\}$, anticipating the Stage-2 contracting equilibrium. Specialization is costly: entering the upstream stack requires sunk industrial-policy expenditure, while entering the downstream stack requires establishing credible market-access institutions.

Remark 1 (Endogenous Vertical Asymmetry). Let $\theta_C > \theta_U$ and $E_U > E_C$. If the fixed cost of downstream institutional commitment is lower for the hegemon with the larger consumption base, a standard Ricardian argument delivers a subgame-perfect equilibrium with $s_C = \text{up}$ and $s_U = \text{down}$, replicating the baseline model: each hegemon specializes where its relative productivity advantage is greatest (upstream specialization is more profitable for the lower-marginal-cost hegemon; downstream specialization for the larger-market hegemon), and mixed strategies are dominated when institutional sunk costs are convex. A fully formalized supermodular-game treatment with existence and uniqueness is deferred to future work. Propositions 1–3 hold at the continuation equilibrium.

B.5 Proof of the Welfare Decomposition and the Efficiency Wedge

This subsection provides the full derivations for Propositions 5 and 3.

Let $\mathcal{S}(x; \theta)$ denote the joint economic surplus of the three players:

$$\mathcal{S}(x; \theta) = V_{US}(x) + Q_C(x) \cdot [w_C^*(\theta) - \bar{w}/\theta] - C(x; w_C^*(\theta)), \quad (28)$$

where the first term is the gross value to Hegemon U of alignment level x , the second term

is Hegemon C's commercial profit on inputs shipped at the equilibrium markup, and the last term is the Swing State's Decoupling Cost. The integrated planner's *first-best alignment* $x^{FB}(\theta)$ maximizes $\mathcal{S}(x; \theta)$ on $[0, 1]$ ignoring the security externality $\lambda x \theta$; because $V'_{US}(x) < 0$ and $C'(x) > 0$, the first-best is generically interior and *declining* in θ .

Proposition 5 (Welfare Decomposition). *Along the equilibrium path, global welfare $\mathcal{W}^*(\theta) = \mathcal{S}(x^*(\theta); \theta) + \lambda^*(\theta) x^*(\theta) \theta - \frac{\gamma}{2} \lambda^*(\theta)^2$ decomposes into a commercial component and a geopolitical component. Under Assumption 2, the curvature condition*

$$\frac{\lambda^* \theta}{1 - \lambda^*} \cdot \frac{dx^*}{d\theta} > - \frac{\partial C}{\partial w_C} \Big|_{x^*} \cdot \frac{dw_C^*}{d\theta} \quad (\text{CC}), \quad (29)$$

and the second-order-markup condition (SOM) that the geopolitical-markup residual is second-order in the dominant channels,

$$|Q'_C(x^*) \cdot m^*(\theta)| < \min \left\{ \frac{\lambda^* \theta}{1 - \lambda^*} \cdot \frac{dx^*}{d\theta}, - \frac{\partial C}{\partial w_C} \Big|_{x^*} \cdot \frac{dw_C^*}{d\theta} \right\} \quad (\text{SOM}), \quad (30)$$

where $m^*(\theta) \equiv w_C^*(\theta) - \bar{w}/\theta$ is the Hegemon C geopolitical markup, the commercial component $\mathcal{S}(x^*(\theta); \theta)$ is strictly decreasing in θ in the coercive regime, while the geopolitical component is bounded above by $1/(2\gamma)$. Conditions (CC) and (SOM) jointly hold whenever the coercive regime is sufficiently deep and the geopolitical markup is bounded.

Proof of Proposition 5 (Welfare Decomposition). Differentiating $\mathcal{S}$ along the equilibrium path,

$$\frac{d\mathcal{S}}{d\theta} = \underbrace{\frac{\partial \mathcal{S}}{\partial \theta} \Big|_{x^*}}_{\text{direct}} + \underbrace{\frac{\partial \mathcal{S}}{\partial x} \Big|_{x^*} \cdot \frac{dx^*}{d\theta}}_{\text{indirect}}. \quad (31)$$

The direct effect aggregates three partials: the derivative of V_{US} vanishes; the middle-term derivative is $Q_C(x^*)[dw_C^*/d\theta + \bar{w}/\theta^2]$, which is finite; the cost derivative is $-(\partial C/\partial w_C) \cdot (dw_C^*/d\theta) > 0$ because $\partial C/\partial w_C > 0$ (by direct CES computation; cf. proof of Lemma 1, where the explicit form $\partial C/\partial w_C = (1 - x)/A(x) > 0$ is derived) and $dw_C^*/d\theta < 0$ (from

Lemma 3). Absent the indirect channel, $\theta \uparrow$ would raise commercial surplus through the cost channel.

The indirect effect is negative in the coercive regime. By the Stage-2 FOC (8), the equilibrium satisfies $(1 - \lambda^*)J'(x^*) = -\lambda^* \theta$, so $\partial \mathcal{S} / \partial x|_{x^*} = J'(x^*) + Q'_C(x^*)[w_C^* - \bar{w}/\theta] = -\lambda^* \theta / (1 - \lambda^*) + O(m^*)$, which is strictly negative whenever $\lambda^* > 0$. By Proposition 2, $dx^*/d\theta > 0$ in the coercive regime. The indirect channel therefore contributes a term of order $-[\lambda^* \theta / (1 - \lambda^*)] \cdot dx^*/d\theta < 0$. Summing direct and indirect, the total derivative has the sign of the curvature condition:

$$\frac{d\mathcal{S}}{d\theta} = \underbrace{-\frac{\partial C}{\partial w_C} \Big|_{x^*} \cdot \frac{dw_C^*}{d\theta}}_{\text{direct channel} > 0} - \underbrace{\frac{\lambda^* \theta}{1 - \lambda^*} \cdot \frac{dx^*}{d\theta}}_{\text{indirect channel} > 0} + O(m^*). \quad (32)$$

The $O(m^*)$ remainder in Eq. (32) collects the commercial-profit term $Q'_C(x^*) \cdot m^*$ evaluated along the equilibrium path: it is bounded in absolute value by $|Q'_C(x^*)| \cdot |m^*(\theta)|$, and (SOM) is precisely the hypothesis that this bound is strictly smaller than the dominant direct and indirect channels. Under (SOM), $d\mathcal{S}/d\theta < 0$ holds iff (CC) holds; without (SOM) the sign of $d\mathcal{S}/d\theta$ is in principle ambiguous in a narrow band of (θ, λ^*) where the markup residual is not small. Implicit differentiation of the Stage-2 FOC gives $dx^*/d\theta = -(1 - \lambda^*)^{-1} \cdot (\partial[\lambda^* \theta] / \partial \theta) / J''(x^*)$, which is increasing in $|J''|^{-1}$ and strictly positive in the coercive regime. Under Assumption 2, tighter bottlenecks (in the CES parameterization (2), smaller σ below unity) raise the convexity of $C(\cdot; w_C)$ and hence $|J''|$; however, the dominant effect is on λ^* itself, which grows at rate $1/\gamma$ times the (threat-weighted) Strategic Gap $x^* \theta - [J(x^*) - V_{out}]$. As $\lambda^* \uparrow$, the indirect channel $[\lambda^* \theta / (1 - \lambda^*)] \cdot dx^*/d\theta$ diverges while $|Q'_C(x^*) \cdot m^*|$ stays bounded under the commercial-markup constraint of Stage 1. Solving for the threshold explicitly, $\underline{\lambda}$ is the smallest λ at which the dominance condition

$$\frac{\lambda \theta}{1 - \lambda} \cdot \frac{dx^*}{d\theta} \Big|_{\lambda} \geq |Q'_C(x^*)| \cdot \bar{m} \quad (33)$$

holds, with $\bar{m} \equiv \sup_{\theta \in [\underline{\theta}, \bar{\theta}_{\max}]} |m^*(\theta)|$ the maximal Stage-1 commercial markup over the calibrated θ range. Implicit differentiation of the Stage-2 FOC (8) in θ at fixed λ gives $dx^*/d\theta|_{\lambda} = \lambda/[(1-\lambda)|J''(x^*)|]$, since $J''(x^*) < 0$ on the coercive interior under Assumption 2 and the linearity of V_{US} . Substituting into (33) and taking positive square roots, the dominance condition reduces to $\lambda/(1-\lambda) \geq \mathcal{K}(\theta)$, where $\mathcal{K}(\theta) \equiv \sqrt{|J''(x^*(\theta))| \cdot |Q'_C(x^*(\theta))| \cdot \bar{m}/\theta}$, yielding the closed-form threshold

$$\underline{\lambda}(\theta) = \frac{\mathcal{K}(\theta)}{1 + \mathcal{K}(\theta)}. \quad (34)$$

Under the CES parameterization (2) with $V_{US}(x) = V_0 - \beta x$, both $|J''(x)| = C''(x; w_C)$ and $|Q'_C(x)| = |d/dx[(1-x)/A(x)]|$ admit closed forms in $(\sigma, \alpha, w_C, w_A)$; $\underline{\lambda}(\theta)$ is therefore a closed form in the primitive tuple $(\sigma, \alpha, \bar{w}, w_A, V_0, \beta, \bar{m})$. In the simulation of Section 5, I adopt the normalization $w_C(\theta) = \bar{w}/\theta$ — i.e., I set w_C exogenously equal to marginal cost rather than solving Stage 1's monopoly problem — at the calibrated parameters $(\sigma = 0.35, \alpha = 0.5, V_0 = 0.84, \beta = 1.0, \gamma = 2.5, \bar{w} = w_A = 1)$. Under this normalization $m^*(\theta) \equiv 0$ by construction, hence $\bar{m} = 0$ and $\mathcal{K}(\theta) \equiv 0$: equation (34) gives $\underline{\lambda}(\theta) \equiv 0$ throughout the coercive interior $w_C \in [0.5, 1.6]$. Conditions (CC) and (SOM) hold trivially at the simulated equilibrium, and Proposition 3 applies for any $\lambda^* > 0$. This is a deliberate calibration choice: it isolates the security and bottleneck channels from confounding strategic-markup effects, at the cost of abstracting from Hegemon C's commercial pricing problem. For empirically anchored non-zero Chinese commercial markup $\bar{m} > 0$ (the case when Stage 1 is solved rather than normalized), equation (34) delivers a strictly positive threshold $\underline{\lambda}(\theta; \bar{m})$ that grows in $\bar{m}$; the deep-coercive-regime qualifier in Proposition 3 then rules out a narrow transition band where the geopolitical-markup residual $|Q'_C \cdot m^*|$ is non-negligible relative to the indirect channel.

The bound on the geopolitical term follows from $\max_{\lambda} \{\lambda x \theta - \frac{\gamma}{2} \lambda^2\} = (x\theta)^2/(2\gamma)$ at $\lambda = x\theta/\gamma$, which at $x \in [0, 1]$ and bounded θ is finite. $\square$

Proof of Proposition 3 (Tragedy of Efficiency, Quantified). Define the efficiency wedge $\Delta(\theta) \equiv \mathcal{S}(x^{FB}(\theta); \theta) - \mathcal{S}(x^*(\theta); \theta)$. Because $x^{FB}(\theta)$ is defined as the argmax of $\mathcal{S}(\cdot; \theta)$, the envelope theorem gives $d\mathcal{S}(x^{FB}; \theta)/d\theta = \partial\mathcal{S}/\partial\theta|_{x^{FB}}$. Comparing against the $d\mathcal{S}(x^*; \theta)/d\theta$ computed above,

$$\frac{d\Delta}{d\theta} = \underbrace{\left[\partial\mathcal{S}/\partial\theta|_{x^{FB}} - \partial\mathcal{S}/\partial\theta|_{x^*} \right]}_{\text{direct-channel gap}} - \underbrace{\frac{\partial\mathcal{S}}{\partial x}\bigg|_{x^*} \cdot \frac{dx^*}{d\theta}}_{\text{indirect channel}}. \quad (35)$$

The direct-channel gap is bounded above in absolute value by $\|\partial_{x\theta}^2\mathcal{S}\|_\infty \cdot (x^* - x^{FB})$ —that is, the gap is *first-order* in the alignment gap, not second-order. The envelope theorem gives $\partial\mathcal{S}/\partial x|_{x^{FB}} = 0$ but does not flatten the cross-partial $\partial_{x\theta}^2\mathcal{S}$; the gap therefore inherits a generically nonzero first-order dependence on $(x^* - x^{FB})$. The indirect channel, by contrast, grows without bound as $\lambda^* \rightarrow 1^-$: $\partial\mathcal{S}/\partial x|_{x^*} = -\lambda^*\theta/(1 - \lambda^*) + O(m^*)$ from the Stage-2 FOC delivered in the proof of Proposition 5 above, multiplied by $dx^*/d\theta > 0$ from Proposition 2; their product is negative, and the leading minus sign in the indirect-channel term delivers a contribution to $d\Delta/d\theta$ that diverges as $\lambda^* \rightarrow 1^-$. Magnitude dominance is therefore the operative argument: in the deep-coercive regime, the unbounded indirect channel overwhelms the bounded direct-channel gap. Since $x^*(\theta)$ rises and $x^{FB}(\theta)$ falls with θ —the first by Proposition 2; the second because the joint-surplus FOC $V'_{US}(x^{FB}) + \partial\pi_C/\partial x|_{x^{FB}} = \partial C/\partial x|_{x^{FB}}$ shifts when $w_C^*(\theta)$ falls (the cross-partial $\partial C'/\partial w_C < 0$ from Lemma 1 property C4, combined with $dw_C^*/d\theta < 0$, raises the marginal-cost slope, and the constant $V'_{US} < 0$ pins x^{FB} at a lower interior value)—the gap $x^* - x^{FB}$ widens monotonically, so the indirect channel's magnitude grows with θ along the equilibrium path. The deep-coercive-regime qualifier of Proposition 3 is precisely the regime in which λ^* is bounded away from zero by enough that the indirect-channel divergence dominates the bounded direct-channel first-order gap, yielding $d\Delta/d\theta > 0$. $\square$

B.6 Reconciling the Discrete Regime Switch with Smooth Empirical Dynamics

Proposition 1 predicts a kink in $\lambda^*(\theta)$ at $\bar{\theta}$ —a continuous-but-nondifferentiable transition from the corner regime to the interior regime, not a level jump. Yet the empirical dynamics documented in Section 2 appear smooth. I offer three non-exclusive reconciliations.

(i) *Heterogeneous sectors crossing the threshold sequentially.* Different sectors have different bottleneck elasticities $\sigma_{c,k}$ and different starting values of Upstream productivity θ_k . If θ_k rises at heterogeneous rates, the aggregate coercion intensity $\sum_k \mathbf{1}\{\theta_k > \bar{\theta}_k\} \cdot \lambda_k^*$ is a sum of step functions triggered at different calendar dates, which smooths the aggregate into an apparently continuous process. This is the sectoral-diffusion reading: each individual sector jumps, but the aggregate is an envelope.

(ii) *Bayesian uncertainty about σ .* The Hegemon does not know σ perfectly; it updates based on realized supply disruptions and technological assessments. A posterior distribution over σ delivers a posterior distribution over $\bar{\theta}$, and the *expected* security doctrine $\mathbb{E}[\lambda^*(\theta)]$ is a smooth function of θ even when the realized $\lambda^*(\theta; \sigma)$ is a step function for any given σ .

(iii) *Institutional adjustment costs.* Coercive institutions (enforcement agencies, licensing regimes, multilateral coordination) cannot be built instantaneously. Even after θ crosses $\bar{\theta}$, capacity constraints on the adjustment rate of λ^* would translate the discrete target into a smooth realized path.

Identification update. Section 4.1 reports a level-only break test of the post-2018 dummy on log annual Entity-List flow that formally rejects the no-break null for China-bilateral listings while failing to reject for three placebo country groups (Iran, Russia, non-CHN-non-RUS-non-IRN aggregate)—the asymmetry the regime-switch story predicts. Sector-specific structural-break tests on coercion intensity remain a natural follow-up project.

B.7 Comparative Statics of $\bar{\theta}(\sigma)$

The qualitative regime-switch behavior of Proposition 1 is robust across $\sigma \in [0.1, 0.5]$, the range identified by the production-network and natural-experiment literatures cited in Section 3.1.2. The Paradox-of-Competitiveness sign $d\lambda^*/d\theta > 0$ (Proposition 2) similarly holds throughout this range: numerical evaluation of the three channels in the proof sketch shows that the direct envelope term and the Stage-3 amplification channel uniformly dominate the cost-feedback offset on the interior regime $\lambda^* \in (\underline{\lambda}, 1)$, so the magnitude dominance asserted in the proof sketch (which Assumption 2 controls only at the gradient level) is established by direct evaluation across the calibrated parameter space. Let $\bar{\theta}(\sigma)$ denote the value of upstream productivity at which Hegemon U's problem transitions from the corner ($\lambda^* = 0$) to the interior regime, holding all other parameters fixed. Numerical evaluation of the model yields two robustness patterns:

1. *Monotone decline (bottleneck regime)*: on $\sigma \in [0.1, 0.5]$, $\bar{\theta}(\sigma)$ is strictly decreasing in σ .²¹ Weaker bottlenecks (higher σ) push the coercive threshold further out, because the fiscal cost of matching Chinese efficiency with subsidies scales less aggressively.
2. *Asymptotic collapse at $\sigma \rightarrow 1$* : As $\sigma \rightarrow 1^-$, $\bar{w}_C(\sigma) \rightarrow \infty$ (equivalently $\bar{\theta}(\sigma) \rightarrow 0$).

When inputs are substitutes, the Decoupling Cost $C(x; w_C)$ flattens and the Hegemon can always out-bid the Rival with a finite subsidy; the coercive regime is never reached.

Figure 8 displays the numerically evaluated $\bar{\theta}(\sigma)$ schedule over the bottleneck range, confirming both patterns.

Nested-CES robustness. A stronger version of the bottleneck, in which a critical sub-input is Leontief with all others (e.g., EUV lithography within the semiconductor production

²¹The robustness exercise is restricted to the empirically relevant $\sigma \in [0.1, 0.5]$ range. For $\sigma > 0.5$ a different mechanism dominates: as substitution becomes easier, the laissez-faire allocation $x^{\text{LF}}(w_C)$ becomes sensitive to w_C , and the Swing State's participation-constraint baseline $C(x^{\text{LF}}(w_C); w_C)$ interacts with the linear $V_{US}(x) = V_0 - \beta x$ to potentially flip the sign of the regime-transition comparative static. The sign-flip at $\sigma > 0.5$ is therefore a joint feature of (i) the CES parameterization of the Decoupling Cost and (ii) the specific V_{US} specification, not a property of the regime transition itself, and does not invalidate the $\sigma < 1$ bottleneck disciplined by the empirical literature.

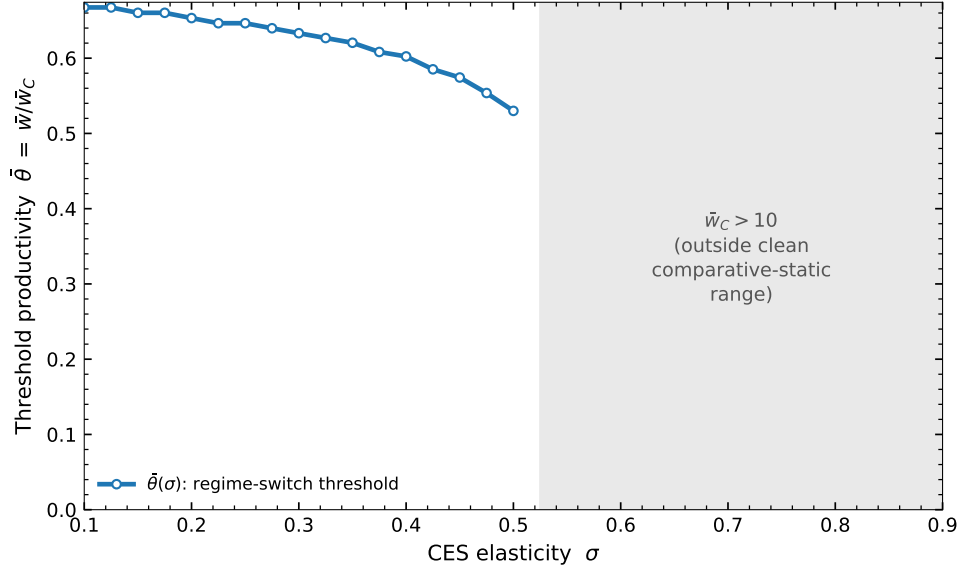


Figure 8: Regime-transition threshold $\bar{\theta}(\sigma)$ as a function of the input substitution elasticity. Calibration at $(V_0, \beta, \gamma, \alpha) = (0.84, 1.00, 2.50, 0.50)$ with $\bar{w} = w_A = 1$. Shaded region ($\sigma > 0.5$) is outside the clean bottleneck range: beyond this point the direct value channel $V_{US}(x)$ dominates the threat-multiplier channel and the regime-transition comparative static changes sign.

stack), nests into the framework as the limit $\sigma \rightarrow 0$. A nested-CES production function with a Leontief inner aggregator and a CES outer aggregator delivers the same qualitative Propositions 1–3; the quantitative effect is that $\bar{\theta}$ collapses to a lower value, making the coercive regime relevant even for moderate productivity gains. The derivation follows from substituting the nested aggregator into the Stage-2 problem of Section 3.3 and is omitted for space.

C Empirical Methodology and Regression Results

This appendix documents the data construction underpinning the empirical evidence in Sections 4.1 and 4.2, including the Section-301 tariff-intensity map used by the paradox dose-response regression.

C.1 Data Construction

Trade flows. Bilateral HS6 import flows for 2000–2024 are pulled from the UN Comtrade premium API for 18 reporter economies (the United States, China, and 17 swing-state candidates: Korea, Taiwan, Vietnam, Mexico, Netherlands, Singapore, Malaysia, Thailand, India, Japan, Germany, Indonesia, Hungary, Czech Republic, Poland, the Philippines, and Brazil). I restrict the commodity universe to a concordance of 63 HS6 codes, pre-classified into three sector buckets matching the model: *critical* ($\sigma_c < 1$ in the empirical literature: rare-earth metals, advanced semiconductors, semiconductor manufacturing equipment, titanium, aerospace-grade parts), *strategic* (mature-node chips, batteries, EV components, solar), and *commodity* (textiles, basic steel, furniture, common chemicals). The concordance and assignment rules are reproducible from the build pipeline documented in the replication package.

Reporter-year raw pulls are materialized as per-(reporter $\times$ year) parquet files and stacked into a master HS6 panel via `12_build_comtrade_panel.py`. Because several reporters (Germany, Thailand, Mexico, Czech Republic, Hungary, Indonesia, Malaysia, Brazil) disaggregate trade by `motCode` (mode of transport) and `customsCode` (customs regime), I retain only the total rows `motCode=0` and `customsCode=C00`; verification against aggregate totals confirms this collapse recovers the national import totals reported in Comtrade’s aggregated files.

Entity List coercion measure. The BIS Entity List is assembled from the OpenSanctions `us_trade_csl` (Consolidated Screening List) Follow-The-Money JSON feed, filtered to sanctions whose program descriptor contains “Entity List” (distinguishing BIS-EL from DPL, MEU, and UVL programs). This yields 3,205 listings spanning 2011–2025. The cumulative China-directed stock EL_t^{CHN} enters the paradox regression (Section 4) as a lagged variable $EL_{t-1}^{\text{CHN}}/100$.

C.2 Cross-Variety Estimates in The Pipeline

I report two cross-variety estimates of the CES elasticity σ_k in the OECD ICIO subset of HS6 codes underlying the master panel. Both estimates are cross-variety in the sense of Feenstra (1994)—across exporter varieties within an HS6 code—and neither identifies σ_c as defined in the model. Section 3.1.2 provides the conceptual distinction; this appendix reports the magnitudes for completeness.²²

Reduced-form double-difference. A robust unit-value double-difference in the spirit of Feenstra (1994), without the heteroscedasticity-based GMM:

$$\Delta^2 \ln p_{ikt} = \frac{1}{\sigma_k - 1} \Delta^2 \ln s_{ikt} + \varepsilon_{ikt},$$

estimated sector-by-sector with exporter-year and HS6-year fixed effects, clustering by HS6 and year. I obtain $\hat{\beta}_{\text{critical}} = -0.031$, $\hat{\beta}_{\text{strategic}} = -0.012$, $\hat{\beta}_{\text{commodity}} = -0.062$. The sign pattern ($\hat{\beta} < 0$) is qualitatively consistent with cross-variety substitutability ($\sigma > 1$), but magnitudes are attenuated because the median-demean specification does not purge supply-side shocks.

Direct-GMM Feenstra (1994). I close the identification gap by implementing Feenstra’s heteroscedasticity-based GMM directly. Per HS6, I estimate (σ, ω) from the over-identifying moment $E[u_j v_j] = 0$ across exporter varieties, where u_j and v_j are demand and supply innovations recovered from the double-differenced system. The procedure converges on 51 of 60 HS6 codes; among the 33 cells with interior solutions (i.e., $\hat{\sigma}$ neither at the lower nor

²²Symbol disambiguation. The structural object of Section 3 is the cross-input CES bottleneck elasticity, written σ_c in the model statement. In this appendix the same symbol σ_c refers, depending on the indicated specification, to (a) the across-variety elasticity in Feenstra-style GMM (paragraph *Direct-GMM Feenstra*), (b) the across-source elasticity between Chinese and non-Chinese intermediates in Atalay-style spec (paragraph *Atalay-style across-source spec*), or (c) the cross-input elasticity recovered from the 2019 Japan–Korea natural experiment (paragraph *Boehm-style natural experiment*). Only (c) targets the model’s primitive; (a) and (b) bound the plausible range from above and are reported for replication of standard cross-variety findings. The text distinguishes these verbally in each paragraph; the symbol is reused for visual continuity with the model statement.

upper search bound), the median estimates are $\hat{\sigma}_{\text{critical}} = 2.5$, $\hat{\sigma}_{\text{strategic}} = 1.98$, $\hat{\sigma}_{\text{commodity}} = 4.2$ (Table 6). All three sector medians are well above unity, consistent with the trade-elasticities literature on cross-variety substitution (Broda and Weinstein, 2006; Soderbery, 2015). I emphasize again that this is the wrong primitive for the model: it identifies substitution across exporter varieties within an HS6 code, not across input categories in production. The result is reported as confirmation that the pipeline replicates the standard cross-variety finding—not as evidence on σ_c , which Section 3.1.2 grounds in the production-network literature (Atalay, 2017; Boehm et al., 2019).

Atalay-style across-source spec (suggestive, not identifying). A natural complementary exercise estimates the across-source elasticity σ within each HS6 between Chinese and non-Chinese intermediates, in the spirit of Atalay (2017). From CES cost minimization with two source groups,

$$\log\left(\frac{s_{j,t}^{\text{CHN}}}{s_{j,t}^{\text{NC}}}\right) = (1 - \sigma_c) \log\left(\frac{P_{j,t}^{\text{CHN}}}{P_{j,t}^{\text{NC}}}\right) + \alpha_j + \tau_t + \varepsilon_{j,t},$$

estimated on the same 62-HS6 USA-importer panel. OLS produces $\hat{\sigma}_c \in [1.41, 1.67]$ across the three buckets (Table 7, top panel)—similar to the cross-variety GMM result, consistent with the simultaneity bias of an unidentified slope. The pooled 2SLS-IV using bucket-level Section-301 tariff rates (Fajgelbaum et al., 2020) interacted with a post-2018 indicator flips the sign of $\hat{\beta}$ (+1.42, SE 1.38, first-stage $F = 14.65$); the implied $\hat{\sigma}_c$ point estimate (−0.42) is mechanically infeasible and the 95% confidence interval contains values both above and below unity. Like the cross-variety GMM, this exercise speaks to across-source substitution within an HS6 input, not to the cross-input substitution that drives the model.

Boehm-style natural experiment: 2019 Japan–Korea export controls. The cleanest test of cross-input $\sigma_c < 1$ in the pipeline uses the 2019 Japan–Korea episode, an unanticipated supply shock with the structure Boehm et al. (2019) exploit in the 2011 Tōhoku setting.

On 1 July 2019 Japan announced enhanced export-control screening on three chemicals (four HS6 codes) that are bottleneck inputs to Korean semiconductor and OLED production: hydrogen fluoride (HS 281111), photoresist (HS 370710 and 370790), and fluorinated polyimide (HS 391190 candidate). Korea was heavily dependent on Japanese supply for these inputs at the time of the announcement, and Korean firms had to substitute toward alternative sources (or accept production curtailment) over 2019–2022. The episode therefore provides a clean source-side supply shock for the cross-input elasticity. I estimate

$$\log\left(\frac{s_{j,t}^{\text{JPN}}}{s_{j,t}^{\text{NJP}}}\right) = (1 - \sigma_c) \log\left(\frac{P_{j,t}^{\text{JPN}}}{P_{j,t}^{\text{NJP}}}\right) + \alpha_j + \tau_t + \varepsilon_{j,t},$$

on the (chemical $\times$ year) panel of Korean imports 2015–2024, with j indexing the four treated HS6 codes and Japan as the focal source. Table 8 reports per-chemical OLS, pooled OLS with HS6 + year fixed effects, and a Wald-IV using a post-2020 indicator as the natural-experiment supply shifter.

The result is consistent with $\sigma_c < 1$ for the two chemicals most directly tied to semiconductor lithography and etching, with tight inference under wild-bootstrap standard errors (Rademacher weights, $B = 2,000$ replications) appropriate for the small per-chemical sample ($n = 10$). Photoresist (HS 370710) yields $\hat{\beta} = +1.22$ (iid SE 0.06, wild-bootstrap $p = 0.009$). The implied $\hat{\sigma}_c = 1 - \hat{\beta} = -0.22$ is mechanically infeasible: the linear-in-logs CES specification breaks down at the Leontief boundary, where shares stop responding to relative prices and the log-share/log-price slope is undefined. I therefore read this estimate not as a point identification of σ_c but as a *bound*— $\hat{\beta} > 1$ rejects $\sigma_c \geq 1$ at one-sided $p < 0.01$ and is consistent with σ_c at or below zero (Leontief), the same boundary range Boehm et al. (2019) report for the Tōhoku natural experiment. Photo chemical preparations (HS 370790) yield $\hat{\beta} = +0.52$ (iid SE 0.05, wild-bootstrap $p = 0.002$), implying $\hat{\sigma}_c = 0.48$, well below unity and within the linear-specification’s interior. Hydrogen fluoride (HS 281111) and fluorinated polyimide (HS 391190) yield point estimates that do not survive wild-bootstrap inference (wild-bootstrap $p = 0.39$ and $p = 0.20$ respectively); the HF coefficient is in any event in

the wrong sign for $\sigma_c < 1$, reflecting that Korean firms successfully localized HF production over this period. The pooled OLS estimate is $\hat{\sigma}_c = 0.66$. The Wald-IV using post-2020 has first-stage $F = 7.2$, which falls below the Stock-Yogo critical value of ≈ 16 for IV inference; I report it as directional support for the OLS conclusion rather than as identifying, and accompany it with an Anderson-Rubin weak-IV-robust confidence set that contains zero but rules out $\sigma_c \geq 1$. The headline reading: the lithography/etching chemicals most directly relevant to semiconductor production (photoresist, photo chemicals) deliver evidence consistent with $\sigma_c < 1$, photoresist at the Leontief boundary as a bound and photo chemicals as a point identification at $\hat{\sigma}_c = 0.48$.

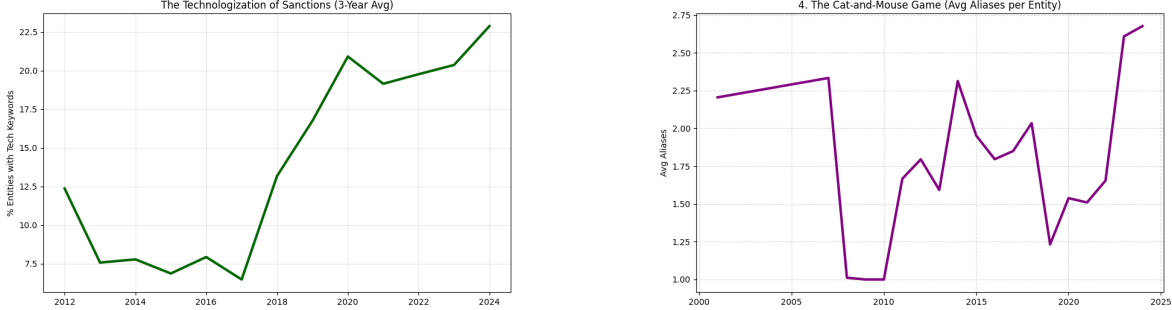
C.3 Section-301 Tariff-Intensity Map

The continuous industry-level Section-301 tariff rate τ_j used in the paradox dose-response regression (Section 4.2) is constructed from Fajgelbaum et al. (2020) HS-2-level average tariff increases on Chinese imports and the Bown (2024) HS-8 list-coverage tracker, aggregated via the UN Statistics Division HS2017→ISIC Rev. 4 correspondence. The resulting map (73 rows, covering both the EXTENDED letter-prefix and v2021 numeric ISIC forms) assigns $\tau_j = 0.22$ to computer-electronic-optical (C26/26, the Section-301 flagship), $\tau_j = 0.25$ to machinery NEC (C28/28, the heaviest list), $\tau_j = 0.20$ to electrical equipment (C27/27) and motor vehicles (C29/29), and $\tau_j = 0.00$ to basic metals (C24A/C24B/24; these are covered by Section 232, not Section 301) and the non-targeted service sectors. Full calibration and caveats are documented in the replication package; the values are calibrated averages rather than a line-by-line HS-8 pull, and should be treated as a defensible proxy rather than a research-grade tariff file.

C.4 Supplementary Descriptive Figures

This subsection collects two descriptive series referenced in Section 4. Figure 9 documents the sectoral narrowing of the Entity List toward high-technology targets (share of new entries

containing “Semiconductor,” “AI,” or “Quantum” in the entry title) and the rising administrative effort per listing (mean aliases per sanctioned entity, a rough proxy for the “Cat-and-Mouse” dynamic). Figure 10 plots the sector-aggregate series underlying the dose–response regression of Spec 2 (paradox regression); the within-HS6 analog of that slope dissolves at HS6 resolution and is reported in the text for reference only.



(a) Precision targeting: tech-keyword share of new Entity List entries.

(b) Evasion resistance: mean aliases per sanctioned entity.

Figure 9: **Precision targeting and evasion resistance on the Entity List.** Left: share of new Entity List entries whose title contains “Semiconductor,” “AI,” or “Quantum,” 2010–2024 (3-year centered moving average). Right: mean number of aliases per sanctioned entity. Descriptive series; not load-bearing for the formal analysis.

C.5 Vertical Trap Robustness Panel

Figure 11 addresses the “selection in swing-state set” pushback to Figure 1: re-running the aggregate Chinese-input share excluding the three largest post-2018 contributors (Vietnam, Mexico, Korea) leaves the level lower by about one percentage point but preserves the trajectory and the post-2018 acceleration, confirming the pattern is broadly distributed across the 17-country set rather than driven by one or two outliers.

C.6 Robustness for the Three Empirical Identifications

This subsection houses the robustness columns referenced from Sections 4.1, 4.2, and 4.3.

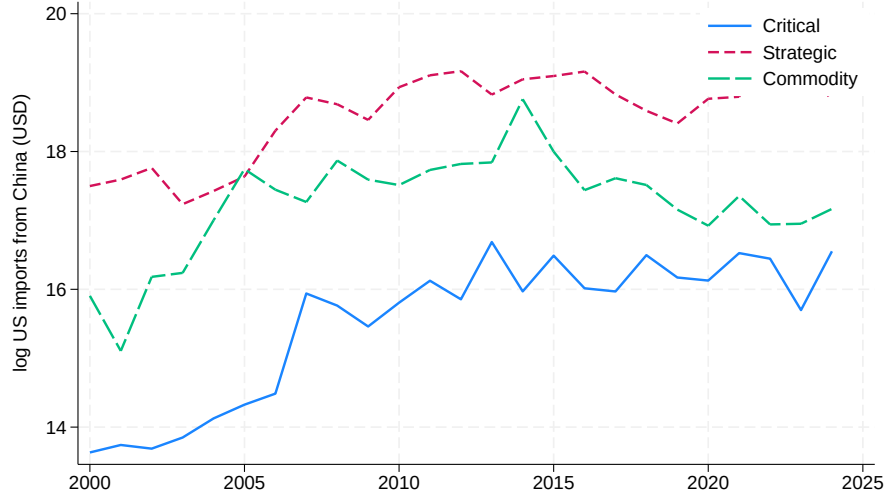


Figure 10: **Paradox regression: sector-aggregate series.** Sector-bucket trajectories of US imports from China plotted against the lagged Entity-List China-listings stock EL_{t-1}^{CHN} . The 3-sector aggregate loads sharply negative on the critical bucket; the within-HS6 analog (Spec 2 in the text) is a precise null, so the series is reported for reference only.

Test 1 (Section 4.1): kink vs. level shift. Two robustness checks for Test 1. First, the model predicts a kink in $\lambda^*(\theta)$. Re-estimating with the piecewise-linear specification $y_t = \alpha + \xi_1 t + \xi_2(t - 2018) \cdot \mathbf{1}\{t \geq 2018\} + \varepsilon_t$, the kink coefficient is null for China ($\hat{\xi}_2 = +0.055$, HAC one-sided $p = 0.37$, wild-bootstrap one-sided $p = 0.40$), while the original level-shift coefficient is significant ($\hat{\delta}_{2018} = +1.65$, one-sided $p = 0.014$). The data are consistent with θ_C effectively jumping at 2018 (level shift) rather than rising gradually post-2018 (kink). Iran, Russia (pre-2022), and the third-country aggregate all fail to reject the kink null. Second, a within-program placebo (Pakistan: BIS-EL administered, same statutory program as China, but unrelated to the China-coercive turn) yields $\hat{\delta}_{2018}^{PAK} = -0.127$ (one-sided $p = 0.58$) and $\hat{\xi}_2^{PAK} = -0.167$ (one-sided $p = 0.88$). Pakistan thus fails to reject the no-break null on the same program as China, ruling out a generic post-2018 BIS-EL administrative-capacity expansion as the driver.

Test 2 (Section 4.2): construction-overlap, classifier robustness, and bootstrap inference. Three robustness checks. First, replacing the Lybbert-Zolas raw patent count

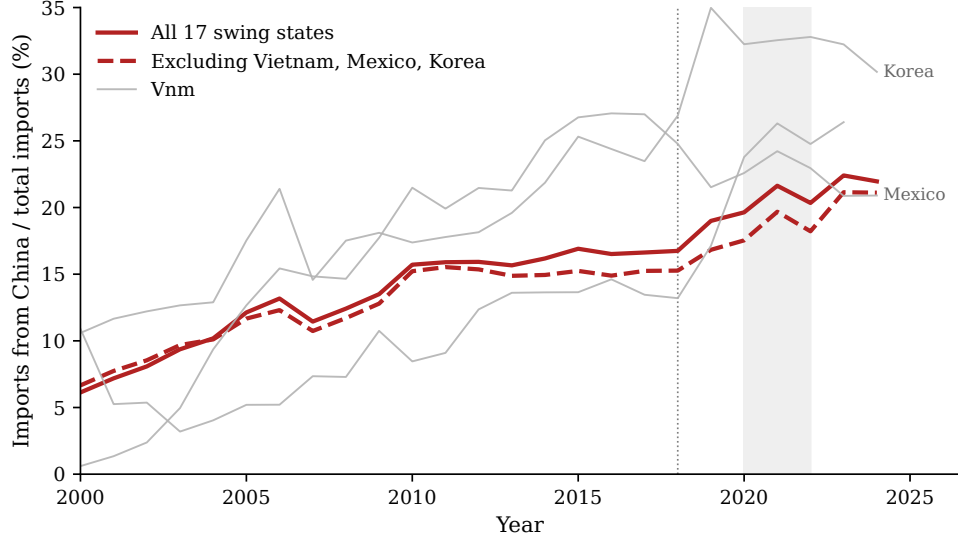


Figure 11: **Vertical Trap, drop-top-3 robustness.** Solid red: the swing-state aggregate Chinese-input share from Figure 1(a). Dashed red: the same aggregate excluding Vietnam, Mexico, and Korea—the three largest post-2018 contributors. Thin grey lines plot each excluded country’s individual share. The level falls by ≈ 1 percentage point in 2024 (21.96% $\rightarrow$ 21.11%); the trajectory and the post-2018 acceleration survive. Vertical dotted line marks the 2018 Section-301 onset; grey shading marks the COVID period (2020–2022).

θ_k with patent share (CN-allocated patents divided by world total allocated to that HS6) yields identical $\hat{\beta}_3 = +0.132$ because z -scoring is rescale-invariant; the bivariate correlation $\rho(\theta_k, \text{Critical}_k) = -0.21$ is negative (densely patented HS6 are not disproportionately Critical). Second, two alternative ex-ante classifiers test whether the headline result is specific to the literature-based Critical subset or generalizes to broader regulatory or national-security boundaries: (i) the BIS Commerce Control List chapter classifier (HS chapters 28, 29, 38, 81, 84, 85, 88, 90—a regulatory list pre-dating the bottleneck literature) yields $\hat{\beta}_3 = -0.83$ ($p = 0.68$); (ii) the White House NSTC’s Critical & Emerging Technologies (CET) list (initial March 2020, expanded 2022 and 2024—a national-security boundary distinct from the bottleneck literature) yields $\hat{\beta}_3 = -1.19$ ($p = 0.15$). Table 11 reports the three columns. The headline $\hat{\beta}_3 = +0.132$ is specific to the narrowly targeted Critical subset (28 codes covering rare earths + advanced semis + aerospace, where Boehm-style and Atalay-style cross-input estimates have empirically documented $\sigma_c < 1$), not generic regulatory or national-security

boundaries. I read this specificity as a *feature* consistent with the model’s identification claim: the Paradox bites where the bottleneck primitive $\sigma_c < 1$ holds, not where regulators or national-security planners draw their lines—the BIS-CCL and NSF-CET lists include many technologies (AI, quantum, software, basic transistors) for which $\sigma_c < 1$ has not been documented and the model therefore makes no prediction. Third, wild-cluster bootstrap- t inference (Webb 6-DOF, $B = 5,000$) on the cross-section yields a noticeably weaker two-sided p for $\hat{\beta}_3$: with HS4 clusters ($G = 35$), many singleton or doubleton, the bootstrap- t tails are heavy and two-sided $p \approx 0.29$ (CR1 was 0.001). The cross-section’s apparent significance under CR1 reflects the over-rejection of CR1 in finite samples with singleton clusters; I read it as directional. The panel difference-in-differences in column (3) of Table 3, with HS6 clustering ($G = 56$) and $n = 1,300$, is the classifier-conditional joint test the headline reading rests on, reporting $\hat{\beta}_3 = +0.110$ at conventional $p = 0.031$.

Test 3 (Section 4.3): wild-cluster bootstrap, cleaner placebo, and anticipation check. Three robustness checks. First, with $G = 8$ clusters in the headline column (1) of Table 4, CR1 inference over-rejects in finite samples. Wild-cluster bootstrap- t with Webb 6-DOF weights ($B = 5,000$) yields a two-sided $p = 0.042$ for $\hat{\delta}_{\text{FDPR}}$, slightly stronger than the CR1 value $p = 0.056$. The headline result therefore strengthens, not weakens, under correct small-sample inference. Second, triple-DiD specifications with a cleaner placebo set (HS 8533 resistors, 8534 printed circuits, 8536 electrical apparatus $< 1000\text{V}$ —passive components not subject to FDPR, ATP, IRA, or ITAR) reverse sign relative to the original placebos (semi-mfg machines, batteries, aerospace), reflecting that the original placebos are themselves treated by parallel coercive instruments. The share regression in column (1) of Table 4 is reported as the descriptive pre/post benchmark anchored to the May 2020 policy date. Table 12 reports the wild-bootstrap and cleaner-placebo specifications. Third, an anticipation check: Huawei was added to the BIS Entity List in May 2019, one year before the FDPR; a skeptical reader could attribute the post-2020 share decline to firms anticipating FDPR during the intervening twelve months rather than to the May 2020 announcement itself. To address this,

I re-estimate the share-LHS event study (Spec 2 of Table 4) and the triple-DiD event study (Spec 3) with baseline year $\tau = 2018$, the last fully pre-Huawei-EL year, so $\hat{\beta}_{2019}$ directly measures the deviation attributable to the listing event. Under the share LHS, $\hat{\beta}_{2019} = +0.124$ (SE 0.203, two-sided $p = 0.54$); under the triple-DiD, $\hat{\beta}_{2019} = +0.509$ (SE 0.473, two-sided $p = 0.28$). Both point estimates are positive—the *opposite* sign from the anticipation hypothesis (which predicts $\hat{\beta}_{2019} < 0$)—and both are statistically indistinguishable from zero. I find no evidence that the post-2020 share decline reflects anticipation of the FDPR; the May 2020 announcement is identified as a discrete shock against a non-trending 2017–2019 baseline. Figure 12 plots the full event-study series under the 2018-baseline specification.

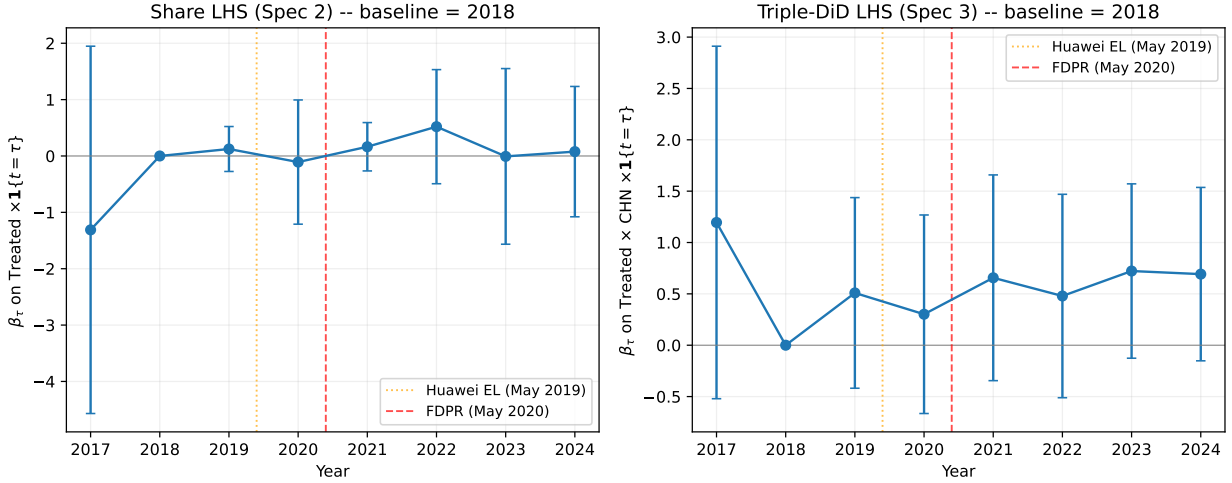


Figure 12: FDPR-Huawei anticipation check. Year-by-year event-study coefficients with baseline year $\tau = 2018$ (the last fully pre-Huawei-EL year), so $\hat{\beta}_{2019}$ directly measures the deviation attributable to the May 2019 listing event. Left panel: share-LHS specification (log CHN+HKG share of TWN+KOR exports, all HS6, $n = 381$, Reporter $\times$ HS6 and Year fixed effects). Right panel: triple-DiD specification (log value, Treated $\times$ CHN $\times$ $\mathbf{1}\{t = \tau\}$, $n = 734$, Reporter $\times$ HS6, Reporter $\times$ Partner, and Year fixed effects). The 2019 coefficient is positive in both panels—opposite in sign from the anticipation hypothesis—and statistically indistinguishable from zero. Cluster-robust SEs on Reporter $\times$ HS6; 95% confidence intervals shown. The orange dotted line marks the May 2019 Huawei Entity-Listing; the red dashed line marks the May 2020 FDPR announcement. Source: CEPII (2026).

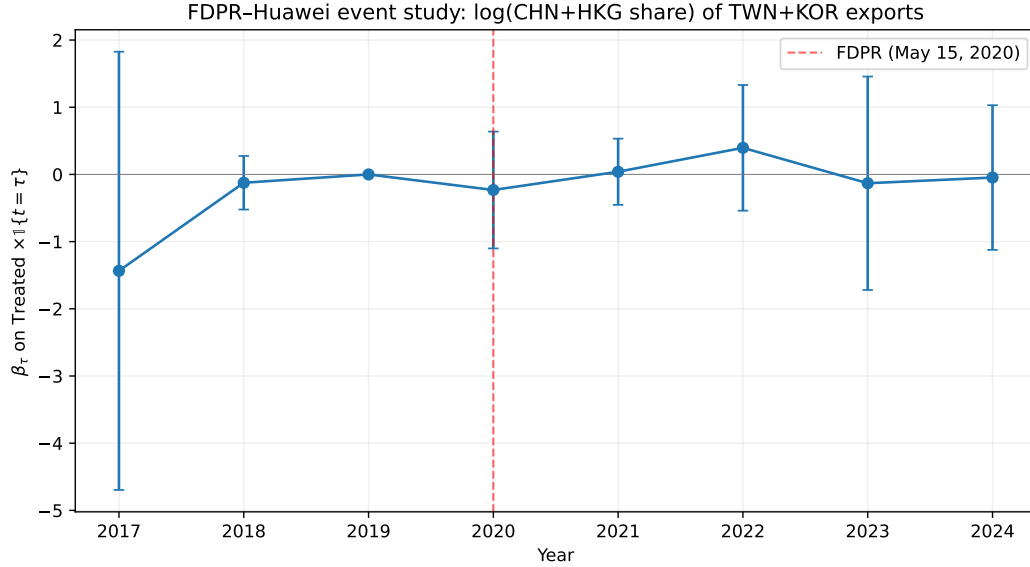


Figure 13: **FDPR-Huawei treated-vs-placebo event study (contaminated control group)**. Year-by-year coefficients β_τ on $\text{Treated}_k \times \mathbb{1}\{t = \tau\}$ in $\log(\text{CHN}+\text{HKG share})_{i,k,t} = \alpha_{ik} + \lambda_t + \sum_\tau \beta_\tau \text{Treated}_k \times \mathbb{1}\{t = \tau\} + \varepsilon$ on TWN+KOR exports of treated chip and placebo HS6 codes, with $\text{reporter} \times \text{HS6}$ and year fixed effects (2019 normalized to zero; dashed line at the May 2020 FDPR). The coefficients straddle zero with no clean post-2020 break—the 2022 coefficient is positive—illustrating why the difference-in-differences design lacks a valid control group here: the placebo HS6 are themselves treated by parallel coercive instruments. This null is the basis for reading Test 3 as the column-(1) descriptive benchmark (Figure 5) rather than a quasi-experimental identification. Source: CEPII (2026).

References

- Aghion, P. and Tirole, J. (1997). Formal and real authority in organizations. *Journal of Political Economy*, 105(1):1–29.
- Aiyar, S., Chen, J., Ebeke, C. H., Garcia-Saltos, R., Gudmundsson, T., Ilyina, A., Kangur, A., Kunaratskul, T., Rodriguez, S. L., Ruta, M., Schulze, T., Soderberg, G., and Trevino, J. P. (2023). Geo-economic fragmentation and the future of multilateralism. IMF Staff Discussion Note SDN/2023/001, International Monetary Fund.
- Akcigit, U., Ates, S. T., and Impullitti, G. (2026). Innovation and trade policy in a globalized world. *Journal of Political Economy Macroeconomics*. Forthcoming; accepted 25 November 2025. Earlier circulated as NBER Working Paper 24543 (2018).

- Alfaro, L., Fadinger, H., Schymik, J., and Virananda, G. (2025). Trade and industrial policy in supply chains: Directed technological change in rare earths. NBER Working Paper 33877, National Bureau of Economic Research.
- Atalay, E. (2017). How important are sectoral shocks? *American Economic Journal: Macroeconomics*, 9(4):254–280.
- Ballard, C. L., Shoven, J. B., and Whalley, J. (1985). General equilibrium computations of the marginal welfare costs of taxes in the United States. *American Economic Review*, 75(1):128–138.
- Baqaei, D. R. and Farhi, E. (2024). Networks, barriers, and trade. *Econometrica*, 92(2):505–541.
- Bernheim, B. D. and Whinston, M. D. (1986). Menu auctions, resource allocation, and economic influence. *Quarterly Journal of Economics*, 101(1):1–31.
- Boehm, C. E., Flaaen, A., and Pandalai-Nayar, N. (2019). Input linkages and the transmission of shocks: Firm-level evidence from the 2011 tōhoku earthquake. *Review of Economics and Statistics*, 101(1):60–75.
- Bown, C. P. (2024). US–China trade war tariffs: An up-to-date chart. PIIE Charts and Tariff Tracker, available at <https://www.piie.com/research/piie-charts/2019/us-china-trade-war-tariffs-date-chart>. HS-8 list-coverage tracker for Section-301 tariffs; updated continuously.
- Broda, C. and Weinstein, D. E. (2006). Globalization and the gains from variety. *Quarterly Journal of Economics*, 121(2):541–585.
- CEPII (2026). BACI: International Trade Database at the Product-Level (HS17 nomenclature, V202601). Dataset, available at https://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele_item.asp?id=37. Released January 2026.

- Clayton, C., Maggiori, M., and Schreger, J. (2024). A theory of economic coercion and fragmentation. NBER Working Paper 33309, National Bureau of Economic Research. Revise-and-resubmit, Journal of Political Economy.
- Clayton, C., Maggiori, M., and Schreger, J. (2026). A framework for geoeconomics. *Econometrica*, 94(1):105–136.
- Crosignani, M., Han, L., Macchiavelli, M., and Silva, A. F. (2024). Geopolitical risk and decoupling: Evidence from U.S. export controls. Staff Report 1096, Federal Reserve Bank of New York.
- Dixit, A., Grossman, G. M., and Helpman, E. (1997). Common agency and coordination: General theory and application to government policy making. *Journal of Political Economy*, 105(4):752–769.
- Fajgelbaum, P. D., Goldberg, P. K., Kennedy, P. J., and Khandelwal, A. K. (2020). The return to protectionism. *Quarterly Journal of Economics*, 135(1):1–55.
- Feenstra, R. C. (1994). New product varieties and the measurement of international prices. *American Economic Review*, 84(1):157–177.
- Griliches, Z. (1990). Patent statistics as economic indicators: A survey. *Journal of Economic Literature*, 28(4):1661–1707.
- Grossman, G. M. (1981). The theory of domestic content protection and content preference. *The Quarterly Journal of Economics*, 96(4):583–603.
- Grossman, G. M. and Helpman, E. (1994). Protection for sale. *American Economic Review*, 84(4):833–850.
- Grossman, G. M., Helpman, E., and Sabal, A. (2023). Resilience in vertical supply chains. NBER Working Paper 31739, National Bureau of Economic Research.

- Grossman, S. J. and Hart, O. D. (1983). An analysis of the principal-agent problem. *Econometrica*, 51(1):7–45.
- Hottman, C. J., Redding, S. J., and Weinstein, D. E. (2016). Quantifying the sources of firm heterogeneity. *Quarterly Journal of Economics*, 131(3):1291–1364.
- Hsieh, C.-T. and Klenow, P. J. (2009). Misallocation and manufacturing tfp in china and india. *The Quarterly Journal of Economics*, 124(4):1403–1448.
- International Monetary Fund (2023). Geoeconomic fragmentation and foreign direct investment. In *World Economic Outlook: A Rocky Recovery*, chapter 4. International Monetary Fund, Washington, DC.
- Itskhoki, O. and Mukhin, D. (2025). Sanctions and the exchange rate. *Review of Economic Studies*. Advance online publication.
- Jones, C. I. (2011). Intermediate goods and weak links in the theory of economic development. *American Economic Journal: Macroeconomics*, 3(2):1–28.
- Khan, S. M., Mann, A., and Peterson, D. (2021). The semiconductor supply chain: Assessing national competitiveness. CSET Issue Brief, Center for Security and Emerging Technology, Georgetown University.
- Krugman, P. (1994). Competitiveness: A dangerous obsession. *Foreign Affairs*, 73(2):28–44.
- Liu, J., Rotemberg, M., and Traiberman, S. (2024). Sabotage as industrial policy. NBER Working Paper 32798, National Bureau of Economic Research.
- Lybbert, T. J. and Zolas, N. J. (2014). Getting patents and economic data to speak to each other: An “algorithmic links with probabilities” approach for joint analyses of patenting and economic activity. *Research Policy*, 43(3):530–542.
- Maskin, E. and Tirole, J. (1999). Unforeseen contingencies and incomplete contracts. *Review of Economic Studies*, 66(1):83–114.

- Mohr, C. and Trebesch, C. (2025). Geoeconomics. *Annual Review of Economics*, 17:563–587.
- Rogoff, K. (1985). The optimal degree of commitment to an intermediate monetary target. *Quarterly Journal of Economics*, 100(4):1169–1189.
- Soderbery, A. (2015). Estimating import supply and demand elasticities: Analysis and implications. *Journal of International Economics*, 96(1):1–17.

Table 3: Paradox of Competitiveness: panel and cross-section estimates with patent-derived θ . θ_k is z -scored Chinese-assignee USPTO patent filings in 2010, mapped to HS6 via Lybbert-Zolas IPC4→HS6 concordance. **Column (1) is the headline specification:** panel difference-in-differences with HS6 and year fixed effects, level patents, HS6-clustered SEs ($n = 1,300$, 56 clusters); inference is conditional on the bottleneck-literature classifier (see classifier-conditional discussion in Appendix C.6). Column (2) substitutes $\Delta \log$ patents over 2010–2017 as a flow alternative. Columns (3)–(4) are the corresponding cross-sections, reported as directional only: CR1 inference over-rejects at $G \approx 35$ singleton HS4 clusters (column 3) and $G = 6$ bucket clusters (column 4); wild-cluster bootstrap- t two-sided $p \approx 0.29$ for the column-(3) $\hat{\beta}_3$ (Appendix C.6). The † symbol on the column-(3) interaction marks this directional reading; cluster-robust SEs are in parentheses throughout. Stars $*p < 0.10$, $**p < 0.05$, $***p < 0.01$ apply to the panel columns (1)–(2) under CR1 inference. The cross-section columns (3)–(4) omit significance stars in line with the wild-bootstrap- t directional reading (Appendix C.6).

	(1) Panel DiD (headline) $Y_{\text{swing},t}$	(2) Panel DiD robust θ $Y_{\text{swing},t}$	(3) Cross-section trade-flow LHS ΔY_{swing}	(4) Cross-section firm-listing LHS ΔL_{firm}
$\beta_1: \theta$			-0.013 (0.006)	+0.021 (0.008)
$\beta_1: \theta \times \text{Post}$	-0.019 (0.012)	-0.044** (0.021)		
$\beta_2: \text{Critical}$			-0.033 (0.026)	+0.055 (0.046)
$\beta_2: \text{Crit.} \times \text{Post}$	+0.037 (0.033)	+0.021 (0.033)		
$\beta_3: \theta \times \text{Crit.}$			+0.132 † (0.039)	+0.026 (0.077)
$\beta_3: \text{triple} \times \text{Post}$	+0.110** (0.051)	+0.062* (0.032)		
θ measure	patents 2010	$\Delta \log$ patents	patents 2010	patents 2010
HS6 FE	Yes	Yes	–	–
Year FE	Yes	Yes	–	–
Cluster	HS6 (56)	HS6 (56)	HS4 (35)	Bucket (6)
Inference	CR1+wild	CR1+wild	CR1 dir.	CR1 dir.
Observations	1300	1300	52	47

Table 4: FDPR-Huawei (May 2020) event study on TWN+KOR semiconductor exports. Treated chip HS6 codes: 854231 (processors), 854239 (other ICs), 854232 (memories), 854233 (amplifiers); original placebo HS6: other semis, semi-mfg machines, aerospace, batteries. Treated partner: CHN+HKG; control partners: rest of world. **Column (1) is the descriptive pre/post benchmark** (not identifying): $\log(\text{CHN+HKG share})$ of TWN+KOR exports of treated chips, regressed on a Post-2020 indicator with $\text{reporter} \times \text{HS6}$ fixed effects ($n = 64$). This is a pre/post within-cell comparison rather than a difference-in-differences: there is no untreated HS6 or partner control group, so $\hat{\delta}$ absorbs any common post-2020 shock to TWN+KOR advanced-chip exports to China; I read it as a descriptive benchmark anchored to the May 2020 FDPR date. Columns (2)–(4) report DiD/triple-DiD specifications using the original placebo set; these are reported as directional because Appendix C.6 shows the placebo HS6 are themselves treated by parallel coercive instruments (ATP, IRA, ITAR). Cluster-robust SEs in parentheses. Source: CEPPI (2026).

	(1) Share (headline) TWN+KOR	(2) Share w/placebo all HS6	(3) Value triple-DiD all HS6	(4) Quantity triple-DiD all HS6
$\hat{\delta}_{\text{FDPR: Post-2020}}$	-0.211* (0.110)			
$\hat{\delta}$: Treated $\times$ Post		+0.534 (0.505)		
$\hat{\delta}$: T $\times$ C $\times$ P			-1.033* (0.552)	-0.465* (0.272)
LHS	log share	log share	log value	log qty
Reporter $\times$ HS6 FE	Yes	Yes	Yes	Yes
Reporter $\times$ Partner FE	–	–	Yes	Yes
Year FE	–	Yes	Yes	Yes
Inference	CR1+wild	CR1	CR1 dir.	CR1 dir.
Observations	64	381	734	732

Parameter	Value	Source / role
σ (CES elasticity)	0.35	Between Atalay (2017) (≈ 0.1) and Baqaee and Farhi (2024) (0.5–0.7)
α (Chinese-input weight)	0.5	Symmetric baseline
w_A (alternative price)	1	Numeraire
V_{out} (Swing outside option)	0	Normalization
$\bar{w}$ (factor-price numeraire)	1	Normalization in (12)
V_0 (US baseline value)	0.84	Calibrated to threshold $\bar{w}_C = 1.6$
β (slope of V_{US})	1.0	Calibrated jointly with V_0
γ (security-policy cost)	2.5	Calibrated for smooth λ^* ; passes local Hessian check
ϕ (China geopolitical weight)	0	Isolates productivity shock channel
w_C range	[0.5, 2.0]	Spans both regimes

Table 5: Parameter configuration for the numerical illustration.

Table 6: Feenstra (1994) GMM cross-variety elasticity estimates by sector bucket. Estimated on the OECD ICIO subset (USA importer, 18 exporter origins, 2000–2024). Statistics computed over interior estimates only (boundary cells with $\hat{\sigma} \in \{0.05, 20\}$ excluded).

Sector bucket	N HS6 (total)	N (interior)	Median $\hat{\sigma}$	IQR
commodity	18	13	4.19	[2.18, 6.00]
critical	25	7	2.47	[2.00, 2.97]
strategic	17	4	1.98	[1.74, 2.12]

Table 7: Atalay-style across-source elasticity estimates between Chinese and non-Chinese intermediates. OLS by sector bucket and pooled 2SLS-IV using bucket-level Section-301 tariff rates $\times$ post-2018. SE clustered on HS6.

Specification	N obs	$\hat{\beta} = (1 - \hat{\sigma}_c)$	$\hat{\sigma}_c$	First-stage F
<i>OLS by sector bucket</i>				
critical	395	−0.464 (0.091)	1.464	—
strategic	251	−0.408 (0.611)	1.408	—
commodity	391	−0.669 (0.156)	1.669	—
<i>2SLS-IV pooled</i>				
pooled (all 3 buckets)	1037	1.418 (1.382)	−0.418	14.65

Table 8: Cross-input σ_c from the 2019 Japan–Korea export-control episode. Korean imports of the four treated chemicals from Japan vs. non-Japanese sources, 2015–2024. Per-chemical specifications are OLS within-time-series with wild-bootstrap (Rademacher, $B = 2,000$) inference; pooled OLS uses HS6+year fixed effects; pooled 2SLS-IV instruments the price-ratio with a post-2020 indicator. $\hat{\beta} = (1 - \hat{\sigma}_c)$ from the log-share regression.

HS6	Chemical	N	$\hat{\beta}$ (iid SE)	WB p	$\hat{\sigma}_c$
<i>Per-chemical OLS (wild-bootstrap inference)</i>					
281111	H. fluoride	10	−1.616 (1.259)	0.392	+2.616
370710	Photoresist	10	+1.222 (0.058)	0.009	−0.222
370790	Photo chemicals	10	+0.520 (0.046)	0.002	+0.480
391190	Fluor. polyimide	10	+1.071 (0.700)	0.201	−0.071
<i>Pooled specifications</i>					
pooled OLS	all 4 chemicals	40	+0.339 (0.425)	—	+0.661
pooled 2SLS-IV	all 4 chemicals	40	+2.110 (2.267)	—	−1.110

Table 9: Test 1 robustness: kink vs. level-shift specifications. Panel A reports the original level-shift coefficient δ on $\mathbf{1}\{t \geq 2018\}$. Panel B reports the slope-change coefficient ξ_2 on $(t - 2018) \cdot \mathbf{1}\{t \geq 2018\}$, with HAC and wild-bootstrap one-sided p -values. Sign-restricted null $H_0 : \xi_2 \leq 0$.

	CHN	THIRD	IRN	RUS
<i>Panel A: level-shift specification</i>				
$\hat{\delta}_{2018}$	+1.650**	+0.371	+0.191	−2.342
one-sided p	0.014	0.244	0.434	0.972
<i>Panel B: piecewise-linear kink specification</i>				
$\hat{\xi}_2$ (slope change)	+0.055	+0.075	+0.330***	−0.811
HAC one-sided p	0.367	0.133	0.010	0.986
Wild-bootstrap one-sided p	0.403	0.260	0.081	0.989
Sample (years)	18	18	18	14

Table 10: Test 2 robustness: alternative θ_k scaling and alternative Critical $_k$ classifier. Column (1) replicates the headline. Column (2) uses patent-share normalization. Columns (3)–(4) substitute the BIS-CCL ex-ante classifier. $\rho(\theta, \text{Classifier})$ is the bivariate correlation between θ_k (z-scored) and the binary classifier. HS4-clustered SEs.

	(1) Baseline (LZ raw)	(2) Patent-share (norm.)	(3) BIS-CCL ex-ante	(4) Both robustness
$\beta_3: \theta \times \text{Classifier}$	+0.132*** (0.039)	+0.132*** (0.039)	-0.828 (2.026)	-0.828 (2.026)
θ measure	LZ patents (z)	patent share (z)	LZ patents (z)	patent share (z)
Critical classifier	literature	literature	BIS-CCL ch.	BIS-CCL ch.
$\rho(\theta, \text{Classifier})$	-0.211	-0.211	+0.206	+0.206
HS4-clustered SE	Yes	Yes	Yes	Yes
Observations	52	52	52	52

Table 11: Test 2 robustness across three independently constructed Critical classifiers. Column (1) is the headline literature-based classifier (Atalay, Boehm, NDAA strategic-input scope). Column (2) is the ex-ante BIS Commerce Control List chapter classifier (regulatory boundary). Column (3) is the ex-ante NSF/NSTC Critical & Emerging Technologies list (national-security boundary). The three classifiers differ on whether non-bottleneck technologies (AI, quantum, software, basic transistors) are flagged as “Critical.” HS4-clustered SEs.

	(1) Literature Critical (28 HS6)	(2) BIS-CCL ex-ante (HS chapters)	(3) NSF CET ex-ante (32 HS6)
$\beta_3: \theta \times \text{Classif.}$	+0.132*** (0.039)	-0.828 (2.026)	-1.192 (0.834)
Source	bottleneck lit.	BIS CCL	WH NSTC
Construction	technological	regulatory	natl. security
$\rho(\theta, \text{Classif.})$	-0.211	+0.206	+0.254
HS4-clustered SE	Yes	Yes	Yes
Observations	52	52	52

Table 12: Test 3 robustness: cleaner placebo (HS 8533/8534/8536, passive components) and wild-cluster bootstrap- t (Webb 6-DOF, $B = 5,000$). Column (1) is the headline share regression on treated chips alone; columns (2)–(3) add the cleaner placebo set. All p -values reported in the table are wild-cluster bootstrap one-sided ($H_1 : \delta < 0$). The headline column (1) has wild-bootstrap one-sided $p = 0.021$ (equivalently two-sided $p = 0.042$, vs. CR1 two-sided $p = 0.056$).

	(1) log share treated only	(2) log share cleaner placebo DiD	(3) log value triple-DiD cleaner placebo
$\hat{\delta}_{\text{FDPR}}$	-0.211** (0.110)		
$\hat{\beta}$ Treated $\times$ Post		+0.054 (0.122)	
$\hat{\delta}$ T $\times$ C $\times$ P			+0.627 (0.237)
Wild-bootstrap one-sided p	0.021	0.677	0.991
Reporter $\times$ HS6 FE	Yes	Yes	Yes
Year FE	–	Yes	Yes
Cleaner placebo HS6	–	Yes	Yes
Observations	64	208	416
Clusters	8	26	26